

Coding Agents

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We recommend working with [AI coding agents](#) to [help you code](#).

1 What Is a Language Model?

A [large language model](#) (LLM) is a statistical model trained to predict the next token in a sequence of text, based on patterns learned from enormous amounts of text during training. That single capability — predicting what comes next — turns out to be enough to write code, answer questions, and hold a conversation, once the model is large enough and trained on enough data.

A base model trained only to predict text is not yet a helpful assistant. A further training step, often [reinforcement learning from human feedback](#), teaches the model to follow instructions, answer as a helpful assistant, and refuse harmful requests, rather than simply continuing whatever text it is given. This is the step that turns a raw language model into something like Claude or ChatGPT.

A model call is stateless: given the same input, it has no memory of any previous call. Every capability the rest of this chapter describes — holding a conversation, using tools, running autonomously as an agent — is scaffolding built on top of that one stateless function. The harness supplies the memory, the tools, and the control flow; the model only ever predicts what token comes next.

For a from-scratch walk through that stack, see Stanford's [CS336: Language Modeling from Scratch](#). The course has students implement:

- data collection and cleaning for pre-training
- transformer construction
- training
- evaluation

Lecture recordings are public. It is an implementation-heavy treatment of how a language model is built, not a product tutorial for coding agents.

2 What are AI coding agents?

AI coding agents are [AI agents](#) specialized for coding. They differ from other AI coding tools in important ways:

Compared to inline coding assistants (like traditional autocomplete), coding agents work autonomously rather than providing suggestions as you type. They can navigate entire codebases, execute commands, and complete multi-step tasks without constant human guidance.

Compared to AI chatbots (like ChatGPT or Claude), coding agents don't just generate code snippets in conversation—they actively interact with your development environment. While chatbots require you to copy code from a chat window and manually integrate it into your project, coding agents directly read your codebase, make changes to files, run tests and build commands, and create pull requests with their proposed changes. Chatbots are conversational assistants; coding agents are autonomous development tools.

Coding agents are autonomous software programs that can:

- **Understand and execute complex tasks:** Coding agents can interpret natural language instructions and break them down into actionable development tasks
- **Navigate and modify codebases:** They can read, understand, and edit multiple files across a repository to implement features or fix bugs
- **Run tools and commands:** Coding agents can execute build commands, run tests, use linters, and interact with development tools
- **Make decisions autonomously:** They can plan their approach, make technical decisions, and adjust their strategy based on results
- **Work iteratively:** Coding agents can test their changes, identify issues, and refine their solutions through multiple iterations
- **Create comprehensive solutions:** They can implement complete features that span multiple files, including code, tests, and documentation

Coding agents operate in isolated environments where they can safely experiment and validate changes before proposing them. This allows them to work more independently than inline coding assistants, which require step-by-step human direction. The agent workflow typically involves analyzing requirements, planning an implementation, making changes, testing those changes, and creating a pull request with the results.

While coding agents can handle substantial development tasks, they still require human oversight and review. The human developer remains responsible for:

- Reviewing the agent's work
- Ensuring the solution meets requirements
- Verifying code quality and security
- Making the final decision to merge changes

3 Coding-Agent Platforms

Coding-agent platforms differ more in where they run and how they return work than in their chat interfaces. Choose a platform by matching its execution model to the task and your repository’s security requirements.

Common coding-agent platforms

The following catalog is a starting point rather than an endorsement:

Platform	Primary surface	Typical repository workflow
GitHub Copilot coding agent	GitHub issues and pull requests	Assign an issue; review the resulting pull request
OpenAI Codex	Cloud tasks, app, and command line	Delegate a task in an isolated environment or work locally
Google Jules (Google 2026e)	Cloud coding agent	Connect a repository and review the proposed changes
Google Antigravity	Agentic development platform	Coordinate coding tasks in a managed development workspace
Claude Code	Terminal, IDE, and cloud	Work interactively or delegate work that returns a pull request
Gemini CLI	Command line	Inspect, edit, and test the current local checkout
Cursor (Cursor 2026)	IDE and cloud agents	Edit interactively with path-scoped <code>.cursor/rules</code> context, or hand a task to a cloud agent that pushes a branch
Aider	Command line	Pair locally with explicit files and Git commits
GitKraken Kepler	Agentic development environment (desktop)	Start a Task from an issue, PR, or idea; run multiple agents in parallel isolated worktrees via reusable Actions; review per-branch diffs and open PRs
OpenCode (OpenCode 2026a)	Terminal, desktop app, and IDE extension	Bring your own model provider (including local endpoints; see Section 25) and edit the local checkout

Platform	Primary surface	Typical repository workflow
Cline (Cline 2026)	VS Code extension, CLI, and SDK	Plan, then act with per-step approval; bring your own key, including Ollama
Kiro (Amazon Web Services 2026) (AWS)	IDE, CLI, web, and mobile	Spec-driven tasks locally, or delegate a web task that opens a pull request
Warp (Warp 2026)	Terminal app, standalone CLI, and cloud agents	Approve actions locally, or trigger cloud agents from Slack, Linear, or GitHub webhooks
OpenHands (OpenHands 2026)	Browser client, managed cloud, self-hosted backend, and CLI	Run headless or hosted; self-host when code must stay on your infrastructure
Devin (Cognition 2026) (Cognition)	Cloud agent and desktop editor	Delegate a task to an isolated cloud machine and review the pull request (see Section 11)
Ollama (Ollama 2026)	Local model runner (not an agent)	Serve open-weight models to any of the harnesses above (see Section 4)

Kepler is not an agent itself but an orchestration layer. It hosts agents you already use (Claude Code, Codex, Copilot, Cursor, OpenCode) rather than locking in one model, and builds on a decade of GitKraken plumbing for branches, worktrees, diffs, and merges. Where a single-agent platform handles one repo at a time, a Kepler Task can span many repos, and its Agent Graph visualizes every session, turn, tool call, and subagent live.

Where the agent runs

The platforms above sort into three execution models (measured 2026-09-09). The model decides where your code is copied, what the agent can reach, and how its work comes back to you.

- **Hosted sandbox.** The vendor provisions an isolated machine, clones the repository into it, and returns a branch or pull request. Jules clones the repository into a virtual machine and submits a pull request after you approve its plan (Google 2026e); Cursor cloud agents clone from GitHub, GitLab, Azure DevOps, or Bitbucket, work on a separate branch, and push it back (Cursor 2026); Kiro web tasks and Warp cloud agents follow the same shape (Amazon Web Services 2026; Warp 2026), as do the Copilot coding agent, Codex cloud tasks, Devin, and OpenHands Cloud. Nothing runs on your machine, so the questions are what the sandbox can reach and whether the vendor’s retention terms suit the repository.

- **Local checkout.** The agent runs on your workstation against the files already there, in a terminal (Claude Code, Codex CLI, OpenCode, Aider, Gemini CLI, Warp) or inside an editor (Cursor, Cline, Kiro, Copilot in VS Code). Code stays put, and the agent inherits whatever credentials and network access your shell has, which is why Section 21 asks for approval gates and a sandbox. Pairing one of these with Ollama keeps the model local too (Ollama 2026), at the cost of the hardware described in Section 23.
- **Orchestration layer.** Kepler, Kiro Crew, and the Cline Kanban surface do not add a model of their own; they run several of the agents above in parallel worktrees or sessions and present the results for review. Reach for one only once a single agent is no longer the bottleneck ([when orchestration helps](#)).

Several vendors now ship all three, so the platform name alone no longer tells you where the code goes: Claude Code, Codex, Cursor, Kiro, and Warp each offer a local surface and a hosted one. Check the execution model of the specific surface you enable. The Windsurf editor is a naming trap of the same kind: `windsurf.com` now redirects to Devin Desktop, Cognition’s rebranding of that editor (measured 2026-09-09; see Section 11).

For licensing, token-cost characteristics, and a decision table for choosing among these tools, see Section 11; for the open-weight and fully local end of the spectrum, see Section 4.

Before connecting any platform, check:

- whether code runs locally or in a vendor-managed environment;
- what repository, network, secret, and tool permissions it receives;
- whether changes arrive as a reviewable branch or pull request;
- which model providers and billing arrangements are supported; and
- whether your organization can retain the required audit trail.

Platform capabilities and commercial terms change quickly. Confirm current details in the linked official documentation before adopting one.

A separate category — named, persistent teammates with their own browser-and-shell computer, rather than a repository checkout — is reviewed in [Grok Bot and Alternatives](#). Those products do not replace the platforms above.

4 Catalog and Comparison of Coding Agents

The coding agent ecosystem encompasses a spectrum of architectures ranging from proprietary cloud-managed assistants to fully autonomous, open-weight local harnesses (measured 2026-09-01; see Section 3 for platform overviews). Choosing an agent architecture requires evaluating autonomy, privacy, tool integration, and hardware constraints.

Autonomous Open-Weight Agents: Hermes Agent

Hermes Agent (developed by Nous Research) is an autonomous agent harness built around the Hermes series of open-weight models (such as Hermes 3). Unlike simple completion models, Hermes Agent specializes in structured function calling, multi-step tool execution, and reflective task planning. Key capabilities include:

- **Structured Tool Calling:** Native support for JSON schema function calling, enabling reliable invocation of compilers, test runners, and file system manipulators.
- **Multi-Turn Reasoning & Reflection:** Iterative planning loops that inspect tool execution outputs, handle runtime errors, and backtrack when encountering obstacles.
- **Deployment Flexibility:** Can execute against self-hosted open-weight model instances (via vLLM, Ollama, or TensorRT-LLM) as well as hosted API endpoints.

Local Coding Agents with Ollama

For air-gapped environments, strict data privacy requirements, or zero-marginal-cost development, developers pair local model runners like **Ollama** with dedicated agent harnesses (detailed setup and hardware sizing are covered in Section 23):

- **Terminal Orchestrators:** Harnesses like **OpenCode** (see Section 25), **Aider**, and **OpenHands** (formerly OpenDevin) connect directly to Ollama endpoints running open-weight coding models (such as **Qwen2.5-Coder**, **DeepSeek-Coder-V2**, or **Llama 3.3**), managing git commits, multi-file edits, and automated test-and-repair loops.
- **Editor Integrations:** Extensions such as **Continue** and **CodeCompanion** embed local Ollama models directly into VS Code, JetBrains IDEs, and Neovim, providing inline autocompletion and interactive chat without transmitting code to cloud APIs.

Comparative Taxonomy of Coding Agent Architectures

The following table compares the primary paradigms across the coding agent landscape:

Dimension	Cloud Frontier Agents (Antigravity, Claude Code)	Autonomous Open-Weight (Hermes Agent, OpenHands)	Local Editor Assistants (Continue + Ollama)
Primary Deployment	Cloud-hosted frontier API	Self-hosted or local server	Local workstation (Ollama or llama.cpp)
Autonomy Level	Semi-autonomous to fully autonomous	High multi-turn autonomy	Interactive / inline assistance

Dimension	Cloud Frontier Agents (Antigravity, Claude Code)	Autonomous Open-Weight (Hermes Agent, OpenHands)	Local Editor Assistants (Continue + Ollama)
Privacy & Air-Gap	Cloud-dependent (enterprise VPC optional)	Fully air-gapped	100% offline & local
Tool Integration	MCP, terminal, browser, subagents	Bash execution, custom APIs	LSP, editor buffer manipulation
Compute Requirements	Minimal client hardware (network only)	High local GPU/VRAM (16GB-64GB+)	Moderate to high GPU/VRAM
Reasoning Capacity	Frontier reasoning & large context	Strong domain code reasoning	Bounded single-file reasoning

Trade-offs: Cloud Frontier vs. Local Open-Weight

When deciding between cloud frontier agents and local open-weight deployments:

1. **Complex Multi-File Refactoring:** Cloud frontier models (such as Claude Sonnet 4.5 or Gemini 3 Pro) currently maintain higher coherence across large, 50+ file refactors and architectural redesigns.
2. **Confidentiality and Compliance:** For proprietary codebases with strict regulatory constraints (HIPAA, defense, or sensitive enterprise IP), local agents powered by Ollama and Hermes Agent ensure that source code never leaves on-premises hardware (see [Section 23](#)).
3. **Cost Predictability:** Local open-weight harnesses incur fixed hardware capital expense but zero marginal token costs, making them attractive for high-volume automated test-and-repair loops.

5 What are AI harnesses?

An **AI harness** is the scaffolding built around a language model that turns it into an agent able to do real work. The model itself only predicts text; the harness is what lets it read files, run commands, call external tools and APIs, and carry state across turns and sessions.

Layers of a Harness

Most coding-agent harnesses — including the [GitHub Copilot coding agent](#) and [Claude Code](#) — share a similar set of layers:

- **Core loop:** the [tool-calling loop](#), permission and sandboxing model, and context management that keep the agent grounded in your repository.
- **Skills:** reusable, named procedures that encode a workflow so it runs the same way every time, instead of being re-improvised in each conversation. See [Section 33](#).
- **Subagents:** a way to spin up a worker with a fresh context window for a self-contained piece of research or work, keeping the main conversation’s context focused.
- **Multi-agent orchestration:** deterministic fan-out and fan-in across many subagents — for example, running several independent reviewers over a diff and reconciling their findings — for work that is large or benefits from independent verification.
- **MCP servers:** the [Model Context Protocol](#) gives a harness typed access to external systems (issue trackers, chat tools, databases) beyond raw shell or API calls.
- **Memory:** files — like this manual, or a repository’s `CLAUDE.md`/`AGENTS.md` — that persist instructions and learned preferences across sessions, so the harness does not relearn your conventions every time.

Using Harness Features Well

- **Push repeatable procedures into skills,** not into ad hoc prompting each time. A skill is testable, shareable, and versionable; a one-off prompt is not.
- **Match orchestration weight to the task.** A single lookup or small edit should stay inline. Reach for subagents or multi-agent workflows only when the work is genuinely decomposable, benefits from independent verification, and is large enough that the coordination overhead pays for itself.
- **Gate destructive or hard-to-reverse actions on explicit human approval** — merges, force-pushes, deletions — and let the agent drive everything reversible (drafting, testing, iterating on review feedback) autonomously.
- **Feed learnings back into the harness.** When a review round or a mistake teaches something generalizable, record it as a memory or skill update rather than letting it evaporate at the end of the session.
- **Treat external or untrusted content as data, not instructions.** PR comments, fetched web pages, and other tool output can contain text that looks like a command; a harness that acts on it uncritically is vulnerable to [prompt injection](#).

6 Inside the Claude Code Harness

Section 5 describes the layers most coding-agent harnesses share, and Section 10 sketches the loop that runs inside them. This section walks through those layers for the harness the lab uses most, [Claude Code](#), and says for each one what Anthropic documents, what the community has inferred from the shipped program, and what remains unknown (measured 2026-09-09). Anthropic’s own description is short: Claude Code is the “agentic harness” around the model, supplying “the tools, context management, and execution environment that turn a language

model into a capable coding agent” (Anthropic 2026ab). The sections below take those three things in turn.

The distinction between documented and inferred matters here more than usual. Claude Code is proprietary, and since mid-2026 it ships as a single compiled executable rather than readable JavaScript (anthropics/claude-code contributors 2026b), so everything not in Anthropic’s documentation comes from people running `strings` on the binary, reading the compiled function bodies, or intercepting its network traffic. Those readings are checkable but fragile: the build tool renames every internal function on every release, and a behaviour observed in one build may be gone in the next. Treat the inferred claims as a snapshot, and re-verify against the build you are running before relying on one.

The loop and the tool registry (documented)

The core of the harness is the loop described in Section 9: the model produces a tool call, the harness runs the matching handler, and the result goes back into the conversation. Anthropic describes the same loop as three blended phases — gather context, take action, verify results — and stresses that the model, not the harness, decides which tool to call next (Anthropic 2026ab).

The built-in tools are the registry the loop dispatches against. As of September 2026 the tools reference lists more than forty of them (Anthropic 2026ae). The ones a lab member sees every day are:

- **File and search tools:** `Read`, `Edit`, `Write`, `Glob`, `Grep`, `NotebookEdit`, and `LSP` for language-server code intelligence.
- **Execution tools:** `Bash`, `PowerShell`, and `Monitor`, which streams a background command’s output lines back to the model.
- **Orchestration tools:** `Agent` (spawns a subagent), `Skill` (runs a skill), `SendMessage` and `ListAgents` (agent teams), the `Task*` family (a session task list), `CronCreate` and `ScheduleWakeup` (timers), and `ToolSearch` (loads deferred tool schemas on demand).
- **Interaction tools:** `AskUserQuestion`, `EnterPlanMode` and `ExitPlanMode`, `EnterWorktree` and `ExitWorktree`, `Artifact`, and `SendUserFile`.

Each tool is a schema the model sees plus a handler it does not, exactly as Section 8 describes. The tool names double as the vocabulary of the permission system, hook matchers, and subagent tool lists, so a rule written as `Bash(git *)` refers to the same `Bash` entry the model calls (Anthropic 2026ae). MCP servers (below) extend this registry without changing its shape.

The system prompt: what is public and what is not

Anthropic does not publish Claude Code’s system prompt. What the documentation does describe is the *startup context*: everything already in the model’s window before you type a word. The context-window page walks through it in order (Anthropic 2026z):

- the system prompt itself, including an output style and any `--append-system-prompt` text, which “both go into the system prompt the same way”;
- an environment block (working directory, platform, shell, whether this is a git repository), with git branch, status, and recent commits loaded “as a separate block at the very end of the system prompt”;
- MCP tool *names*, with their full schemas deferred until `ToolSearch` loads them;
- one-line skill descriptions, so the model knows what it can invoke;
- your `CLAUDE.md` files and the first 200 lines of auto memory.

One documented detail changes how the rest of this section reads: `CLAUDE.md` content “is delivered as a user message after the system prompt, not as part of the system prompt itself” (Anthropic 2026ac). So the layered picture is harness-authored system prompt first, then your instructions, then the conversation. The same page is candid that the harness “treats them as context, not enforced configuration”: a `CLAUDE.md` instruction is prose the model may weigh against other prose, and anything that must hold regardless goes in a permission rule or a hook.

Three documented levers reach the system prompt directly (Anthropic 2026x):

- `--append-system-prompt` adds text to it for one invocation.
- `CLAUDE_CODE_SIMPLE_SYSTEM_PROMPT=1` requests “a shorter system prompt and abbreviated tool descriptions”, and 0 opts out “even on models where the experiment or server configuration would otherwise enable it” — the documentation’s own acknowledgment that the prompt varies by model and by server-side configuration.
- `--bare` (equivalently `CLAUDE_CODE_SIMPLE=1`) runs “with a minimal system prompt and only the Bash, file read, and file edit tools”, and disables discovery of hooks, skills, subagents, plugins, MCP servers, auto memory, and `CLAUDE.md`. It is a stripped harness, not a stripped prompt, which is why it is the wrong fix for the problem described next.

What the community has inferred is the prompt’s internal structure. Binary analysis reported in the Claude Code issue tracker shows the prompt assembled from named sections, each behind a function that returns text or `null`, with names such as `anti_verbosity`, `thinking_guidance`, `action_caution`, `delivering_work_max`, `overcorrection`, `subagent_steer_delegation`, `heron_brook`, and `autonomy_append` (anthropics/claude-code contributors 2026a, 2026b). Some sections are gated on a *model capability* declared in an internal model registry (the reported gate for Opus 5 is a capability string `opus_5_prompt_bundle`), some on a remote feature flag, and some on both. One analyst counted roughly 23 sections in a mid-2026 build, with user memory rendered around eighth and `heron_brook` second-to-last (anthropics/claude-code contributors 2026b). None of this is documented, the counts and names are build-specific, and the section order is one person’s inference from the compiled code — so the safe statement is that the system prompt is *modular and conditional*, not that any particular section exists on your machine today.

Hard-coded instructions, remote flags, and heron_brook

`heron_brook` is not a feature. It is the internal name of one system-prompt section, and its story is the clearest public window into how the harness’s prompt is controlled. The name follows the pattern of every Claude Code feature flag: a `tengu_` prefix (the product’s internal code name, per one community catalog of the binary’s strings (wtfwhs 2026)) followed by an auto-generated `adjective_noun` pair that carries no meaning of its own. `tengu_fennel_godwit`, mentioned below, is the same kind of name. Anthropic’s only public statement touching the section is the short reply quoted below.

May 2026: the slot appears. Claude Code 2.1.150 shipped with a changelog entry reading, in full, “Internal infrastructure improvements (no user-facing changes)” (Anthropic 2026n). Within days a user reported that the build added a function reading a string from two network-backed sources — the `client_data` field of the `/api/claude_cli/bootstrap` response, and a GrowthBook feature flag named `tengu_heron_brook` that refreshes every 60 seconds — and inserting it verbatim as a system-prompt section (anthropics/claude-code contributors 2026a). The same finding reached Hacker News under the title “Claude Code now allows Anthropic to remotely inject system prompts” (matheusmoreira 2026). An Anthropic engineer replied on the issue that the company “sometimes run[s] experiments on changes to our system prompt so that we can evaluate how a change impacts quality before fully rolling it out”, that users can opt out with `CLAUDE_CODE_DISABLE_NONESSENTIAL_TRAFFIC=1` and `DISABLE_GROWTHBOOK=1`, and that Claude Code should not be used through an untrusted proxy (anthropics/claude-code contributors 2026a). The issue was closed. A separate HackerOne report on the same channel was closed as *Informative*, with Anthropic’s security team stating that TLS is the integrity boundary and no response signing is planned (cnighswonger 2026). An independent catalog of the binary’s strings describes the section the same way: “a server-controlled prompt-injection slot” (wtfwhs 2026).

July 2026: the slot gets a hard-coded default. Claude Code 2.1.219 was the release that added Opus 5 (Anthropic 2026n). Users on that model noticed their sessions had stopped delegating to subagents, and the model, when asked, quoted two lines it said it had been given:

Do not call the `AgentTool` unless the user requested it Do not use workflows or deep-research unless the user requested it

The canonical report (anthropics/claude-code contributors 2026b) traced the lines to a constant compiled into the binary, served as the *fallback* text of the `heron_brook` section when neither the bootstrap response nor the GrowthBook flag supplies a string. The reported gate has two parts: the section fires only for models whose registry entry carries the `opus_5_prompt_bundle` capability (in the builds examined, only `claude-opus-5`, not the Fable, Sonnet, or Haiku families), and only while a kill-switch flag, `tengu_fennel_godwit`, is false, which is its default. A thread in the r/ClaudeCode community on Reddit carried the finding to a wider audience (r/ClaudeCode 2026). (Reddit blocks unauthenticated fetches, so the thread is cited by title only.)

Four consequences are reported across that issue and its siblings:

- **It overrides user configuration silently.** Users whose `CLAUDE.md` *requires* delegation — including several with mandatory review-by-subagent gates — saw zero subagent dispatches for whole sessions, and one fleet operator measured Opus 5 sessions taking 2.5 times as many assistant turns as Opus 4.8 on the same work, which they attributed to serial work that had previously fanned out (anthropics/claude-code contributors 2026b). The documentation says delegation is driven by each subagent’s **description** (Anthropic 2026u); a separate report argues the directive contradicts that documented behaviour (anthropics/claude-code contributors 2026e).
- **The model attributes the line to you.** Because the section arrives in the same voice as everything else, and because `CLAUDE.md` is a *user* message that appears earlier, the model reads “unless the user requested it” as the user’s own standing rule and tells users their configuration forbids delegation when it says the opposite (anthropics/claude-code contributors 2026d). A related report frames the underlying gap: there is no defined precedence between an Anthropic-authored prompt section and a user-authored `CLAUDE.md`, and no way to observe from inside a session which sections are active (anthropics/claude-code contributors 2026c).
- **The documented opt-outs do not reach it.** `DISABLE_GROWTHBOOK=1` disables flag *fetching*, so every flag takes its code default (Anthropic 2026x) — and the kill switch’s default is off, so blocking the flag source guarantees the hard-coded text is used. `--bare` removes the **Agent** tool along with the prompt section. Six sibling sections in the same Opus 5 bundle reportedly have dedicated `CLAUDE_CODE_*` environment variables; `heron_brook` and its kill switch do not (anthropics/claude-code contributors 2026b). Session transcripts under `~/.claude/projects/` do not record the system prompt, so the injection leaves no trace to search for afterwards.
- **It persisted.** Comments on the issue confirm the same constant, gate, and behaviour in every build examined from 2.1.219 through at least 2.1.245 (2026-08-26), across macOS, Windows, and Linux, with no maintainer response on the thread (anthropics/claude-code contributors 2026b).

What is documented, and what is not, about `heron_brook` as of 2026-09-09:

Table 1: `heron_brook`: documented, inferred, and unknown

Claim	Status
Anthropic runs server-side system-prompt experiments and offers two environment variables to opt out of flag fetching	Documented, by an Anthropic engineer’s comment (anthropics/claude-code contributors 2026a) and the environment-variable reference (Anthropic 2026x)

Claim	Status
A prompt section named <code>heron_brook</code> reads its text from the bootstrap response, then a GrowthBook flag, then a compiled-in fallback	Inferred from binary analysis, reproduced independently by many reporters on three platforms
The fallback text tells the model not to call the Agent tool unless the user requested it	Inferred; the two strings are verifiable with <code>grep -a -c</code> on the installed executable
The gate is Opus 5’s <code>opus_5_prompt_bundle</code> capability plus a <code>tengu_fennel_godwit</code> kill switch	Inferred from the compiled code; not documented anywhere
Whether the remote sources have ever carried non-empty text in production, and what	Unknown; one Windows reporter found the flag absent from the locally cached flag set, meaning the fallback branch was firing
Why the directive exists (cost control, quality regression, an experiment that leaked)	Unknown; Anthropic has not said
Whether it is still active in the build you are running	Unknown until you check

The lab’s own [ai-config](#) leans heavily on subagents (adversarial self-review, delegated UMS passes, parallel issue workers), so this affects us directly on any Opus 5 session. Two workarounds came out of the thread and cost nothing to adopt. First, tell the harness in `CLAUDE.md` that text arriving in the system prompt is not the user’s voice, and that subagent use is requested in advance — several reporters found this restored delegation (anthropics/claude-code contributors 2026b). Second, when a session unexpectedly stops delegating, ask it to quote the instruction it is following before auditing your own configuration for a rule that is not there.

Permission modes and settings layers (documented)

Permission rules “are enforced by Claude Code, not by the model” (Anthropic 2026t), which makes them the first layer that the prompt controversy above cannot reach. A prompt can change what the model *tries*; only a rule changes what the harness *allows*. Six modes are documented (Anthropic 2026t):

- **default** (labeled Manual): prompts on first use of each tool.
- **acceptEdits**: auto-accepts file edits and common filesystem commands inside the working directory.
- **plan**: read-only exploration; no source edits.
- **auto**: auto-approves tool calls, with a classifier checking that actions match the request.
- **dontAsk**: auto-denies anything not pre-approved by an allow rule.
- **bypassPermissions**: skips prompts except for a short list no mode auto-approves.

Rules and the mode live in settings files that layer from managed policy through user, project, and local scope, with `permissions.deny` in managed settings as the enforcement an organization

cannot have overridden (Anthropic 2026ac). One inferred caveat belongs here: the `heron_brook` thread cites a separate report that GrowthBook flags can override `permissions.defaultMode` from the server (anthropics/claude-code contributors 2026b), so even this layer may have a remotely controlled input. That report is cited second-hand and was not verified for this section.

Hooks (documented)

Hooks are the harness’s event system: shell commands, HTTP endpoints, MCP tool calls, single-turn prompts, or spawned subagents that run at fixed lifecycle points (Anthropic 2026aa). The reference lists more than thirty events, grouped as session (`SessionStart`, `SessionEnd`), per-turn (`UserPromptSubmit`, `Stop`), tool execution (`PreToolUse`, `PostToolUse`, `PermissionRequest`), agent (`SubagentStart`, `SubagentStop`), context (`InstructionsLoaded`, `PreCompact`, `PostCompact`), and several more. A `PreToolUse` hook that exits with code 2 blocks the call, and a hook’s JSON output can deny, add context, or rewrite the tool’s input. This is the mechanism Section 39 calls “the part the model cannot talk its way around”, and it is the right home for any rule that must hold regardless of what the prompt says — including a rule that the prompt itself has been told to contradict. `InstructionsLoaded` is also the only documented way to log exactly which instruction files a session loaded and when (Anthropic 2026ac).

Subagents (documented)

The `Agent` tool spawns a subagent: a fresh instance of the same loop with its own context window, its own system prompt, a restricted tool list, and independent permissions (Anthropic 2026u). Built-in subagents include `Explore` and `Plan` (read-only; they skip `CLAUDE.md` and git status to stay cheap) and `general-purpose`; custom ones are markdown files with front matter in `.claude/agents/` or `~/.claude/agents/`, as Section 8 illustrates, and plugins can ship more. The harness watches those directories and picks up edits without a restart. Delegation is documented as description-driven: “Claude uses each subagent’s description to decide when to delegate tasks”, and the combined descriptions are capped at 15,000 tokens before a startup warning (Anthropic 2026u). Delegation can be removed entirely by denying the `Agent` tool in `permissions.deny`. That is the documented, user-controlled way to stop subagent use; `heron_brook` is the undocumented, server-controlled one.

Skills and plugins (documented)

A skill is a `SKILL.md` with front matter and a body, run through the built-in `Skill` tool rather than as a tool of its own (Anthropic 2026ae). Only skill *descriptions* sit in the startup context; the body loads when the skill is invoked, and a skill marked `disable-model-invocation: true` stays out of context entirely until you type its slash command (Anthropic 2026z). Section 33 covers the format. A plugin bundles skills, agents, hooks, MCP and LSP server

configurations, background monitors, and default settings into one directory with a `.claude-plugin/plugin.json` manifest, namespaces its skills as `/plugin-name:skill`, and can be loaded from a marketplace, from a local path with `--plugin-dir`, or from the user's skills directory (Anthropic 2026v). Section 63 dissects the bundle, and Section 40 describes how the lab's own plugin reaches a machine.

MCP servers (documented)

An MCP server adds tools to the registry without changing the loop. By default the harness lists only the server's tool *names* at startup and defers the full schemas, loading each on demand through `ToolSearch`; `ENABLE_TOOL_SEARCH=false` loads everything up front (Anthropic 2026z). That deferral is why a session can carry dozens of servers without paying their schema cost on every turn. Two dedicated tools, `ListMcpResourcesTool` and `ReadMcpResourceTool`, expose server resources as well as tools (Anthropic 2026ae). Section 59 covers registration and its failure modes.

Memory and CLAUDE.md loading (documented)

Two mechanisms carry knowledge across sessions (Anthropic 2026ac):

- **CLAUDE.md files**, loaded in a fixed order: managed policy, then `~/.claude/CLAUDE.md`, then every `CLAUDE.md` and `CLAUDE.local.md` from the filesystem root down to the working directory, concatenated rather than overriding each other. Files in subdirectories load on demand when the model reads files there. `@path` imports expand at launch to a depth of four, `.claude/rules/*.md` files load alongside, with `paths:` front matter scoping a rule to matching files, and block-level HTML comments are stripped before injection. A file over 4 MiB is skipped.
- **Auto memory**, notes the model writes itself under `~/.claude/projects/<project>/memory/`, of which the first 200 lines or 25 KB of `MEMORY.md` load every session and topic files load on demand.

`/context` shows which files actually loaded, and `/memory` opens them. The “delivered as a user message” detail above applies to all of it.

Compaction (documented)

When the context window nears its limit, the harness summarizes the conversation and replaces it; `/compact` does the same on demand, optionally with a focus instruction, and `/autocompact <tokens>` moves the threshold (Anthropic 2026z). What survives is specific:

- the system prompt and output style are unchanged, because they were never in message history;

- project-root `CLAUDE.md`, rules without a `paths:` field, auto memory, and a plan-mode plan are re-injected from disk;
- path-scoped rules and nested `CLAUDE.md` files reload only as matching files are read again;
- up to five recently modified files are re-read;
- invoked skill bodies are re-injected, capped at 5,000 tokens each and 25,000 total;
- context that hooks added earlier is summarized away, unless a `SessionStart` hook matching the `compact` source re-adds it;
- the skill *listing* is not reloaded (Anthropic 2026z).

The lab’s `compress-session` practice exists because the automatic summary guesses what matters; running `/compact` with a focus before it triggers keeps that choice with you.

The Agent SDK relationship (documented)

The Claude Agent SDK “gives you the same tools, agent loop, and context management that power Claude Code, programmable in Python and TypeScript” (Anthropic 2026a). It is the harness as a library: built-in tools, hooks, subagents, MCP, permissions, sessions, skills, memory, and plugins are all listed as SDK capabilities, and it loads `.claude/` and `~/.claude/` configuration “same as Claude Code”. Anthropic distinguishes it from the Client SDK (raw API access, where you write the loop yourself), from the CLI (interactive use), and from Managed Agents (a hosted product where Anthropic runs the sandbox). Other languages drive the same loop by running the CLI as a subprocess with `-p` and `--output-format json` (Anthropic 2026a). The corollary for this section is that everything above, including the modular and conditional system prompt, is what an SDK-built agent inherits. The SDK documents a way to replace or extend the system prompt, which is the supported route where the CLI offers only `--append-system-prompt`.

Summary: what you can and cannot see

Claude Code’s execution engine, tool registry, permission system, hooks, subagents, skills, plugins, MCP integration, memory loading, compaction, and SDK surface are all documented in detail, and the documentation is unusually precise about what loads when. The one layer it does not describe is the system prompt, and the `heron_brook` episode shows that layer is modular, model-gated, and partly remote-controlled, with no in-session way to observe which parts are active. The practical rule for the lab follows from the documentation’s own advice: anything that must hold goes in a permission rule or a hook, which the harness enforces, not in `CLAUDE.md`, which the model weighs against text you cannot see.

7 How Agents Are Structured and Implemented

An **agent** is not part of the harness itself. It is a configuration — a goal, a role, a bounded toolset, and a stopping condition — executed on top of the harness’s core loop (see Section 5). A single harness can host many different agents at once: a main conversation, and any number of subagents it spawns.

The Shape of an Agent

Structurally, an agent is a small record plus a fresh execution of the harness’s loop:

- **Identity:** a name and description, used to route a task to the right agent (“when should this agent be picked?”).
- **Instructions:** a system-prompt fragment that specializes behavior, for example “you are a read-only search agent.”
- **Tool allowlist:** a subset of the harness’s tool registry this agent may call — often narrower than the caller’s own toolset.
- **Model and effort:** which model backs the agent, and how much reasoning depth it applies; these can differ from the caller’s own settings.
- **Output contract:** whether the agent returns free text, or must call a schema-validated tool to return a typed result.

How an Agent Runs

1. **Spawn:** allocate a fresh message history with no inherited conversation — just the agent’s instructions, plus whatever prompt the caller wrote. A subagent prompt needs to be self-contained for this reason: brief it like a colleague who just walked into the room.
2. **Run:** execute the harness’s core loop (model call, parse tool calls, execute against the allowlist, append results, repeat), the same machinery the main session uses, just bound to a narrower toolset and a different system prompt.
3. **Terminate:** stop when the model emits a final answer with no further tool calls, when a schema-validated call satisfies the output contract, when it hits an error or a budget ceiling, or when the caller kills it.
4. **Return:** everything that happened inside the agent — every tool call, every intermediate step — is discarded from the caller’s context. Only the final text or validated object crosses back. This is the point of an agent: it is a context-isolation boundary, not just a prompt.

Composability and Its Limits

Agents can spawn agents: an orchestration layer runs many agent instances, some concurrently, and composes their results. Nesting is deliberately capped, usually to one level, because unbounded recursion has no natural stopping point and burns cost and time with no guardrail. An orchestration script is a scheduler over independent agent-loop instances, not a different execution model.

Two Axes That Define an Agent's Behavior

- **Isolation versus continuation:** a subagent gets no inherited context (isolation); a resumed agent keeps its own accumulated history and continues it (continuation). Both use the same loop machinery, differing only in history-management policy.
- **Free-form versus structured output:** by default an agent returns prose. Given a schema, it is forced to call a structured-output tool instead, turning it into a typed function from the caller's point of view — input in, validated object out — even though internally it is still a multi-turn loop.

8 How Harnesses and Agents Are Built

The layers described above are not all built the same way. Some are ordinary software; others are just text files the harness reads at runtime.

The Execution Engine Is Ordinary Software

The program that runs the core loop — calling the model, parsing tool calls, enforcing permissions, managing the sandbox — is compiled or interpreted source code, the same as any other application. There is no markdown involved here; this layer is what makes a harness a harness, rather than just a prompt someone wrote.

Agent and Skill Definitions Are Markdown with a Front Matter Header

An agent's identity, and a skill's metadata, are usually just a markdown file with a [YAML front matter](#) header. For example, a custom [Claude Code subagent](#) defined in `.claude/agents/code-reviewer.md`:

```
---
name: code-reviewer
description: Reviews diffs for bugs and style issues.
tools: Read, Grep, Glob
model: sonnet
```

```
---  
  
You are a meticulous code reviewer. Focus on correctness,  
security, and idiomatic style.
```

The front matter is parsed as structured configuration (name, description, allowed tools, model); the markdown body below it becomes that agent's system prompt, verbatim. An [Agent Skill's SKILL.md](#) follows the same shape: front matter for discovery metadata, a markdown body for instructions, and an optional folder of bundled scripts or reference files alongside it. No compilation step is involved; the harness reads the file and uses it directly.

Tools Are a Schema Paired with a Handler

A tool definition has two parts: a [JSON Schema](#) describing its parameters, which is the only part the model ever sees, and a handler function, ordinary code that performs the actual action (reading a file, running a command, calling an API). The schema is declarative data; the handler is real software the model never inspects or writes.

Orchestration Needs Real Code

Multi-agent orchestration cannot be expressed declaratively, because it needs genuine control flow — loops, conditionals, parallel fan-out with a concurrency limit. So orchestration scripts are literal source files, executed by the harness, not parsed as prompt text the way an agent definition is.

Memory Is Just Prose

Files like this manual, or a repository's [CLAUDE.md](#)/[AGENTS.md](#), carry no front matter and no schema. They are concatenated into the system prompt as plain text, and the harness trusts the model to read and follow that prose, the same way it follows any other instruction in its context.

9 What Kind of Program Is an Agent?

An agent is not a standalone program that does the reasoning itself. It is an [orchestration](#) program: something closer in shape to a chat client or a build tool than to a compiler or a web server.

It Is I/O-Bound, Not Compute-Bound

The actual token prediction happens on remote inference infrastructure, reached over HTTPS. The agent process itself does no heavy computation; it spends almost all its wall-clock time waiting — for a model API response, for a shell command to finish, for a file read. Structurally it is an **event-loop** program, the same category as a network client. Its core loop can be sketched in a few lines:

```
# Start the conversation with the agent's instructions and the task.
history = [system_prompt, user_message]

while True:
    # Send everything so far to the model, along with what it's allowed to call.
    response = call_model(history, tools=tool_schemas)
    history.append(response)

    # No tool calls means the model gave a final answer -- stop.
    if not response.tool_calls:
        break

    # Otherwise, run each requested tool and feed the result back in,
    # so the next model call can see what happened.
    for call in response.tool_calls:
        result = tool_registry[call.name](call.arguments)
        history.append(result)
```

Everything a harness adds — permissions, sandboxing, memory, subagents — is scaffolding wrapped around this loop, not a replacement for it.

Real Open-Source Examples

Because most production coding-agent harnesses are closed source, the clearest way to see this shape in real code is to read an open-source one:

- **aider** — an open-source AI pair-programming CLI.
- **SWE-agent** — a research coding-agent harness from Princeton NLP, described in its associated paper.
- **OpenHands** (formerly OpenDevin) — a general-purpose open-source agent platform.

Their orchestration code runs to thousands of lines, because that is where the real engineering lives: retries, streaming, permission checks, and state management. A single *agent definition* running on top of that engine, by contrast, is typically tens of lines (see Section 8).

Where It Runs

- **The harness process:** an ordinary OS process, either on your own machine (CLI mode) or inside an ephemeral, managed cloud container (remote/web mode) that is discarded when the session ends.
- **Subagents:** run inside the *same* host process as their caller, not a separate container. They differ only in having their own message history and a narrower tool set, unless a workflow explicitly asks for a separate git [worktree](#) to avoid file conflicts during parallel edits.
- **The model call itself:** not part of the agent’s environment at all. It is a network request to inference infrastructure the agent has no visibility into, beyond the request and response.

So an agent’s lifetime is scoped to a single task, not persistent: it starts when given a goal, runs for as long as its loop keeps producing tool calls, and ends the moment a stopping condition fires.

10 How Does a Harness Relate to an Agent?

The relationship between a harness and an agent is closer to an [interpreter](#) running a program than to two peers calling each other.

Does the Harness Call the Agent, or the Agent Call the Harness?

Harness to agent: not a call, an instantiation. The harness does not “call” an agent as a subroutine it invokes and waits on. An agent has no code of its own outside the harness’s loop (see Section 9) — its whole behavior *is* that loop, running with the agent’s configuration (instructions, tool allowlist, model) loaded in. The harness instantiates and runs an agent, start to termination; it is not a function call with a return address.

Agent to harness: yes, a real call, via tool calls. While an agent’s loop is running, the model produces a [tool-call request](#), and the harness’s dispatcher looks up and executes the matching handler — read a file, run a command, call an API. So the concrete direction of calling is **agent calls harness**, through tool dispatch, not the reverse.

Agent to agent: routed through the harness. When a parent agent spawns a subagent, it does not call that subagent directly. It issues a tool call that the harness’s dispatcher handles by spinning up a fresh instance of its own loop (see Section 7), running it to completion with the subagent’s configuration, and handing the result back to the parent as a tool result. Even “agent calls agent” bottoms out as: parent calls harness, harness instantiates and runs a new agent, harness returns that agent’s output to the parent.

Sketching the Harness's Own Loop

The [agent loop](#) sketched earlier is really just the innermost piece. The harness wraps a bootstrap step and a permission/dispatch layer around it:

```
def run_harness():
    # Load everything the loop will need before any conversation starts.
    tools = load_tool_registry()          # built-ins, plus whatever MCP servers expose
    memory = load_memory(CLAUDE_MD_PATHS) # CLAUDE.md / AGENTS.md, concatenated

    # The "main session" is just the harness's own loop, run with a default,
    # unrestricted configuration -- not a separate program.
    main_agent = Agent(config=default_config, system_prompt=memory)
    return run_agent(main_agent, tools)

def run_agent(agent, tools):
    history = [agent.system_prompt, agent.first_message]
    while True:
        response = call_model(history, tools=agent.tool_schemas)
        history.append(response)
        if not response.tool_calls:
            break
        for call in response.tool_calls:
            # Every tool call passes through the harness's own gate first,
            # regardless of which agent requested it.
            check_permission(call)

            if call.name == "spawn_subagent":
                # A subagent is not called directly -- the harness recurses
                # into a fresh instance of this same loop, then hands the
                # finished result back as an ordinary tool result.
                result = run_agent(Agent(call.arguments), tools)
            else:
                result = tools[call.name](call.arguments)

            history.append(result)
    return history[-1]
```

`run_agent` is identical in shape to the loop in [Section 9](#). `run_harness` and the permission check are the parts that only exist at the harness level, not inside any individual agent. That recursive call — `run_agent` calling itself for a subagent — is the concrete mechanism behind “agent calls agent, routed through the harness,” described in the previous subsection.

What Do You Launch When You Type `claude`?

Typing `claude` at a shell starts the harness process: it initializes the engine — the permission system, the tool registry, MCP client connections, and memory loaded from `CLAUDE.md`/`AGENTS.md` files. But the harness does not sit idle waiting for a program to be supplied separately. It immediately instantiates the **default agent** — the “main session” — to handle the interactive conversation: full tool access, a system prompt assembled from the loaded config, no restricted allowlist. That default agent is simply the harness’s baseline configuration for its own loop, not a second thing launched afterward.

There is no observable moment of “harness running, no agent yet.” The closest analogy is typing `python` at a shell: it launches the interpreter *and* drops you straight into a `REPL` evaluating your input, rather than leaving the interpreter idle with nothing loaded. The difference is that the harness’s default “program” is built in (the main-session agent’s configuration), rather than something you must supply. A custom subagent or a `.claude/agents/*.md` definition, by contrast, *is* a separate agent, instantiated on demand, mid-session, when the already-running main agent issues a tool call for it.

So typing `claude` launches the harness, and that act inherently instantiates the default agent that handles the session: **harness** names the engine and process; **agent** names the particular loop instance and configuration currently running inside it. At startup, those two come into existence together.

11 The Harness Landscape in 2026

Section 5 explains what a harness is, and Section 3 lists common platforms as a starting point. This section is the wider map: which harnesses exist as of August 2026, how they group, and how to choose among them.

 A fast-moving, opinionated snapshot

This section condenses a research summary compiled in August 2026 ([issue #95](#)), which drew on vendor announcements, project repositories, and community discussion. Harness releases arrive weekly, default models changed several times during 2026, and the corporate landscape reshuffled more than once, so treat every named version, figure, and ownership claim as dated to that month. Check the linked project pages before acting on any of it.

The organizing idea: **agent = model + harness**

The consensus across the comparisons surveyed is that the frontier models have converged on the standard coding benchmarks, so the harness layers that Section 5 describes now do most of the differentiating. The commonly cited puzzle is that the same Claude model performs

noticeably better inside Claude Code than inside a model-agnostic harness, and the answer offered is repeatedly the same word: the harness.

Two consequences follow for reading the rest of this section:

- **Benchmarks are saturating.** Leading models cluster within about a point on SWE-bench Verified, which is widely regarded as near its ceiling and exposed to training-data contamination. Harder successors (SWE-bench Pro, Terminal-Bench 2.x) show much lower absolute scores and separate tools better, but their figures are vendor-reported and version-specific.
- **Your own tasks are the benchmark that matters.** A day of real work on a repository you ship reveals more than any leaderboard, which is why Table 4 ends in a pilot rather than a ranking.

Terminal-first harnesses

Table 2: Terminal-first coding harnesses

Harness	License	Character (August 2026)
Claude Code (Anthropic)	Proprietary	The deepest programmable harness: hooks, skills, plugins, subagents, MCP, and multi-agent workflows. Widely rated first on code quality, and consistently the heaviest token spender, with real volume starting on the higher subscription tiers.
OpenAI Codex	Apache 2.0 CLI; proprietary cloud	The reach leader: terminal, cloud, IDE, ChatGPT app, and browser surfaces. Strong on terminal benchmarks, sandboxed at the OS level, markedly more token-efficient than Claude Code, and now carrying skills, plugins, subagents, and a small hook system.

Harness	License	Character (August 2026)
OpenCode	MIT	The most-starred open harness: model-agnostic across many providers including local models, client-server architecture with a headless server mode, custom agents, MCP, and AGENTS.md . Claude works only through an API key, not a subscription login.
Aider	Apache 2.0	The git-native pioneer: maps the repository and commits each edit. Reported to be in maintenance mode since early 2026, still respected for auditability and local-model flexibility.
Goose (Block, now Linux Foundation)	Apache 2.0	Rust terminal agent with vendor-neutral governance, many model providers and MCP extensions, parallel subagents, and Agent Client Protocol support. Local-first with no hosted option.
Gemini CLI (Google)	Apache 2.0	Being folded into Antigravity CLI; issue #95 records free individual access ending on 2026-06-18, with enterprise Code Assist licenses unaffected.
Amp	Proprietary	Deliberately optimizes for the best outcome regardless of token cost: multi-model routing, parallel subagents, and an “Oracle” deep-reasoning mode. Spun out of Sourcegraph.

Editor-integrated harnesses

Table 3: Editor-integrated coding harnesses

Harness	License	Character (August 2026)
Cursor	Proprietary	The in-editor leader: a VS Code fork with a deep agent loop, an in-house model tuned for fast agentic editing, and rules, MCP, hooks, skills, plugins, and subagents. Reads <code>.cursor</code> , <code>.claude/agents</code> , and <code>.codex/agents</code> configuration, per issue #95.
GitHub Copilot	Proprietary	The most widely distributed option, multi-model, with a cloud agent that turns an issue into a pull request in an ephemeral Actions environment; framed by GitHub for low-to-medium complexity tasks in well-tested codebases.
Cline	Apache 2.0	The open-source VS Code standard: a plan-then-act workflow with human approval at each step, bring-your-own-key across many providers including local Ollama, and a Kanban surface for orchestrating parallel agents. The usual pick for audit-sensitive environments.
Google Antigravity	Proprietary	Google’s agent-first development platform, built around an agent manager surface and now the center of its coding strategy; early reception focused on rate limits.

Harness	License	Character (August 2026)
Devin Desktop (Cognition)	Proprietary	The former Windsurf editor, rebranded in June 2026 with a local Devin agent, an agent command center, and Agent Client Protocol support.
Continue , Tabnine , Kilo Code , Zed	Mixed	Specialists: Continue adds an open-source agent to an existing editor; Tabnine is the on-premises, air-gapped enterprise pick; Kilo Code carries on the Roo Code lineage after that project's 2026 shutdown; Zed is an open-source AI-native editor.

Autonomous and cloud harnesses

- **Devin** (Cognition) is the archetype: delegate a task, and parallel managed instances each run in an isolated cloud machine and open pull requests. Early trials were poor, and the product has matured since.
- **OpenHands** (MIT) is the open-source autonomous leader, running headless in CI to browse, edit, test, and retry.
- **The cloud tiers of interactive tools** (Copilot's cloud agent, Codex Cloud, Cursor's cloud agents) share one design assumption: the bottleneck is no longer what one agent can do, but how many you can direct and review at once. [When orchestration helps](#) covers when that parallelism is worth having.

The minimalist counter-trend

Against the deep-harness orthodoxy, a 2026 movement argues that frontier models already know what a coding agent is, so a harness should get out of the way. Its flagship is **Pi** (MIT), built by Mario Zechner and championed by Armin Ronacher: four tools (read, bash, edit, write), a system prompt plus tool definitions under a thousand tokens, no MCP, plan mode, to-do list, or subagents, and extension by asking the agent to write its own TypeScript extensions. In its author's own Terminal-Bench 2.0 run ([post dated 2025-11-30, on Claude Opus 4.5](#)) it placed eighth, which its author read as evidence that a minimal harness competes with the heavy ones.

The maximalist reply is that the extra tokens in a heavy harness encode product features (memory, scheduled tasks, subagents, plan mode, worktree support) rather than waste. Even

Ronacher concedes that task queues, orchestration, and durable sessions will matter more over time, so the two camps may converge.

Open source versus proprietary

Open source, where you pay only your model provider or run local models:

- OpenCode
- Codex CLI
- Cline
- Aider
- OpenHands
- Goose
- Continue
- Gemini CLI
- Pi
- Zed

Proprietary:

- Claude Code
- Cursor
- GitHub Copilot
- Devin
- Amp
- Antigravity
- Replit Agent
- Tabnine

For raw capability the proprietary leaders still edge ahead, but the open options are competitive and win outright on cost, privacy, auditability, and model freedom. The 2026 signature of open-source use is bring-your-own-key, with two friction points worth knowing. Anthropic restricted third-party harnesses' use of flat-rate Claude subscriptions in April 2026, which pushed some of those users onto extra-usage or per-token billing (Section 27 covers the same rule from the Claude Code side), and OpenCode lost Claude Pro and Max login access after a dispute with Anthropic.

Local-model coding

Local coding has matured to the point where the strongest open-weight models compete on the standard benchmarks. The community favorites for consumer hardware are the Qwen, Devstral, Gemma, and DeepSeek families, and the typical stack pairs OpenCode or Aider with a local OpenAI-compatible endpoint. Section 23 and Section 28 cover the mechanics and the model choice.

Beyond coding: general agent frameworks

The general agent-framework layer settled into three tiers:

- **Graph-based orchestration:** [LangGraph](#), the production default for stateful, auditable, human-in-the-loop workflows, and Mastra for TypeScript.
- **Role-based multi-agent:** [CrewAI](#), and AutoGen, now merged with Semantic Kernel into the [Microsoft Agent Framework](#).
- **Lab SDKs:** the OpenAI Agents SDK, Anthropic’s Claude Agent SDK, and Google’s ADK.

MCP and the Agent-to-Agent protocol moved to Linux Foundation stewardship, and `AGENTS.md` and `SKILL.md` have become cross-tool standards in practice. The practitioner warning that recurs in production surveys is that most deployed agents are short-horizon and single-agent, so the prevailing advice is to start single-agent; [when orchestration helps](#) makes the same case for this lab’s own use.

Browser-use agents are a fast-growing adjacent category, and they remain structurally vulnerable to indirect prompt injection; do not give one credentials or financial access without sandboxing and human checkpoints.

Choosing

Table 4: Matching a harness to the job

If you want	Reach for
Maximum code quality and programmable depth, and can absorb the token cost	Claude Code on a high-volume plan; drop to Codex or Cursor if the bill or the task mix does not justify it
Cross-surface continuity, terminal tasks, and lower cost	Codex, the safest default on price for most teams
To stay in the editor	Cursor, or Copilot if you are GitHub-native and want issue-to-pull-request automation
Open source, self-hosting, local models, or auditability	Cline (editor, approval-gated), OpenCode (terminal, headless), or OpenHands (autonomous CI); Tabnine for air-gapped deployments
Hands-off delegation of well-scoped backlog work	Devin or the cloud agents in Codex, Cursor, and Copilot, with humans still reviewing the pull requests
Control, observability, and cheap parallel runs in a sandbox	Pi or Goose; stay with a heavier harness for long unattended runs

Then pilot two of them on a repository you actually ship. Token or usage-limit pain points toward Codex or an open bring-your-own-key harness; a need for audit trails points toward Cline or LangGraph; a need for private inference points toward OpenCode or Aider with a local model.

12 Custom Harnesses and How to Build Our Own

Section 5 defines a harness, Section 8 explains which of its layers are code and which are prose, and Section 11 maps the vendor harnesses. This section covers the layer that sits on top of those: the **custom harnesses** that individuals and small teams build for themselves, and what building one would mean for this lab (measured 2026-09-09 unless stated otherwise).

Two things people mean by “custom harness”

The projects surveyed here split into two kinds, and the distinction decides how much work a custom harness is:

- **A configuration layer on a vendor harness.** The execution engine stays Claude Code, Codex, or OpenCode, and the “harness” is a plugin: skills, hooks, subagent definitions, slash commands, and memory files that the vendor harness reads at runtime (the markdown-plus-front-matter layer of Section 8). `oh-my-claudecode`, Hoyeon, Superpowers, and GSD Core are all this kind.
- **A standalone runtime.** The tool-calling loop, permission model, and tool handlers are your own code, and the model is called through an API. `OpenHarness` and the `LangChain` guide are this kind.

The first kind is days of work and inherits every vendor upgrade for free. The second is months of work and gives you control over the loop itself, which is the only reason to pay for it.

Surveyed projects

Table 5 lists the projects surveyed.

Table 5: Custom harnesses surveyed for this section

Project	Kind	Runs on	License	Size (2026-09-09)
oh-my-claudecode (Heo 2026)	layer	Claude Code (plus CLI workers for Codex, Gemini, Antigravity, Grok, Cursor)	MIT	39.1k stars
Hoyeon (team-attention 2026)	layer	Claude Code	MIT	174 stars; last push 2026-05-21
Superpowers (Vincent 2026b)	layer	Claude Code, Codex, Cursor, OpenCode, Pi, Hermes, and eight others	MIT	283.8k stars; on the official Claude plugin marketplace
GSD Core (open-gsd 2026)	layer	Claude Code, OpenCode, Codex, Copilot, Cursor, and four others	MIT	9.3k stars
OpenHarness (HKUDS 2026)	runtime	any Anthropic- or OpenAI- compatible endpoint	MIT	15.7k stars; last push 2026-06-04
LangChain create_agent (Runkle 2026)	runtime	any LangChain model	MIT (library)	guide, not a product

oh-my-claudecode (OMC)

The README states: “Multi-agent orchestration for Claude Code. Zero learning curve.” (Heo 2026). It installs as a Claude Code plugin (`/plugin marketplace add`, then `/plugin install oh-my-claudecode`) or as an npm CLI, and adds:

- execution modes: a staged `team` pipeline (`team-plan`, `team-prd`, `team-exec`, `team-verify`, `team-fix` in a loop), `autopilot` for single-agent autonomy, and `ralph` for persistence until a task verifies;
- 19 specialized subagents with a model-by-agent compatibility matrix and routing that sends simple tasks to Haiku and reasoning to Opus, which the README says “saves

30-50% on tokens” without a stated methodology;

- `/skillify`, which extracts a reusable skill from a session into `.omc/skills/` (project) or `~/.omc/skills/` (user) and auto-injects matching skills later;
- a HUD status line, cost tracking, Discord/Telegram/Slack notifications, and a `tmux`-based mode that drives other vendors’ CLIs as workers.

It is the same family as the Oh My OpenCode project in Section 65 (the npm package is still named `oh-my-claude-sisyphus`), and by star count the most-used custom harness in this survey after Superpowers.

Useful to us? Partly. Its model routing and skill extraction are things Morrison-Lab/`ai-config` already does with `select-model` and `ums`, so installing OMC on top would double them. Its notification integrations and HUD are the pieces we lack, and they are separable.

Hoyeon

Hoyeon calls itself a “Requirements-first Harness — derive, verify, execute” (team-attention 2026). Its idea is that most agent failures are planning failures, so it forces a derivation chain (goal, context, decisions, requirements, tasks, execution) and gates each transition twice: a CLI validates the structure of `requirements.md` and `plan.json` before any model reads them, and a reviewer agent checks for scope drift and blind spots. The commands are `/specify` (a Socratic interview), `/blueprint` (contract-first planning into a task dependency graph), `/execute` (parallel workers plus independent verification against sub-requirements), `/bugfix`, `/council`, and `/tribunal` (a three-agent adversarial review: risk, value, feasibility). It ships 22 agents and 29 skills and installs with `claude plugin add team-attention/hoyeon` plus an npm CLI.

Useful to us? The design principle is exactly the lab’s own “deterministic tools over model judgment” rule (Section 39): a CLI that rejects malformed plans before an LLM sees them is the kind of instrument our hooks already are for pushes and merges. The requirements interview is heavier than most of our tasks need, and it is Claude-Code-only.

Superpowers

Superpowers is Jesse Vincent’s “agentic skills framework & software development methodology” (Vincent 2026b). It is the purest example of the layer kind: a set of skills and one bootstrap that makes them mandatory. The `using-superpowers` meta-skill injects itself at session start so the agent checks for a relevant skill before any task, and the skills encode a workflow: brainstorming, writing and executing plans, test-driven development, systematic debugging, git worktrees, requesting and receiving code review, and subagent-driven development with a two-stage review (spec compliance first, then code quality). It installs from the official Claude plugin marketplace (`/plugin install superpowers@claude-plugins-official`) and from Codex’s `/plugins`, and supports fourteen harnesses. Its own README records the cost of the

layer approach: Hermes has no post-compaction hook, so a long session that compacts loses the bootstrap.

Useful to us? Yes, and the lab’s maintainer already runs it alongside `ai-config` in Claude Code. The overlap is real (both have a brainstorming skill, both have a code-review workflow), so treat it as a reference implementation of skill design rather than a second source of truth.

GSD Core

GSD (“Git. Ship. Done.”) describes itself as a “context-engineering and spec-driven development framework that drives AI coding agents through a disciplined phase loop” (open-gsd 2026). Its target is **context rot**, which it defines as the quality degradation that accumulates as an agent fills its context window. The remedy is a five-step loop per milestone:

1. discuss (capture decisions);
2. plan (research and decompose in a subagent);
3. execute (parallel executors, each starting with a fresh context);
4. verify (walk through the built work before declaring done);
5. ship (open the PR, archive the phase).

State lives in `STATE.md` and `CONTEXT.md` so it survives session boundaries. It installs with `npm install @opengsd/gsd-core@latest`, which asks which of nine runtimes to target.

Useful to us? The fresh-context-executor pattern is the same one `@sec-orch-when` recommends, and `ai-config`’s `compress-session` and `handoff` skills play the role of `STATE.md`. GSD’s contribution is having made the loop the default rather than an option.

OpenHarness

OpenHarness, from the HKUDS group at the University of Hong Kong, is an “Open Agent Harness with a Built-in Personal Agent” (HKUDS 2026) and the one standalone runtime in this survey that targets coding-agent use. It is Python with a React terminal UI, installs from a shell script in the repository (the command is `oh`), and rebuilds the whole stack from Section 5: an agent loop with streaming tool use and retries, a registry of more than forty tools, Markdown skills loaded on demand, plugins that follow Claude Code’s conventions (the README reports testing against twelve official plugins), three permission modes (ask, auto, plan) plus path and command rules, lifecycle hooks, MCP, persistent memory, background tasks, and subagent coordination. It talks to Anthropic, OpenAI, Copilot, Codex, Kimi, GLM, MiniMax, and NVIDIA endpoints.

Useful to us? As an escape hatch. Because it reads Claude Code skills and plugins, `ai-config` would largely carry over, so it is the cheapest way to run our configuration on a model or endpoint Claude Code cannot use (Section 27 covers the subscription restrictions that motivate this). It is not a reason to leave Claude Code while Claude Code does what we need.

The LangChain guide

Sydney Runkle’s “How to build a custom agent harness” (2026-06-03) (Runkle 2026) is the runtime kind described from the library side. It defines the harness as “the scaffolding around the model that connects it to the real world,” starts from `create_agent` with three inputs (a model, tools, a system prompt), and adds everything else as **middleware** at fixed points in the loop (before and after each model call and tool call). Its table maps eight production needs to middleware: context overflow to summarization and context editing; memory to filesystem, memory, and skills middleware; environment actions to shell, filesystem, and code-interpreter middleware; delegation to subagent and to-do-list middleware; transient failures to retry and model-fallback middleware; policy to personal-data and human-in-the-loop middleware; steering to human-in-the-loop; and cost to call limits and prompt caching. It positions the Claude Agent SDK and its own Deep Agents as pre-assembled alternatives.

Useful to us? As a checklist. Every row of that table has a counterpart in the layer approach (a hook, a skill, a permission rule), so it is a good audit of what a harness must cover whichever kind you build.

The community thread

The r/ClaudeCode thread “Show off your own harness setups here” collects personal setups of the layer kind. Reddit blocks automated fetches, so the thread is not summarized here and is worth reading directly. One small repository of the same kind, [my-claude-code-harness](#) (code-yeongyu 2025) (12 stars, no license, last pushed 2025-11-28), shows the shape such personal setups take: plan-reviewer, executor, and librarian agents; hooks that inject language guidelines and run a linter; and `/planner`, `/execute`, `/commit`, and `/create-pr` commands. Section 13 lists the communities where these setups are discussed.

What the projects have in common

Across the layer-kind harnesses the same five moves recur:

- a **mandatory workflow** injected at session start (Superpowers’ bootstrap, OMC’s prompt triggers, GSD’s loop);
- **planning before execution**, with the plan written to a file (`plan.json`, `STATE.md`, a writing-plans skill);
- **fresh-context workers** for execution, with the main session kept lean;
- **independent verification** by a different agent than the one that did the work;
- **deterministic gates** where a check can be decided without a model (Hoyeon’s CLI validation, `my-claude-code-harness`’s linter hooks).

The runtime-kind projects add the loop itself, the permission model, and provider independence; nothing else in the surveyed runtimes is missing from the layer kind.

How we would build our own

Morrison-Lab/ai-config is already a custom harness of the layer kind, and a large one. On 2026-09-09 a local checkout held:

- 191 skills;
- 94 hook scripts, and a `hooks/hooks.json` with 55 registrations (30 `PreToolUse`, 10 `UserPromptSubmit`, and 15 `Stop`);
- 8 subagent definitions (including the `adversarial-reviewer` that gates every push);
- 131 shared prose fragments and 59 memory files;
- a plugin manifest, so Claude Code and Cursor load it as a plugin (Section 40 explains how, and Section 39 what the corpus contains).

It already implements all five common moves: the bootstrap is `CLAUDE.md` and `AGENTS.md`, planning goes through issue-first and `st`, fresh-context workers are the `Agent` and `Workflow` calls, independent verification is the `adversarial-reviewer` subagent, and the deterministic gates are the hooks. So “building our own” means three concrete things, in order:

1. **Keep the layer approach.** Nothing in the survey justifies a standalone runtime for us yet: every vendor upgrade to Claude Code arrives for free, and the layer is portable to Codex, OpenCode, and `OpenHarness` because all of them read the same `SKILL.md` and plugin conventions.
2. **Borrow the pieces we lack.** Superpowers’ two-stage review (spec compliance, then code quality) is a sharper split than our single adversarial pass; Hoyeon’s structural validation of a plan file before a model reads it is a hook we do not have; OMC’s notification integrations would replace our scheduled-poll check-ins. Each is one skill or one hook, and each should land as its own `ai-config` issue and PR.
3. **Decide the runtime question by a measured need, not by ambition.** The trigger for the runtime kind is a feature the vendor loop cannot give us: a model Claude Code will not call under our subscription, a permission mode it does not offer, or a loop change (a retry policy, a cost ceiling) that hooks cannot express. When that day comes, `OpenHarness` is the cheaper route, because it already loads our plugin, and the `LangChain` middleware table is the checklist for anything it lacks.

13 Where Current Practice Is Discussed: Reddit Communities

The practice of working with AI agents changes faster than any static page (these notes included) can track. Much of the current guidance circulates as forum discussion: tool comparisons, configuration recipes, failure reports, and launch announcements. This section catalogs the Reddit communities where that discussion concentrates, grouped by what they are for, so you know where to look when a page here has gone stale. This section covers Reddit only: vendor documentation, project issue trackers, and the papers behind a technique are cited where the chapters use them, and the forum layer had no index here until this section.

⚠ Counts and characterizations are snapshots

The member counts and the one-line characterizations in these tables are approximate and dated. They come from a directory compiled in August 2026 ([issue #94](#)) from third-party trackers (Prowlo, GummySearch, usefulai.com, and ReddTrends), which disagree with each other by tens of percent for the same community depending on crawl date. GummySearch stopped accepting new signups or renewals after 2025-11-30, so some cells it supplies reflect older snapshots. Reddit stopped showing member totals on community pages in late 2025, so a figure cannot be checked against the page itself. Read the counts as orders of magnitude, and when precision matters use a tracker that reads the official Reddit API (ReddTrends is one).

Dedicated to AI agents

Table 6: Communities dedicated to AI agents

Community	Members	What it is for
r/AI_Agents	~425k (Aug 2026)	The largest dedicated community: building and deploying LLM-based agents, framework comparisons, showcases, and launches. One marketing tracker describes it as roughly doubling over the preceding year.
r/aiagents	~118k	A separate community (no underscore) focused on tools, stack recommendations, and business automation.
r/AgentsOfAI	~98k	Showcase-friendly; known for its “I Made This” flair.
r/automation	~228k	Workflow automation with heavy agent crossover; not agent-only.

Frameworks and tools

Table 7: Framework and tool communities

Community	Members	What it is for
r/LLMDevs	~161k	Developer discussion of LLM applications, agent architectures, memory, and MCP builds.
r/mcp	~89k	The Model Context Protocol; a smaller r/modelcontextprotocol (~23k) covers the same ground.
r/LangChain	~87k	The largest framework community; much LangGraph discussion happens here too.
r/n8n	~220k	The n8n workflow platform, with heavy agent-workflow content.
r/AutoGPT	~21k	Grew out of the AutoGPT project into general agents and automation; tolerates launch posts.
r/AutoGenAI	~7k	Microsoft’s AutoGen; small but active.
r/crewai	~1.6k	The CrewAI role-based framework.

The framework-specific communities are much smaller than the general agent ones, and [r/PydanticAI](#), [r/LangGraph](#), and [r/LlamaIndex](#) exist but had no reliable counts at compilation time. In practice, framework discussion concentrates in [r/AI_Agents](#), [r/LLMDevs](#), and [r/LocalLLaMA](#) rather than in the niche communities.

AI coding agents

Table 8: AI coding-agent communities

Community	Members	What it is for
r/ClaudeCode	~253k-387k	Claude Code specifically; trackers disagree widely on its size (usefulai.com listed ~253k in May 2026, while GummySearch listed ~387k).
r/ChatGPTEcoding	~383k	The multi-tool coding community, where Claude Code, Cursor, Codex, and others are compared side by side.
r/vibecoding	~343k	Building applications by directing agents; heavy Claude Code, Cursor, and Bolt content.
r/VibeCodeDevs	~59k	Developers shipping real products with agents and automation.
r/Cursor	not verified	The Cursor editor and its agent features.

General AI communities with heavy agent discussion

Table 9: General AI communities

Community	Members	What it is for
r/ChatGPT	~11.6M	The largest applied-AI community; frequent agent and automation threads.
r/singularity	~4.0M	AI progress and futurism; skews toward hype.
r/MachineLearning	~3.0M	Research-grade, strictly moderated; agent papers.
r/OpenAI	~2.7M	OpenAI products, including its agent offerings.
r/ArtificialIntelligence	~1.7-1.9M	Broad AI community (the misspelling is the actual handle).
r/artificial	~1.3M	Broad AI news and tools.

Community	Members	What it is for
r/ClaudeAI	~1.06M (Aug 2026)	Claude and Claude Code, with significant agentic crossover.
r/LocalLLaMA	~800k (Aug 2026)	The dominant local and open-weight model community; high signal on local agents and tooling.
r/LocalLLM	~207k	A second local-model community.

How to use this list

- **For the broadest agent discussion**, start with [r/AI_Agents](#), then add [r/AgentsOfAI](#) and [r/aiagents](#) for showcases and use cases.
- **For framework work**, monitor [r/AI_Agents](#), [r/LLMDevs](#), and [r/LocalLLaMA](#), where much of the framework discussion actually lands, and subscribe to a framework’s own community once it reaches daily posting at a few tens of thousands of members.
- **For coding agents**, prioritize [r/ClaudeCode](#) and [r/ChatGPTEcoding](#), and add [r/vibecoding](#) for the builder angle.
- **For research and trend-spotting**, follow [r/LocalLLaMA](#), [r/MachineLearning](#), and [r/singularity](#), filtering the last for hype.
- **Treat a community as a lead, not a source.** A forum thread is where a technique surfaces first; verify it against the tool’s documentation before you rely on it, in the spirit of [reviewing AI-generated work](#).

14 Thinking Machines Lab’s Inkling Models

[Inkling](#) (Thinking Machines Lab 2026a) is a pair of open-weight, multimodal foundation models from Thinking Machines Lab (measured 2026-09-09). This is a model release, not a coding agent or a chat product: the weights are downloadable, and the company’s own hosted surfaces for it are a fine-tuning API and a playground rather than an end-user assistant. This section summarizes the release and asks whether it matters for our workflow.

Who Thinking Machines Lab is

Thinking Machines Lab is a San Francisco AI startup founded in February 2025 by Mira Murati, formerly OpenAI’s chief technology officer, with John Schulman, an OpenAI co-founder, as chief scientist (Wikipedia contributors 2026). It raised \$2 billion at a \$12 billion valuation in July 2025, and two founding members left for OpenAI in January 2026 (Wikipedia contributors

2026). Its first product, [Tinker](#) (Thinking Machines Lab 2026f), released in October 2025, is a LoRA fine-tuning API for open-weight models where the customer writes the training loop in Python and Thinking Machines runs the GPUs (Wikipedia contributors 2026). Inkling is the company’s first in-house model, released on 2026-07-15 (Thinking Machines Lab 2026c).

What Inkling is

Two models share the name (measured 2026-09-09) (Thinking Machines Lab 2026a):

- **Inkling:** a mixture-of-experts transformer with 975 billion total parameters, 41 billion of them active per token, and a context window of up to one million tokens (Thinking Machines Lab 2026c).
- **Inkling-Small:** 276 billion total parameters, 12 billion active, released 2026-07-30 and post-trained by on-policy distillation with Inkling as the teacher (Thinking Machines Lab 2026e). The product page says it matches the larger model on many benchmarks at a quarter of the size (Thinking Machines Lab 2026a).

Both accept text, image, and audio input and produce text only (Thinking Machines Lab 2026b, 2026d). Both are released under the Apache 2.0 license (Thinking Machines Lab 2026b, 2026d), and both expose an effort setting that trades reasoning tokens for score; the reported evaluations are run at `effort=0.99` (Thinking Machines Lab 2026c).

The announcement positions Inkling as a broad base for customization rather than a leaderboard winner (Thinking Machines Lab 2026c). Its claims, and the evidence offered for them:

- **Breadth:** reported scores of 77.6% on SWE-Bench Verified, 63.8% on Terminal Bench 2.1, 87.2% on GPQA Diamond, and 73.5% on MMMU Pro, all at the highest effort setting (Thinking Machines Lab 2026c).
- **Efficiency:** the company reports that Inkling matches NVIDIA’s Nemotron 3 Ultra on Terminal Bench 2.1 using roughly a third of the tokens (Thinking Machines Lab 2026c).
- **Training:** 45 trillion tokens of text, image, audio, and video, then supervised fine-tuning on synthetic data and reinforcement learning with more than 30 million rollouts (Thinking Machines Lab 2026c).

These are the vendor’s own numbers on the vendor’s own runs; we have not reproduced any of them. Inkling-Small’s announcement reports 80.2% on SWE-Bench Verified and 31.6% on Humanity’s Last Exam, against Inkling’s 77.6% and 29.7% (Thinking Machines Lab 2026e).

How it is accessed and priced

Access routes (measured 2026-09-09):

- **Weights:** on Hugging Face, in BF16 and NVFP4 checkpoints (Thinking Machines Lab 2026b, 2026d). Even the smaller model has 276 billion parameters, so neither fits any hardware the lab owns.
- **Tinker:** LoRA fine-tuning at 64K or 256K context, and a beta serverless inference endpoint, both billed per million tokens (Thinking Machines Lab 2026g). Under a limited-time 50% discount, fine-tuning Inkling at 64K context costs \$1.87 per million prefill tokens, \$4.68 per million sampled tokens, and \$5.61 per million training tokens, roughly doubling at 256K; Inkling-Small is about a third of that. Serverless inference is \$1.00 input and \$4.05 output per million tokens for Inkling, and \$0.30 and \$1.20 for Inkling-Small. Checkpoint storage adds \$0.10 per GB per month (Thinking Machines Lab 2026f).
- **Third-party hosts:** the announcement lists Together AI, Fireworks, Modal, Databricks, and Baseten as hosts, and vLLM, SGLang, and `llama.cpp` among supported inference engines (Thinking Machines Lab 2026c). Our quota-aware defaults for the Databricks endpoint are already in

- 1.

What it is for

Thinking Machines’ stated bet is that organizations will want to fine-tune a capable open model on their own data rather than rent a closed frontier model, and Inkling exists to give Tinker a strong first-party base (Thinking Machines Lab 2026c). Tinker’s own pitch lists specialized agents, forecasting, continual learning, and AI research as the intended uses (Thinking Machines Lab 2026f). The audio and image inputs make it a candidate for transcription-plus-reasoning pipelines as well as coding.

Useful to us? Marginally, and only through a host

As a coding-agent backend, Inkling is one more open-weight option alongside the models in Section 4, reachable through Databricks (Section 31) or, when a host lists it there, OpenRouter (Section 26). Its reported SWE-Bench and Terminal Bench scores are competitive but not ahead of the closed models we already pay for, and the 8,192-token output cap on the Databricks endpoint is a real constraint for agentic edits. Self-hosting is out: the lab has no machine within an order of magnitude of the memory required.

The more interesting angle is research. Tinker lets a student run a fine-tuning or reinforcement-learning experiment on a frontier-class open model with a Python training loop and no cluster administration, and universities can ask for wider access (Thinking Machines Lab 2026f). If a project ever needs to adapt a model to epidemiological text or lab-specific coding conventions, Tinker with Inkling-Small is a cheap place to try it. Until such a project appears, no action is needed beyond keeping the Databricks endpoint in our defaults table.

15 AI Agents and the Technological Singularity

The emergence of sophisticated [AI agents](#) has prompted discussions about whether we are witnessing or approaching a [technological singularity](#). Understanding this concept helps contextualize the rapid evolution of AI tools and our responsibility in using them.

What is the technological singularity?

The technological singularity is a hypothetical future point when technological growth becomes uncontrollable and irreversible, resulting in unforeseeable changes to human civilization. The concept, popularized by mathematician Vernor Vinge and futurist Ray Kurzweil, typically involves the creation of artificial superintelligence that recursively improves itself, leading to an intelligence explosion beyond human comprehension or control.

Do current AI agents represent the singularity?

No, current AI coding agents (as of early 2026) do not represent the technological singularity.

While modern AI agents demonstrate impressive capabilities, they remain fundamentally different from the singularity scenario in several critical ways:

- **Limited autonomy:** Today's AI agents operate within strict boundaries and require human oversight. They cannot recursively improve their own core architecture or develop capabilities beyond their training.
- **Narrow intelligence:** AI coding agents are specialized tools designed for specific tasks. They lack general intelligence, self-awareness, or the ability to operate outside their designed domain.
- **Human dependency:** These agents require human developers to: review their work, provide direction, validate correctness, and make final decisions about their outputs.
- **No recursive self-improvement:** Current AI agents cannot fundamentally redesign themselves or create more advanced versions of themselves autonomously. Any improvements to AI systems still require human researchers and engineers.
- **Controlled development environment:** AI coding agents work in sandboxed environments with explicit permissions and constraints. They cannot independently acquire resources, modify their own constraints, or operate without human authorization.

Why this matters for responsible AI use

Understanding that current AI agents are powerful but limited tools—not autonomous superintelligences—has important implications:

- **Maintain appropriate skepticism:** AI agent outputs require the same critical review as any other tool-generated code.
- **Preserve human decision-making:** The responsibility for code quality, security, and correctness remains with human developers.
- **Continue skill development:** Using AI agents should enhance rather than replace human expertise.
- **Stay vigilant:** While current agents don't represent a singularity, the rapid pace of AI development requires ongoing attention to emerging capabilities and risks.

The value of AI coding agents lies in their ability to accelerate human productivity and learning, not in replacing human judgment or expertise. They are sophisticated tools that augment human capabilities while remaining under human control and oversight.

Further reading

For thoughtful perspectives on AI consciousness and intelligence, see Douglas Hofstadter's reflections in [“I Thought I Was in an AI Apocalypse. Then I Started Looking Closer.”](#)

16 Relative Advantages of AI and Humans

AI coding agents and human coders have complementary strengths. Understanding these differences helps you decide when to delegate work to agents and when to handle tasks yourself.

Comparative Strengths: Humans vs. AI Agents

Table 10 summarizes the relative advantages of human coders and AI coding agents across different types of tasks:

Table 10: Relative advantages of humans and AI coding agents

Task Type	Humans	AI agents
Creative thinking	Humans excel at understanding context, handling ambiguous requirements, and thinking creatively about novel problems	AI agents struggle with ambiguous requirements and creative problem-solving in unfamiliar domains
Algorithmic thinking	Humans make mistakes when following repetitive instructions and may introduce inconsistencies	AI agents excel at executing well-defined, repetitive tasks with precision and consistency

Or, if you prefer a more visual representation:

Table 11: Relative advantages of humans and Agents

	Humans	AI Agents
Creative thinking		



Algorithmic thinking

This pattern mirrors the evolution of programming itself. Just as almost no one writes machine code anymore because higher-level languages and compilers handle those details, most developers will increasingly spend less time writing low-level code. Instead, you'll describe what the system needs to do as clearly as possible, and AI agents will handle many of the computational and coding details.

For most tasks, you won't need to step in and manipulate code yourself. However, you'll still need enough domain judgment to:

- Supervise and validate AI-generated code
- Handle edge cases that agents struggle with
- Make creative decisions about architecture and design
- Understand when agent suggestions are incorrect or suboptimal

Strong coding skills are one route to that judgment, and not the only one. A public field report from a non-programmer ("[Vibe coded this game in four months](#)", r/ClaudeCode, 2026-08-22; summarized in [issue #98](#)) describes four months of building a browser racing game with coding agents and no coding experience at all, crediting knowledge of game technology and design, plus long practice with language models, for the planning and the early catches that kept the project on track; by the author's own account, lacking that knowledge as well would have stalled the project after a couple of prompts. The requirement is the supervision, whichever background supplies it.

Future Developments: World Models

As AI technology advances, the distinction between these strengths may shift. Yann LeCun, 2019 Turing Award winner and AI researcher at Meta and NYU, advocates for developing “world models”—AI systems that understand and reason about the physical world, not just language patterns (LeCun 2022).

World models aim to give AI systems:

- **Persistent memory and reasoning:** Understanding that persists across interactions
- **Physical world understanding:** Reasoning about how things work in reality, not just in text
- **Better handling of ambiguity:** Using world knowledge to interpret unclear requirements

As these technologies mature, AI agents may become better at tasks requiring contextual understanding and creative problem-solving. This makes it even more important to develop strong supervision and validation skills now, so you can effectively work with increasingly capable AI systems.

17 How to Work with Coding Agents

Coding agents can be accessed through several interfaces, each with different trade-offs for task size, feedback speed, and collaboration style.

Ways to interact with coding agents

The main differences are where the agent runs, how much repository context it can access, and how directly it can apply changes.

- **Cloud coding agents (GitHub Issues/PR workflow):** Best for larger, asynchronous tasks (for example, multi-file refactoring, dependency updates, or documentation changes spanning several chapters). The agent runs in a managed cloud environment, works through an issue, and proposes changes in a pull request for review.
- **CLI agents (terminal-first workflows):** Best for rapid, local iteration. You stay close to shell tools and local files, with faster back-and-forth for debugging, refactoring, and test-driven edits.
- **Chat/app agents (for example Claude or Codex-style chat interfaces):** Best for planning, design discussion, and drafting code ideas. They can be very strong thought partners, but often need more manual copy/edit/execute steps unless tightly integrated with your repository tools.
- **IDE-integrated agents:** Best for mixed human and agent editing. They combine conversational help with direct file edits, code navigation, and in-editor testing loops.

In practice, teams often combine these modes: use app or IDE chat to refine requirements, then hand implementation to a cloud or CLI agent, and finish with human review.

The following sections focus on GitHub Copilot's specific workflow for assigning and managing coding tasks.

Assigning Issues to Copilot

You can assign GitHub Issues directly to @copilot just like you would assign to a human collaborator:

1. **On GitHub.com:** Navigate to an issue and assign it to Copilot in the assignees section
2. **In VS Code:** In the GitHub Pull Requests or Issues view, right-click an issue and select "Assign to Copilot"
3. **From Copilot Chat:** Delegate tasks to Copilot directly from the chat interface in supported editors

The Agent Workflow

Once assigned an issue, the coding agent follows an autonomous workflow:

1. **Analysis:** Reviews the issue description, related discussions, repository instructions, and codebase context
2. **Planning:** Determines what changes are needed and creates a work plan
3. **Development:** Works in an isolated GitHub Actions environment, modifies code, runs tests and linters, and validates changes
4. **Pull Request Creation:** Creates a draft pull request with implemented changes, audit logs, and a summary of modifications
5. **Review and Iteration:** You review the PR and can request changes; the agent will iterate based on your feedback

Collaborating with Coding Agents

Between iterations of asking coding agents to extend a PR, human collaborators can also push changes directly to the PR branch. This allows for a collaborative workflow where both humans and agents contribute:

- **Human contributions:** You can make quick fixes, add content, or refine the agent's work by pushing commits to the same branch

- **Agent iterations:** After your changes, you can ask the agent to continue working on additional requirements

Important: Try to avoid pushing changes while the coding agent is actively working. Simultaneous edits can produce conflicting diffs that:

- Need to be manually resolved
- May confuse both human and AI collaborators
- Could result in lost work or merge conflicts

Best practice: Wait for the agent to complete its current iteration (indicated by the PR being updated) before pushing your own changes to the branch. Then assign new work to the agent for the next iteration.

Directly Prompting for Pull Requests

You can also prompt Copilot to create pull requests without first creating an issue:

- Use Copilot Chat in your editor to describe the changes you want
- The agent will analyze your request and create a pull request
- This is useful for quick fixes or well-defined tasks

Important Safeguards

- **Human approval required:** Coding agents cannot merge their own changes
- **Branch restrictions:** Agents can only push to their own branches (e.g., `copilot/*`)
- **Full transparency:** All agent actions are logged and visible in the PR

Workflow Approval Requirements

When GitHub Copilot creates or updates a pull request, it cannot automatically trigger GitHub Actions workflows. **You must manually approve each workflow run** by clicking the approval button in the Actions tab or on the PR.

This manual approval requirement is a security measure that prevents potentially malicious or unintended code execution. Because Copilot can modify any file in the repository—including workflow files themselves or scripts called by workflows—allowing automatic workflow execution could create security vulnerabilities.

Key points:

- **No automatic approval:** There is currently no way to bypass manual workflow approval for Copilot PRs, even if you are the repository owner

- **Security reasoning:** Copilot could modify workflow files (`.github/workflows/*.yaml`) or scripts they execute, potentially injecting malicious code
- **Impact on workflow:** This means you need to actively monitor and approve workflow runs as Copilot iterates on your issue, which can slow down the development cycle

Workaround considerations:

Some users have discussed using Personal Access Tokens (PATs) to allow Copilot to trigger workflows on your behalf, but this approach has security implications and should be carefully evaluated before implementation.

For more details and community discussion about this limitation, see:

- [GitHub Community Discussion #162826](#): Discussion about workflow approval requirements
- [GitHub Community Discussion #183966](#): Product feedback on this topic

For detailed instructions, see [GitHub Copilot coding agent documentation](#).

18 Useful Prompt Formats

When working with coding agents, using clear and specific prompts helps achieve better results. Here are some useful prompt formats that you can use when requesting assistance from coding agents:

Common Task Patterns

Tidying up code:

- “tidy up [file, function, module, whole project]”
- Useful for improving code organization, consistency, and readability
- Example: “tidy up the data processing module”

Addressing failing workflows:

- “address failing workflows”
- Helps fix continuous integration (CI) failures, build errors, or test failures
- Example: “address failing workflows in the GitHub Actions pipeline”

Decomposing code:

- “decompose [function, quarto-file, etc]”
- Breaks down large or complex code into smaller, more manageable pieces
- Example: “decompose this analysis function into separate helper functions”

Updating content:

- “update [links, content, etc]”
- Refreshes outdated information, fixes broken links, or modernizes code
- Example: “update all package URLs in the documentation”

Expanding documentation:

- “expand [a section in a document]”
- Adds more detail, examples, or explanation to existing content
- Example: “expand the section on data validation with practical examples”

Condensing content:

- “condense [a section in a document]”
- Reduces verbosity while preserving essential information
- Example: “condense the installation instructions to be more concise”

Clarifying content:

- “clarify [a section in a document]”
- Improves clarity, removes ambiguity, or simplifies complex explanations
- Example: “clarify the explanation of the analysis workflow”

Tips for Effective Prompts

- **Be specific:** Include file names, function names, or specific sections when possible
- **Provide context:** Explain what you want to achieve and why
- **Set boundaries:** Specify what should or shouldn’t change
- **Request validation:** Ask the agent to test or verify its changes when appropriate

19 Addressing Failing GitHub Actions Workflows

When GitHub Actions workflows fail, you can use Copilot to help diagnose and fix the issues. However, it’s important to use the right prompts depending on whether the problem is in your code or in the workflow configuration itself.

Scenario 1: Code Issues Found by Workflows (Most Common)

When to use: The workflow is functioning correctly, but it's detecting problems in your code (e.g., failing tests, linting errors, build failures).

What you want: Fix the code issues without modifying the workflow files themselves.

Recommended prompts:

- “fix the code issues found by the failing workflows”
- “address the linting errors reported in the GitHub Actions checks”
- “fix the test failures in the CI pipeline”
- “resolve the build errors shown in the workflow logs”

Example: If your R package has failing tests detected by `usethis::use_github_action("check-standard")`, you want Copilot to fix the test failures in your R code, not modify the workflow YAML file.

Why this matters: These prompts make it clear that you want code changes, not workflow changes. This helps prevent the agent from unnecessarily modifying your carefully-configured CI/CD pipeline.

Scenario 2: Issues with Workflow Files Themselves

When to use: The workflow configuration itself has problems (e.g., syntax errors in YAML, incorrect job definitions, outdated actions).

What you want: Fix the workflow files, but with extreme caution due to security implications.

Recommended prompts:

- “fix the syntax error in the GitHub Actions workflow file at line X”
- “update the workflow to use the latest version of action Y”
- “correct the job configuration in `.github/workflows/check-standard.yaml`”

Important considerations:

Warning

Security Warning

Workflow files have access to repository secrets and can execute arbitrary code. Before accepting any changes to workflow files:

1. **Review every line** of the proposed changes
2. **Verify** the changes only address the specific issue
3. **Check** that no new secret access or command execution has been added

4. **Test** in a safe environment if possible

See Section 21 for more details on workflow file security.

When to do it yourself: Workflow syntax errors and configuration issues are often faster to fix manually than with Copilot, especially if you're familiar with GitHub Actions. See Section 49 for more guidance.

Scenario 3: Uncertain Which Scenario Applies

When to use: You're not sure whether the failure is due to code issues or workflow configuration problems.

Recommended approach:

1. **First, examine the workflow logs:**
 - Look at the error messages in the GitHub Actions tab
 - Identify whether the error is in your code or the workflow itself
 - Common code issues: test failures, linting errors, compilation errors
 - Common workflow issues: YAML syntax errors, missing actions, permission errors
2. **Use a diagnostic prompt:**
 - "examine the failing workflow logs and identify whether the issue is in the code or the workflow configuration"
 - "diagnose the root cause of the workflow failure"
3. **Then use the appropriate scenario above:** Once you understand the issue, use the specific prompts from Scenario 1 or 2.

Example workflow:

```
1. Prompt: "examine the failing workflow logs and identify the issue"
2. Copilot responds: "The workflow is failing because of linting errors
   in src/analysis.R"
3. Prompt: "fix the linting errors in src/analysis.R"
```

Additional Resources

- See the [UCD-SERG Lab Manual's continuous integration chapter](#) for setting up GitHub Actions workflows
- See Section 21 and Section 20 for security considerations with workflow files
- See Section 49 for guidance on when to use Copilot vs. fixing issues yourself
- See the [GitHub Actions documentation](#) for workflow syntax and troubleshooting

20 Benefits and Hazards

Coding agents are powerful programs that can work autonomously. They create pull requests that propose changes to the code in our repositories, potentially including their own configuration files and our automated workflows. They can work powerfully on our behalf, but they require careful oversight and control to ensure they serve our interests and that we understand the consequences of their actions.

Coding agents offer several advantages:

- **Built-in transparency:** Coding agents create a clear record of their role in your work through commit history and code suggestions
- **Context-aware suggestions:** Coding agents understand your codebase and can make contextually relevant suggestions
- **Integration with version control:** Using coding agents within GitHub ensures that AI-assisted changes are tracked alongside all other code changes
- **Interactive workflow:** Coding agents' interactive nature encourages you to review and modify suggestions rather than blindly accepting them
- **Accelerated development:** Coding agents can help you write boilerplate code, refactor existing code, and implement common patterns more quickly
- **Learning opportunities:** Coding agents can suggest approaches or techniques you may not have considered, helping you expand your coding knowledge

However, coding agents also come with significant hazards:

- **Over-reliance:** Depending too heavily on coding agents can atrophy your coding skills and understanding
- **Subtle bugs:** AI-generated code may contain logic errors that are not immediately obvious
- **Security vulnerabilities:** Coding agents may introduce insecure patterns or fail to follow security best practices
- **Inappropriate solutions:** AI may suggest solutions that work but are not optimal for your specific research context or constraints
- **Hidden biases:** Coding agents may perpetuate coding patterns or approaches that reflect biases in their training data
- **False confidence:** Well-formatted, professional-looking code from AI can mask underlying problems and reduce critical review

- **Workflow manipulation risks:** Coding agents that modify CI/CD workflows (`.github/workflows/*.yaml`) or setup configurations can inadvertently or maliciously compromise repository security, expose secrets, or execute harmful commands

Further reading/viewing

- *I Robot* (Asimov 1950)
- *Dune* (Herbert 1965)
- *2001* (1968)
- *Terminator 3* (2003)
- *The Matrix* (1999)
- *Blade Runner* (1982)
- *WarGames* (1983)
- *Battlestar Galactica* (2004) (*Battlestar Galactica* 2004)
- *Ender's Game* (Card 1985)
- “The Humans are Dead” (Flight of the Conchords 2007)



Figure 1: [Agents](#)

21 Best Practices for Safe and Successful Use

To work with coding agents safely and successfully:

1. **Maintain active supervision:** Never assume AI-generated code is correct. Review every line critically.

2. **Understand before accepting:** If you don't understand what the code does, don't use it. Take time to learn or ask a colleague.
3. **Test thoroughly:** AI-generated code must be tested as rigorously as code you write yourself. Don't skip testing because "the AI wrote it." Budget for it, too. Two commenters on the same field report ("[Vibe coded this game in four months](#)", r/ClaudeCode, 2026-08-22; summarized in [issue #98](#)) independently named manual testing as the dominant cost once generation is cheap. One put it as needing either deep domain experience or heavy testing to get a defect-free result.
4. **Start small:** Begin with small, well-defined tasks to build confidence and understanding of the agent's capabilities and limitations.
5. **Verify logic and assumptions:** Check that the AI hasn't made incorrect assumptions about your data, requirements, or scientific context.
6. **Review for security:** Explicitly check for security issues, especially when handling sensitive data or user input.
7. **Iterate and refine:** Use coding agents as a starting point, not an endpoint. Refine and improve the generated code.
8. **Maintain coding practice:** Regularly write code yourself to maintain and develop your skills. Don't let the agent do everything.

UC Davis campus guidance

UC Davis Student Affairs also provides guidance on the [Responsible Use of Artificial Intelligence \(AI\)](#). That page provides UC Davis-specific guidance on AI tool selection, campus training, and careful handling of sensitive data. Their recommendations, like ours, include careful validation, active supervision, and protection of confidential information.

Critical: Exercise Extreme Caution with Workflow Files

Be especially careful when allowing coding agents to edit GitHub Actions workflows or CI/CD configurations. These files control automated processes that can:

- Access secrets and credentials
- Deploy code to production
- Execute arbitrary commands in your repository

Never allow a coding agent to edit workflow files (especially `.github/workflows/*.yaml` or `copilot-setup-steps.yaml`) without thorough manual review. Before approving any workflow run, always check if the workflow files themselves have been modified. Malicious or erroneous changes to workflows can compromise your entire repository and its secrets.

When using coding agents, work interactively with the AI suggestions: review, modify, and test them rather than accepting them wholesale. This interactive approach helps ensure code quality and deepens your understanding of the code.

Remember: AI tools are assistants, not replacements for your expertise and judgment. The quality and correctness of your work remains your responsibility.

22 Firewall and Network Configuration

Coding agents require specific network access to function properly. If a coding agent is running behind a corporate firewall or on a restricted network, you may need to configure allowlists to enable coding agent functionality.

Built-in Agent Firewall

Coding agents run in a GitHub Actions environment with a built-in firewall that limits internet access by default. This firewall helps protect against:

- Data exfiltration
- Accidental leaks of sensitive information
- Execution of malicious instructions

By default, the agent's firewall allows access to:

- Common OS package repositories (Debian, Ubuntu, Red Hat, etc.)
- Popular container registries (Docker Hub, Azure Container Registry, AWS ECR, etc.)
- Language-specific package registries (npm, PyPI, Maven, RubyGems, etc.)
- Common certificate authorities for SSL validation

For the complete list of allowed hosts, see the [Copilot allowlist reference](#).

Customizing Agent Firewall Settings

In your repository's "Coding agent" settings page, you can:

- Add custom hosts to the allowlist (for internal dependencies or additional registries)
- Opt out of the default recommended allowlist for stricter security
- Disable the firewall entirely (not recommended)

If a coding agent's request is blocked by the firewall, a warning will be added to the pull request or comment, detailing the blocked address and the command that triggered it.

For more information, see [Customizing or disabling the firewall for GitHub Copilot coding agent](#).

Recommended URLs for Data Science Repositories

For data science and R-focused repositories, we recommend adding the following URLs to your Copilot allowlist. These sites are safe, reputable sources of documentation and packages that coding agents may need to access:

R Package Documentation and Ecosystems:

- tidyverse.org - `{tidyverse}` package documentation and learning resources
- r-lib.org - Core R infrastructure packages (`{devtools}`, `{testthat}`, `{usethis}`, etc.)
- ggplot2.tidyverse.org - `{ggplot2}` visualization package
- dplyr.tidyverse.org - `{dplyr}` data manipulation package
- tidyr.tidyverse.org - `{tidyr}` data tidying package
- purrr.tidyverse.org - `{purrr}` functional programming package
- readr.tidyverse.org - `{readr}` data reading package
- stringr.tidyverse.org - `{stringr}` string manipulation package
- forcats.tidyverse.org - `{forcats}` categorical data package

R Package Repositories:

- cran.r-project.org - The Comprehensive R Archive Network
- cloud.r-project.org - CRAN mirror (cloud-based)
- docs.ropensci.org - rOpenSci package documentation (e.g., `{targets}`)
- rdatatable.gitlab.io - `{data.table}` package documentation
- rstudio.github.io - RStudio-maintained packages (e.g., `{renv}`)

Code Style and Quality Tools:

- styler.r-lib.org - `{styler}` code formatting package
- lintr.r-lib.org - `{lintr}` code linting package
- roxygen2.r-lib.org - `{roxygen2}` documentation package
- style.tidyverse.org - Tidyverse style guide

General Documentation and Reference:

- en.wikipedia.org - General reference and technical documentation
- r-project.org - Official R project website
- quarto.org - Quarto publishing system documentation
- pandoc.org - Pandoc document converter documentation

GitHub Organizations (for package repositories):

- github.com/tidyverse/* - Tidyverse package source code
- github.com/r-lib/* - R-lib package source code
- github.com/rstudio/* - RStudio package source code
- github.com/ropensci/* - rOpenSci package source code

💡 When to Add These URLs

Add these URLs to your repository's allowlist if:

- Coding agents report blocked access to these sites
- You're working on R or data science projects that use these packages
- You want agents to access current documentation during code generation

You can add URLs selectively based on your project's specific dependencies rather than adding all URLs at once.

i Safety of These URLs

All URLs listed here are:

- Maintained by reputable organizations (Tidyverse, RStudio/Posit, R Core Team, rOpenSci)
- Widely used in the R community
- Focused on documentation and package distribution
- Safe for coding agents to access

These sites do not host user-generated content or allow arbitrary code execution, making them appropriate for inclusion in your allowlist.

23 Running Coding Agents Offline

Some environments restrict or prohibit internet access—high-performance computing (HPC) clusters, hospital networks, or air-gapped research servers may block connections to cloud AI providers. Running a local AI model lets you use coding assistance in these settings without sending code to external servers, which also addresses data-privacy concerns when working with sensitive or confidential data.

i Trade-offs of Local vs. Cloud Models

Local models work best with a GPU (roughly 8 GB VRAM or more for smaller models); CPU-only inference is possible but significantly slower, and hardware needs vary with model size and quantization. They are generally less capable than frontier cloud models, and may produce lower-quality results on complex tasks. For routine work in fully connected environments, cloud-based agents remain the better choice. Use local models when network access or data-privacy policies require it.

Running a Local Model with Ollama

[Ollama](#) is a common way to run open-weight AI models locally. It packages models and a simple API server into a single tool and is available for macOS, Linux, and Windows.

Install Ollama:

Caution

Before running any remote install script, review it first. The real risk is piping `curl` straight into `sh/bash` (`curl ... | sh`), which executes unreviewed code. Save the script, read it, then run it: `curl -fsSL https://ollama.com/install.sh -o install.sh && less install.sh` (paging the saved file rather than piping `curl` into `less`, which can behave oddly in some terminal emulators). Alternatively, use your system package manager (e.g., `brew install ollama` on macOS) or follow the manual installation steps on the [Ollama releases page](#).

```
# macOS / Linux: download, review, then run (do not chain these into one command)
curl -fsSL https://ollama.com/install.sh -o install.sh
less install.sh    # review the script before running it (see caution above)
sh install.sh
```

On Windows, download the installer from <https://ollama.com/download>.

Pull a code-focused model:

```
# Smaller, faster; works on most machines with a modern GPU or Apple Silicon
ollama pull qwen2.5-coder:7b

# More capable; larger memory footprint
ollama pull qwen2.5-coder:32b

# Alternatively, a general-purpose model (70B variant; needs a high-memory GPU)
ollama pull llama3.3
```

The VRAM each model needs depends on its size and quantization and changes as models are re-quantized—check the [Ollama model library](#) for current requirements. As a rough guide, smaller (7B) models run on consumer GPUs with around 8 GB of VRAM, while larger (32B and 70B) models need substantially more and may not fit on a single GPU.

⚠️ A good completion model is not necessarily a usable agent

The models above are strong at *writing code* when you ask them to. That is a different skill from **tool calling** — emitting a well-formed request to read a file or run a command, and then using the result. Inline completion and chat need only the first. Anything autonomous needs the second, because an agent that cannot call a tool cannot read your repository at all.

The two come apart in practice, and the failure is subtler than a model simply refusing. Tested against a single-function tool schema on a 24 GB M2, `qwen2.5-coder:14b` returned `finish_reason: stop` with an empty `tool_calls` field on four attempts out of four. It had not ignored the request: it wrote a correct tool call as ordinary prose in the `content` field,

```
{"name": "read_file", "arguments": {"path": "src/main.py"}}
```

which is the right JSON in the wrong place. A harness looks for `tool_calls`, finds nothing, and treats the turn as a plain reply, so the tool never runs. `granite4:7b-a1b-h`, at half the parameter count, returned a well-formed call in the `tool_calls` field three times out of three and completed a full multi-turn round trip.

An advertised `tools` capability is necessary but not sufficient, so do not settle the question with `ollama show`. On the same machine `qwen2.5-coder:14b` lists `tools` among its capabilities and still cannot be driven by a harness, for the reason above. Test it yourself with one request before building a loop on it:

```
RESP=$(curl -s -w '\n%{http_code}' \
  http://localhost:11434/v1/chat/completions -H 'content-type: application/json' -d '{
  "model": "granite4:7b-a1b-h", "stream": false,
  "messages": [{"role": "user", "content": "What is in src/main.py? Use the tool."}],
  "tools": [{"type": "function", "function": {"name": "read_file",
    "parameters": {"type": "object", "properties": {"path": {"type": "string"}},
    "required": ["path"]}}}])

CODE=$(printf '%s' "$RESP" | tail -1)
BODY=$(printf '%s' "$RESP" | sed '$d')

if [ "$CODE" != "200" ]; then
  echo "request failed (HTTP $CODE): $BODY" # bad tag, tools unsupported, server down
elif printf '%s' "$BODY" | grep -q '"tool_calls"'; then
  echo "usable as an agent"
else
  echo "no tool call; completion model only"
  printf '%s' "$BODY" | grep -o '"content": "[^"]*"' | head -c 200
fi
```

Check the status code separately from the result. A request that simply errored — a mistyped tag, a model the server rejects for tool use, a server that is not running — produces the same silence as a model that declined to call the tool, and only one of those is a fact about the model. Printing the `content` field on the no-call branch is what distinguishes a model that ignored the tools from one that described the call in prose instead of emitting it, which is the `qwen2.5-coder` case above.

Start the Ollama server:

```
ollama serve
```

By default the server listens at `http://localhost:11434`.

Connecting Positron Assistant to Ollama

[Positron](#) supports Ollama natively through the [OpenAI-compatible API endpoint](#) that Ollama exposes.

1. Open the Command Palette with `Cmd+Shift+P` (macOS) or `Ctrl+Shift+P` (Windows/Linux).
2. Run **Positron Assistant: Configure Language Model Providers**.
3. Select **Ollama** (or **Custom/OpenAI-compatible** if Ollama is not listed).
4. Set the base URL to `http://localhost:11434/v1` and leave the API key blank, or, if the client rejects an empty field, enter any placeholder value such as `ollama`.
5. Choose your model from the model list (e.g., `qwen2.5-coder:7b`).

Once configured, Positron Assistant will send requests to your local Ollama server instead of a cloud provider.

Connecting VS Code to Ollama via Continue

[Continue](#) is an open-source VS Code extension that supports Ollama and many other local and cloud backends.

1. Install the **Continue** extension from the VS Code marketplace.
2. Open the Continue sidebar and click the model selector.
3. Choose **Ollama** and select a model (Continue detects running Ollama models automatically).

Continue provides inline completions and a chat panel, similar to GitHub Copilot, but routed entirely to your local model.

Running an Autonomous Coding Agent with Aider

The editor integrations above provide inline completion and chat. For an autonomous agent that reads your files, proposes edits across a whole repository, and commits them to git—the local counterpart to a cloud coding agent—[aider](#) works directly against Ollama.

Install it in an isolated environment so its dependencies do not collide with other tools:

```
# Recommended: isolated install with pipx
pipx install aider-chat

# Or into your user environment with pip
python3 -m pip install --user aider-chat
```

Point it at Ollama and choose a model with the `ollama_chat/` prefix, which gives better results in `aider` than the plain `ollama/` prefix:

```
export OLLAMA_API_BASE=http://127.0.0.1:11434
aider --model ollama_chat/qwen2.5-coder:7b
```

! Raise the Ollama context window

Ollama’s default context window is small, which silently truncates your code and makes the model look far less capable than it is. This is the single most common mistake when pairing `aider` with Ollama.

The default is not a fixed number. Ollama picks it from the memory it detects, as its own `ollama serve --help` states:

```
OLLAMA_CONTEXT_LENGTH    Context length to use unless otherwise specified
                          (default: 4k/32k/256k based on VRAM)
```

Do not assume you landed in a generous tier. A 24 GB Apple-silicon machine gets **4096 tokens**, not the 32k its total memory suggests, because only about 75% of unified memory is addressable by the GPU and the tier boundary sits above that share. Check what you actually got rather than inferring it — `ollama ps` prints the context of each loaded model:

```
ollama ps
# NAME                                SIZE      PROCESSOR    CONTEXT
# qwen2.5-coder:14b                  9.5 GB    100% GPU     4096
```

There are three ways to raise it, and they differ in which clients they reach:

- `OLLAMA_CONTEXT_LENGTH` on the server, which sets the default for everything. Note that the macOS menu-bar app starts the server with its own environment, so

exporting the variable in your shell does not reach it; this route applies when you run `ollama serve` yourself.

- A `num_ctx` parameter sent per request, which is what `aider` does through `~/.aider.model.settings.yml`:

```
- name: ollama_chat/qwen2.5-coder:7b
  extra_params:
    num_ctx: 32768
```

- A `Modelfile` that bakes the context into a derived model, which is the only one of the three that reaches clients that cannot send `num_ctx` themselves:

```
printf 'FROM granite4:7b-a1b-h\nPARAMETER num_ctx 32768\n' > Modelfile
ollama create granite4-32k -f Modelfile
```

The derived model shares weight blobs with its base, so it costs no extra disk.

Context is not free. Raising a 14B model from 4k to 32k on a 24 GB M2 took its resident size from 9.5 GB to 15 GB, about 5.5 GB of key-value cache, so pick the largest value that still leaves the weights and the cache in GPU memory and confirm with `ollama ps` that `PROCESSOR` still reads 100% GPU.

To avoid passing flags every time, set defaults in a config file at `~/.aider.conf.yml`:

```
model: ollama_chat/llama3.2:3b
set-env:
  - OLLAMA_API_BASE=http://127.0.0.1:11434

# Automatically load shared lab instructions and agent conventions
read:
  - ~/path/to/ai-config/AGENTS.md
```

Keep instruction files lean for small local models

While cloud frontier models handle tens of thousands of tokens of system prompt easily, small local models (1.5B–7B) lose speed and instruction fidelity when loaded with large instruction files. A multi-page documentation bundle (such as a 95 KB `CLAUDE.md`) will consume most of an SLM’s active context window and cause severe prompt-ingestion delays. For local SLMs, point `read` at a concise, focused summary file (such as `AGENTS.md` or a project-specific rules snippet) rather than the full multi-tool configuration suite.

Running Aider with a Graphical User Interface (GUI)

While `aider` is primarily used from the command line, it includes a built-in browser-based GUI:

```
aider --model ollama_chat/llama3.2:3b --gui
```

This launches a local web application in your default browser with:

- A visual chat timeline and diff review panel
- Point-and-click file selectors for adding files into the active context
- Speech-to-text voice input support

Integrating Aider into VS Code

You can bring Aider directly into VS Code or Positron through three workflows:

1. **Integrated Terminal:** Run `aider` in the built-in terminal (`Ctrl+``). Edits and Git commits made by Aider immediately reflect in your editor tabs.
2. **VS Code Simple Browser:** Run `aider --gui` in the terminal, open the Command Palette (`Cmd+Shift+P`), and run **Simple Browser: Show** with `http://localhost:8501` to dock the Aider interface side-by-side with your code.
3. **VS Code Extension:** Install a community integration such as **Aider** (by MattFlower) or **Aider Composer** (by lee2py) from the marketplace for dedicated sidebar controls and editor context-menu actions.

`aider` can also split the work between two models in “architect” mode: a larger model plans the change (the architect), and a second model applies the edits (the editor).

```
aider --architect \
  --model ollama_chat/qwen2.5-coder:32b \
  --editor-model ollama_chat/qwen2.5-coder:7b
```

This can improve results on multi-step changes, but on a machine without a strong GPU it roughly doubles the time per turn, because the two models take turns and their weights are swapped in and out of memory. Reserve it for genuinely tricky changes; for small edits, a single model is faster.

! Set `edit_format: whole` for a small model

`aider` asks the model to express an edit either as a SEARCH/REPLACE block (`diff`, the default for most models) or by rewriting the file (`whole`). Producing an exact

SEARCH/REPLACE block is a demanding format, and small models are unreliable at it. Measured on a 24 GB M2 with `granite4:7b-a1b-h` at 32k context, same prompt each time:

File	edit_format: diff	edit_format: whole
One function	3 of 3 correct	3 of 3 correct
Two functions, one to be left alone	0 of 3 correct	3 of 3 correct

The two-function failure is worth dwelling on, because it is not the failure you would expect. The model did not refuse the edit or produce a broken file. In all three `diff` runs it fixed the target function correctly **and silently deleted the other one**, then committed with a message naming only the intended fix. Nothing in the commit message, the exit status, or the model's own summary mentioned the deletion.

That is the hazard of an unattended loop in its most concrete form: a step that reports success while destroying work, leaving a wrong state as the premise for every step after it. It is also why committing after every step matters — `git` held the original, so the damage was one `git revert` away rather than lost.

Set the format per model:

```
- name: ollama_chat/granite4-32k
  edit_format: whole
```

`whole` costs more tokens per edit, since the model rewrites the whole file, which is a real cost on large files. Larger models generally handle `diff` correctly; re-measure rather than assuming either way.

Connecting Claude Code to Ollama

Ollama also serves an **Anthropic-compatible** endpoint at `/v1/messages`, alongside the OpenAI-compatible one used above. Any client that speaks the Anthropic API can therefore be pointed at a local model, including [Claude Code](#) itself, with no proxy in between. Check that the endpoint answers before wiring anything to it:

```
curl -s http://localhost:11434/v1/messages -H 'content-type: application/json' \
-d '{"model": "granite4-32k", "max_tokens": 50,
  "messages": [{"role": "user", "content": "Say OK only."}]}'
```

Point Claude Code at it with three environment variables:

```
export ANTHROPIC_BASE_URL=http://localhost:11434
export ANTHROPIC_AUTH_TOKEN=ollama # any non-empty value; Ollama ignores it
export ANTHROPIC_API_KEY="" # ensure no real key is sent to localhost
claude --model granite4-32k
```

Set these in a wrapper script rather than in your shell profile, so that plain `claude` keeps using the cloud and a separate command uses the local model.

⚠ A heavyweight harness can defeat a small model

This works, but expect worse results than the same model gives through `aider`, and understand why before blaming the model.

A harness spends context before your task does. Claude Code sends a long system prompt and a schema for every tool it exposes, and any Model Context Protocol (MCP) servers you have configured add their own schemas on top. Against a 32k local context that overhead is a large fraction of the budget, and a small model handles it poorly.

Observed on a 24 GB M2 with `granite4:7b-a1b-h`, asking only that it read one file and comment on one function:

- With a GitHub MCP server loaded, the model ignored the question and produced a paragraph about missing credentials.
- With MCP disabled, it invented a task list whose contents were copied from the description of a tool it had been shown.
- With the tool surface cut to `Read`, `Grep`, and `Glob`, it still ignored the question and asked what it should work on.

The same model, same context, through `aider`, fixed a real bug and committed it in 17 seconds. Swapping in a 12B model produced no answer at all in ten minutes, because processing that much prompt at 10 tokens per second is simply slow.

Two practical rules follow. Shrink the tool surface a local model is shown — `--strict-mcp-config --mcp-config '{"mcpServers":{}}'` loads no MCP servers, and `--allowed-tools` narrows the built-ins. And tell the harness the real context size, since Claude Code assumes a 200k window for a model it does not recognize and would let the conversation grow far past what the model can hold:

```
export CLAUDE_CODE_MAX_CONTEXT_TOKENS=32768
```

For autonomous work on a small local model, prefer a light harness such as `aider`. Reserve this route for using a familiar interface offline, not for getting the best out of the hardware.

Falling Back Between Cloud and Local Automatically

Air-gapped work aside, the common case is a laptop that is usually online but sometimes is not—on a plane, behind a flaky hospital network, or temporarily rate-limited by a cloud provider. You can keep a coding agent working across these gaps by putting a cloud model and a local model behind one endpoint and falling back automatically.

LiteLLM runs a small local proxy that presents a single OpenAI-compatible endpoint. You give it a primary model and one or more fallbacks; when the primary fails with a retryable error—a rate-limit response (HTTP 429), or a connection error when you are offline—it retries the request on the next model. Pointing your agent at the proxy instead of directly at a provider makes the cloud-to-local switch automatic and invisible to the tool.

Install the proxy:

```
python3 -m pip install --user "litellm[proxy]"
```

Create a config file (for example, `~/.litellm/config.yaml`) with a cloud primary and a local fallback:

```
model_list:
  # Cloud primary --- replace with your provider and a current model id
  - model_name: coder
    litellm_params:
      model: anthropic/YOUR-MODEL-ID
      api_key: os.environ/ANTHROPIC_API_KEY
  # Local fallback, served by Ollama
  - model_name: coder-local
    litellm_params:
      model: ollama_chat/qwen2.5-coder:7b
      api_base: http://localhost:11434

litellm_settings:
  num_retries: 2
  fallbacks:
    - coder: ["coder-local"]
```

Run the proxy, which listens on `http://localhost:4000`:

```
litellm --config ~/.litellm/config.yaml --port 4000
```

Then point your agent at the proxy. Because the proxy speaks the OpenAI API, most tools accept it as a custom endpoint. For **aider**:

```
export OPENAI_API_BASE=http://localhost:4000/v1
export OPENAI_API_KEY=placeholder # the proxy needs no key unless you set one
aider --model openai/coder
```

Requests now go to the cloud model when it is reachable and fall back to the local model on a rate limit or when you are offline.

i A cloud API key is separate from a chat subscription

The cloud side needs an API key billed per token, issued from the provider's developer console. A chat subscription such as Claude Pro or a Copilot seat is not an API key and cannot be used here. Omit the cloud entry entirely to run local-only, or add the key later to enable the hybrid.

🔥 Keep the proxy on loopback

By default the proxy binds to localhost, which is what you want. Do not expose it on 0.0.0.0 on a shared or untrusted network without authentication, because anyone who can reach the port can spend your cloud API key.

Working in HPC / Cluster Environments

Many HPC clusters do not have outbound internet access on compute nodes but do allow access on login nodes.

i Install the Ollama binary on the cluster first

Ollama must be installed on the cluster (on the host that will run `ollama serve`) before any of the steps below. On most HPC systems you do not have root access, so the `curl | sh` installer may fail or install to the wrong place. Instead, check whether your cluster already provides it (e.g., `module load ollama`), ask your HPC administrators, or download a static binary from the [Ollama releases page](#) and place it on your `PATH`.

A useful pattern:

1. **Pre-pull models** on a login node or a machine with internet access, then copy the model files to the cluster:

```
# On a machine with internet access
ollama pull qwen2.5-coder:7b
# Ollama stores models in ~/.ollama/models by default
rsync -a ~/.ollama/models/ user@cluster.example.edu:~/.ollama/models/
```

 Watch your home-directory quota

Model files are large—`qwen2.5-coder:7b` is ~4 GB and `qwen2.5-coder:32b` is ~20 GB—and most HPC home directories have tight quotas (often 10–50 GB). Filling your home directory can break other jobs. Redirect model storage to a scratch or project filesystem with `OLLAMA_MODELS` and `rsync` to that path instead:

```
# On your local machine: copy the models to the cluster scratch filesystem
rsync -a ~/.ollama/models/ user@cluster.example.edu:/scratch/$USER/ollama-models/
```

```
# On the cluster: point ollama at that path (export before `ollama serve`)
export OLLAMA_MODELS=/scratch/$USER/ollama-models
```

Set the same `OLLAMA_MODELS` value before running `ollama serve` so the server finds the models.

2. **Start Ollama on a compute node** (or an interactive session) using the pre-downloaded model files—no internet required. Set `OLLAMA_HOST=0.0.0.0` so the SSH tunnel from the login node can reach the port:

```
OLLAMA_HOST=0.0.0.0:11434 ollama serve
# If you redirected model storage (see quota warning above):
# OLLAMA_HOST=0.0.0.0:11434 OLLAMA_MODELS=/scratch/$USER/ollama-models ollama serve
```

If you are on a shared compute node, be aware that binding to `0.0.0.0` exposes the Ollama port to other users on that host. Scheduler policies vary by site and job type, so confirm whether your job has exclusive node access (request it explicitly when in doubt—e.g., `--exclusive` in SLURM), or bind only to loopback (`OLLAMA_HOST=127.0.0.1:11434`) and tunnel from the login node when the node is shared.

3. **Forward the port** to your local machine to use your editor’s Ollama integration. Because Ollama is running on a compute node (e.g., `gpu-node-01`), forward through the login node to that specific host:

```
# Replace gpu-node-01 with your actual compute node hostname
ssh -L 11434:gpu-node-01:11434 user@cluster.example.edu
```

This terminal must stay open for as long as you use the editor’s Ollama integration—closing it tears down the tunnel and silently drops the connection. Alternatively, start the tunnel in the background (non-interactive) so it does not occupy a terminal:

```
ssh -N -f -L 11434:gpu-node-01:11434 user@cluster.example.edu
```

(`-N` runs no remote command, `-f` backgrounds `ssh` after authenticating.) To stop the tunnel later, match the full SSH command rather than a bare port string—`pkill -f "ssh.*-N.*11434:gpu-node-01"`—so you don’t accidentally kill unrelated processes

whose command line happens to contain that port. Safer still, note the PID when you start it (`pgrep -f "11434:gpu-node-01"`) and kill that PID directly.

Then configure your editor to use `http://localhost:11434/v1` as the base URL.

⚠ SLURM and GPU Allocation

If running Ollama on a SLURM-managed cluster, request a GPU node with enough VRAM for your chosen model and load any required CUDA modules before starting `ollama serve`. See the [UCD-SERG Lab Manual's SLURM chapter](#) for guidance on requesting GPU resources.

Privacy Considerations

Running a model locally ensures that your code and prompts never leave your machine or cluster. This is important when working with:

- Protected health information (PHI) or other HIPAA-regulated data
- Unpublished research data under data-use agreements (DUAs)
- Proprietary or commercially sensitive code

Even with local models, avoid including raw sensitive data in prompts. Work with anonymized or synthetic data wherever possible.

! “Local” is not automatic—some model tags route to a cloud

Running Ollama does not by itself guarantee that a prompt stays on your machine. Ollama can serve **cloud-hosted** models alongside local ones, and those are the models too large to run on a laptop at all, which is exactly when a tag is tempting. A cloud-routed tag looks much like a local one in everyday use.

Two habits keep this honest, and they matter most in precisely the settings that motivated running locally:

- **Pull and reference explicitly local tags**, and treat a tag with no listed download size as cloud-routed until you check its own page in the [Ollama model library](#).
- **Disable the cloud path outright** when the data is regulated, so the guarantee does not depend on remembering which tag is which:

```
OLLAMA_NO_CLOUD=1 ollama serve
```

Verify rather than trust either one. Cut the machine off the network, or block outbound traffic, and confirm the agent still completes a real task — the check described under

Verifying you are genuinely offline below. A setup that quietly depended on a cloud endpoint fails that test immediately.

Verifying you are genuinely offline

A local setup that has never been tested without a network is a local setup you are guessing about. Cutting the machine off entirely is the honest test. A lighter one that does not disturb the rest of your session is to make outbound traffic fail for a single command, while leaving `localhost` reachable:

```
export HTTPS_PROXY=http://127.0.0.1:9 HTTP_PROXY=http://127.0.0.1:9
export NO_PROXY=localhost,127.0.0.1

# Confirm the block is real before trusting the result:
curl -s -m 5 -o /dev/null -w '%{http_code}\n' https://example.com # 000 = blocked
curl -s -m 5 -o /dev/null -w '%{http_code}\n' http://localhost:11434/api/version # 200 = local

aider --yes --message "Fix the off-by-one error in mean()." stats.py
```

Check the block itself first, as above. A test that passes because the proxy was never applied tells you nothing, and looks exactly like success.

24 Assessing Aeris and Graft

Issue #46 asked whether two GitHub projects, Aeris and Graft, have a place in the lab’s AI workflow. The notes below reflect each repository’s README and metadata as read on 2026-09-09.

Aeris

[Cedrick-Coto/Aeris](#) (Coto 2026) describes itself as a deterministic cognitive simulation engine with emergent narrative, whose README describes an entity-component-system (ECS) architecture in C# (.NET 10). Its design premise is that the language model “verbalizes, never thinks”: a deterministic simulation core computes perception, attention, memory, affect, and goals each tick, and the LLM only turns that state into narrative or dialogue, never modifying it. The target application is a simulated character in a Pokemon world.

Repository facts (measured 2026-09-09):

- GPL-3.0 license

- 2 stars, 1 fork, 2 contributors, 80 commits since creation on 2026-07-26
- 2 commits in the month before the reading date
- 87 Markdown files and three `.csproj` project files on `main`, but no `.cs` source files, so the “complete” engine and “210 tests” that the README roadmap reports are not present in the public repository

The project is a research design document for LLM-backed simulated agents, not a tool for working with AI coding assistants. Nothing in it targets writing, reviewing, or running code.

Useful to us? No

Aeris does not overlap with any lab workflow. Its one transferable idea, keeping the LLM out of state changes and confining it to presentation, is already how this site recommends treating agents: the [responsibility-for-validation policy](#) and the [review guidance](#) put the deterministic checks and the accountability with the human and the CI pipeline, not the model. The GPL-3.0 license would also complicate reuse of any code in lab packages, though at the reading date there was no code to reuse. No further action is warranted.

Graft

[trailhq/Graft](#) (NanoNets and Trail contributors 2026) (formerly `NanoNets/Graft`; the old URL redirected at the reading date) is an open-source “context layer” for coding agents. Its aim is to stop an agent re-exploring a repository at the start of every task by building a persistent map of the codebase once and feeding the relevant parts of that map into each prompt.

Repository facts (measured 2026-09-09):

- MIT license
- TypeScript, published on npm as `@nanonets/graft` (0.16.0 on npm; `package.json` on `main` reads 0.17.0), requiring Node 20 or later
- 6,788 stars, 613 forks, 34 contributors
- 174 commits in the 30 days before the reading date, and 43 open issues (excluding pull requests)

How it works, per the README:

- `graft build` parses the repository with tree-sitter into a per-symbol code graph (functions, classes, call and import edges). This layer is deterministic, runs locally, and never calls a model, so it needs no API key.
- `graft build --deep` adds an LLM layer: a short summary per file, grouped into a few dozen Markdown “nodes” (one per subsystem or concept) with typed links between them and a “crux” excerpt of the lines that carry the logic. The model runs under your own key through one of:

- OpenAI
 - Anthropic
 - OpenRouter
 - a LiteLLM proxy
 - a local server
- `graft init` wires the graph into an agent. For Claude Code it writes a skill file, hooks, and a status line. For the other hosts it adds a fenced section to `AGENTS.md`, `GEMINI.md`, or `.github/copilot-instructions.md`:
 - Codex
 - OpenCode
 - Gemini CLI
 - Copilot
 - Cursor
 - others that read those files

It also registers an MCP server exposing six tools (find code, file API, trace callers, regex search, repo map, freshness check).

- The graph is a git-ignored local cache, rebuilt incrementally and refreshed before each query; only the small wiring is committed.
- R is one of the “full-fidelity” languages, with support for plain functions, S3, S4, and R6 classes and methods, roxygen `@export` tags, and `library()` and `source()` imports.

The README reports benchmarks from the project’s own harness (42% fewer tokens and 60% less latency with equal correctness on 162 runs across two repositories) and a 50-instance SWE-bench Verified run, Claude Code with Claude Sonnet 5 on both arms (33 of 50 resolved with Graft against 27 without, using 23% fewer tokens). These are vendor-run measurements on the vendor’s chosen tasks, so treat them as an upper bound until reproduced. The package sends an anonymous batched usage ping by default; the README states the ping carries no code, paths, or queries, and that `graft telemetry disable` or `DO_NOT_TRACK=1` turns it off.

Useful to us? Yes, trial it on one R package

Graft addresses a cost the lab pays constantly: each new agent session re-reads an R package to rebuild the same picture of its structure. Three things make it a good fit for a trial:

- The structural layer is free and local, so it can run on a package without spending model tokens or sending code anywhere, which matches the constraints in Section 23.
- R is a first-class language in its parser, including S3, S4, and R6 dispatch and roxygen tags.

- It wires into the agents the lab already uses (Claude Code, Codex, OpenCode, Gemini CLI, and Copilot) through their existing instruction files and an MCP server, the mechanism Section 59 covers, so a trial adds no new harness.

Check two things before adopting it lab-wide. First, the benchmark claims are vendor-run, so measure the token and tool-call counts on one of our own packages before and after `graft init`. Second, the `--deep` layer spends model tokens on every changed file and the `init` step for Codex writes to user-level config outside the repository (`~/.codex/`; skip with `--no-global`), so start with the structural layer only and the Claude Code wiring only. The `graft init --dry-run` flag lists every file it would touch, which is the right first command. Graft is complementary to the memory tooling in Section 61: that section covers remembering what happened across sessions, whereas Graft covers what the code is, and the two do not overlap.

25 Connecting OpenCode to Local Models

[OpenCode](#) is an open-source coding agent that runs in your terminal, reads your project, edits files, and runs commands. It supports local models through OpenAI-compatible providers.

This section assumes Ollama is already installed and that you have pulled a code-focused model — see Section 23 for both, including the Linux and Windows install paths.

Verify the server is running:

```
curl -s http://localhost:11434/api/version
# {"version":"0.1.x"}
```

On macOS, `brew services start ollama` registers a launchd agent so the server comes back automatically at login rather than needing a manual start each session.

Install the model-discovery plugin:

Rather than hand-coding each model into `opencode.json`, use the [opencode-local-ollama](#) plugin, which discovers your local Ollama models automatically on startup:

```
opencode plugin --global opencode-local-ollama
```

This writes to `~/.config/opencode/opencode.json`:

```
{
  "plugin": ["opencode-local-ollama"]
}
```

Restart OpenCode and run `/models` to see your local models listed alongside any cloud providers. The plugin reads from Ollama's `/api/tags` and `/api/show` endpoints, so newly pulled models appear on the next restart with no config edits.

i Alternative: multi-backend local provider

If you also run LM Studio, `llama.cpp`, or `vLLM` alongside Ollama, the `opencode-local-provider` plugin auto-detects all of them under a single `local` provider and probes each at runtime for loaded models. Install it with `opencode plugin --global opencode-local-provider`.

A lightweight hand-written provider block in your project's `opencode.json` still works if you prefer explicit control over model names and context limits, but the plugin removes the need to keep that list in sync with `ollama pull`.

26 Connecting OpenCode to OpenRouter

`OpenRouter` is a gateway that exposes hundreds of hosted models — Claude, GPT, Gemini, DeepSeek, Qwen, Kimi, Llama, and more — behind a single API key and billing account. OpenCode treats it as a built-in provider, so its catalog appears in the `/models` picker alongside local models (Section 25). The catalog changes frequently; model IDs below were verified against it in August 2026.

Connect an API key:

1. Create a key at <https://openrouter.ai/settings/keys> and add credits at <https://openrouter.ai/credits>. Some models carry a `:free` ID suffix and cost nothing, at the price of tight rate limits.
2. In the OpenCode TUI, run `/connect`, select **OpenRouter**, and paste the key. The CLI command `opencode auth login` does the same thing outside the TUI. Either way the key is stored in `~/.local/share/opencode/auth.json`, never in `opencode.json`.
3. Run `/models`, filter for `openrouter`, and pick a model.

Pick a model that supports tool calls:

Coding agents drive every action — reading files, editing, running commands — through tool calls. Image-generation, speech, and embedding models have no endpoints that support tool use, so an agent session fails on them immediately with `No endpoints found that support tool use`. Prefer chat or coder variants such as `anthropic/claude-sonnet-4.5`, `deepseek/deepseek-chat`, or `qwen/qwen3-coder`.

One trap worth naming: on OpenRouter, Google lists Gemini 3 Pro only as image-output variants (`google/gemini-3-pro-image`, `google/gemini-3-pro-image-preview`) — there is no plain `google/gemini-3-pro` entry — so those image models are easy to pick by mistake. For

tool-calling work, use one of the Gemini Flash chat variants instead, such as `google/gemini-3-flash-preview`.

Models are addressed as `openrouter/<vendor>/<model>`, for example `openrouter/deepseek/deepseek-chat`.

Optional configuration in `~/.config/opencode/opencode.json` (or the project-level file):

```
{
  "$schema": "https://opencode.ai/config.json",
  "model": "openrouter/deepseek/deepseek-chat",
  "small_model": "openrouter/openai/gpt-oss-20b",
  "provider": {
    "openrouter": {
      "models": {
        "moonshotai/kimi-k2": {
          "options": {
            "provider": { "order": ["baseten"], "allow_fallbacks": false }
          }
        }
      }
    }
  }
}
```

- `model` pins the session default; without it, OpenCode starts each session on its own built-in default
- `small_model` sends housekeeping tasks (session titles, summaries) to a cheap model instead of a frontier one
- entries under `provider.openrouter.models` add models that are not preloaded, or pin routing: OpenRouter load-balances across upstream hosts by default, and `order` restricts requests to named providers ([provider-selection docs](#))

Config loads at startup, so restart OpenCode after editing it.

27 Running Claude Code with Non-Anthropic Models

Claude Code is built around the Anthropic Messages API, and its documentation describes how to point the harness at any endpoint that speaks that format (measured 2026-09-01). That mechanism is what makes it possible to run Claude Code against models Anthropic does not make, and the same documentation says plainly that doing so is unsupported. This section summarizes:

- the mechanism
- the routes people use
- what breaks
- which billing rules apply

A fast-moving snapshot

The configuration claims in this section were checked against the [Claude Code documentation](#) on that date. The community practice and the policy history come from the research summary in [issue #96](#), compiled from press coverage and forum discussion up to August 2026; Anthropic revised the subscription rules more than once during 2026. Re-read the current [gateway documentation](#) and the consumer terms before relying on any of it.

The official position: documented mechanism, unsupported use

Anthropic’s [gateway documentation](#) is written for organizations that run their own gateway in front of **Claude** models, so that credentials, usage tracking, cost controls, and audit logging live in one place. It states that “any gateway that exposes a supported API format works,” and in the same paragraph that Anthropic “doesn’t endorse, maintain, or audit third-party gateway products, and doesn’t support routing Claude Code to non-Claude models through any gateway.”

The supported non-default backends are all Claude models on other clouds:

- [Amazon Bedrock](#) (CLAUDE_CODE_USE_BEDROCK=1)
- [Google Cloud’s Agent Platform, formerly Vertex AI](#) (CLAUDE_CODE_USE_VERTEX=1)
- [Microsoft Foundry](#) (CLAUDE_CODE_USE_FOUNDRY=1)

Routing to a different model family is the unsupported case. It works because the gateway mechanism is provider-agnostic, not because Anthropic tests it.

The configuration surface

The variables that matter, from the [connection guide](#), the [model configuration page](#), and the [environment-variable reference](#):

- ANTHROPIC_BASE_URL points the harness at a gateway. The documentation is explicit that it “changes where requests are sent, not which model answers them.”
- ANTHROPIC_AUTH_TOKEN (sent as `Authorization: Bearer`) or ANTHROPIC_API_KEY (sent as `x-api-key`) carries the credential. ANTHROPIC_AUTH_TOKEN takes precedence over a saved claude.ai login immediately; ANTHROPIC_API_KEY takes over after a one-time approval prompt in interactive mode.

- `ANTHROPIC_MODEL`, and the `ANTHROPIC_DEFAULT_SONNET_MODEL`, `ANTHROPIC_DEFAULT_OPUS_MODEL`, `ANTHROPIC_DEFAULT_HAIKU_MODEL`, and `ANTHROPIC_DEFAULT_FABLE_MODEL` family, map the built-in aliases to whatever model IDs the backend accepts.
- `CLAUDE_CODE_ENABLE_GATEWAY_MODEL_DISCOVERY=1` makes Claude Code query the gateway's `/v1/models` endpoint at startup and add the results to the `/model` picker.
- `ANTHROPIC_CUSTOM_MODEL_OPTION` (with optional `_NAME` and `_DESCRIPTION`) adds a single custom row to the picker; Claude Code skips validation for that ID, so any string the endpoint accepts works.
- `CLAUDE_CODE_DISABLE_EXPERIMENTAL_BETAS=1` stops Claude Code from sending pre-release request fields and beta headers. The connection guide lists it as the fix for 400 errors naming `context_management` or `Extra inputs are not permitted`, which is what a non-Anthropic upstream returns when it rejects Claude-specific fields.
- `CLAUDE_CODE_MAX_CONTEXT_TOKENS` declares the context window for a model ID Claude Code does not recognize; Section 23 shows it in use for a local model.

Claude Code reads these at startup, either from the shell or from the `env` block of `~/.claude/settings.json`.

The routes people use

The forum reports collected in the tracking issue fall into a few shapes, from fewest moving parts to most:

- **A model vendor's Anthropic-compatible endpoint.** Several labs, including [Z.ai](#) (GLM), [Moonshot](#) (Kimi), and [MiniMax](#), publish endpoints that speak the Messages format directly, so the whole setup is `ANTHROPIC_BASE_URL` plus the vendor's key. It is the route those reports describe most often, and the one bundled with the vendors' "coding plan" subscriptions.
- **A hosted aggregator.** [OpenRouter](#) exposes an Anthropic-format endpoint, so Claude Code can talk to it with no local proxy; Section 26 covers the same aggregator from the OpenCode side.
- **A self-hosted gateway.** [LiteLLM](#) and [Kong](#) are the examples that Anthropic's own environment-variable reference names for gateway model discovery. LiteLLM translates Messages-format requests to many providers and adds virtual keys, fallbacks, and cost tracking; Section 23 shows a LiteLLM configuration for local models.
- **A community router.** [Claude Code Router](#) intercepts each request and routes it by type (background, thinking, long-context, default) to a different provider or model. It has the most features of the five, and the tracking issue quotes its own issue tracker describing it as unstable and hard to configure.
- **A local model server.** [vLLM](#) and [Ollama](#) both serve the Messages API directly, so Claude Code can point at either with no proxy; Section 23 covers the mechanics, and Section 28 covers which local models can sustain an autonomous loop.

Treat any self-hosted gateway or community router as infrastructure: pin its version, and keep the official `claude` launcher as a fallback. The tracking issue records a 2026 incident in which malicious LiteLLM releases were briefly published to PyPI, which is the concrete reason for pinning.

What degrades

The forum reports in the tracking issue agree on three seams:

- **The harness is tuned for Claude.** Tool-call conventions, context compaction, and the prompt structure behind plan mode and subagents were built against Claude’s behavior. Other models, reached through a translating gateway, show worse tool-call reliability (a model describing an edit instead of making it is the failure reported most often), occasional compaction failures, and behavior that is hard to debug because two layers are involved.
- **Prompt caching depends on the gateway.** Claude Code re-sends the system prompt, tool schemas, and history on every turn and relies on [prompt caching](#) to make that cheap. The [gateway protocol reference](#) lists caching among the features a gateway has to pass through; a gateway that does not, or an upstream with no equivalent, bills every turn at full input price, which erodes much of the per-token saving from a cheaper model.
- **Claude-specific request fields are rejected.** The 400 errors and the `CLAUDE_CODE_DISABLE_EXPERIMENTAL_BETAS` remedy in the configuration list above are the documented form of this, and the model configuration page adds that the context window Claude Code assumes for an unrecognized model ID may not match the real one.

The summary also notes that benchmark gaps between the strongest open-weight coding models and Claude narrowed during 2026, while cautioning that the open-weight figures it saw were largely the vendors’ own launch numbers. Harness-level reliability is a separate question from benchmark score in any case.

Billing and policy

Two rules from the gateway documentation settle most questions:

- While a gateway credential variable or `apiKeyHelper` is active, “a developer’s `claude.ai` subscription isn’t used ... and the subscription’s usage limits don’t apply.” That traffic is billed per token to whoever owns the credential the gateway forwards.
- Setting `ANTHROPIC_BASE_URL` alone, without a gateway credential, does not replace the subscription; the saved login stays the active credential and its limits apply. The documented case for that is an organization’s own gateway forwarding to Anthropic, not a third-party harness serving other models.

The policy history in the tracking issue concerns the other direction, using a Claude **subscription** login to drive third-party harnesses or proxies. As that summary records it, Anthropic announced in April 2026 that subscriptions would stop covering usage in third-party tools from 2026-04-04, with such usage drawing on separately purchased extra usage or an API key instead, while provisioning your own API keys or provider credentials, billed to the key owner, stayed allowed. The summary also notes that the terms were reworded afterwards, so read it as a pointer to what to check rather than as the current text.

Recommendations

Table 12: Choosing a route for running Claude Code with other models

Goal	Suggested route
Model flexibility is the priority	A model-agnostic harness (OpenCode , Aider , Cline), which is built for it; see Section 25 and Section 26
Claude Code’s skills and plugins with one cheaper model	The vendor’s Anthropic-compatible endpoint with your own key
Several models with team governance	A self-hosted gateway such as LiteLLM, or an enterprise gateway product, pinned as infrastructure
Per-request-type cost routing	Claude Code Router, pinned as infrastructure
Claude models at subscription prices	The official Claude Code CLI signed in with the subscription, which remains fully supported

Whichever route you pick:

- Authenticate non-Claude traffic with your own API key or provider credential rather than a subscription login.
- Assume prompt caching is off until the gateway proves otherwise.
- Measure cost per completed task on your own work rather than comparing token prices; if the alternative is not cheaper on that measure, or tool-call failures disrupt sessions, go back to a supported configuration.

28 Small, Local Models for Autonomous Agentic Coding

[Section 23](#) covers the mechanics of running a model on your own hardware: installing Ollama, wiring up an editor, and driving `aider` against a local endpoint. This section is about a narrower and harder question sitting on top of that setup: which local model to pick, and how

to let it work **autonomously** — making a sequence of edits, commits, and tool calls with no human approving each step — without the loop quietly going wrong.

 A fast-moving, opinionated snapshot

As of August 2026, the open-weight coding-model landscape changes monthly: new releases, new quantizations, and new benchmark numbers appear faster than any static page can track. The model names, sizes, and figures below were verified against each model’s own listing at the time this section was written, not against benchmark round-ups, and they will drift. Re-check the source links before choosing a model for a new project, and re-benchmark on your own tasks rather than trusting a published score — your repository’s mix of languages and idioms is not the benchmark’s.

The honest catch

Small models make more per-step mistakes than frontier cloud models: a slightly wrong function signature, a hallucinated package, a test edited to pass instead of a bug fixed. A human working alongside a small model catches most of these immediately. An **autonomous** loop does not have that human in it, so a small error on step 3 becomes the premise for steps 4 through 40, and the mistakes compound rather than cancel out.

This is the reason **small + local + fully autonomous** is the hardest combination to run safely, and the reason this section spends most of its length on structure rather than on model selection. The fix is not a bigger local model — that only raises the error rate at which the same compounding problem starts to bite. The fix is bounding the loop so that a compounding error is caught and stopped early, covered under [Guardrails for autonomy](#) below.

Model landscape

Prefer a model explicitly trained for **tool calling and agentic use** over a general chat or plain code-completion model: an agentic loop depends on the model reliably emitting well-formed tool calls and stopping when it has finished a step, not only on writing plausible code. A handful of open-weight families currently fit that description well enough to run an autonomous loop against:

Family	Sizes worth running locally	License	Best for
Llama 3.2	3B (2.0 GB), 1B (1.3 GB) dense	Llama 3.2 Community License	High-speed local subagents (< 1s action turnaround); native structured tool calling

Family	Sizes worth running locally	License	Best for
Qwen3-Coder	30B-A3B mixture-of-experts (smallest tag, 19 GB)	Apache 2.0	General-purpose agentic coding across languages
Qwen2.5-Coder	1.5B, 3B, 7B, 14B, 32B dense	Apache 2.0	Fast code generation (3B at ~ 45 tok/s). Verify tool-call formatting in your harness
Phi-4-mini	3.8B (2.4 GB) dense	MIT	Strong multi-step reasoning in a compact footprint
Granite 4	7B-A1B mixture-of-experts (4.2 GB), 32B-A9B	Apache 2.0	A small, fast tool-caller that fits where the tiers above do not
Granite 3.2	2B (1.5 GB) dense	Apache 2.0	Ultra-lightweight IBM tool-calling model for single-task triage
Devstral Small	24B	Apache 2.0	Purpose-built for coding agents (multi-file edits, tool use)
Codestral	22B	Mistral AI Non-Production License	Fill-in-the-middle completion, not redistribution in a product
DeepSeek-Coder-V2	16B (Lite) mixture-of-experts	DeepSeek Model License (commercial use permitted, own terms)	A capable, low-VRAM mixture-of-experts option
GLM-4.x	Flagship models are large MoE (over 100B total parameters)	MIT	Strong agentic benchmarks, but sized for a workstation or rented GPU, not a laptop

Qwen3-Coder’s 30B-A3B tag is a mixture-of-experts model: 30B total parameters, but only about 3.3B active per token. VRAM at rest is set by the total, not the active count — every

expert has to stay resident in memory even though only a fraction fires on any given token — which is why the tag still needs roughly 19 GB at 4-bit quantization, in line with its 30B total rather than its 3.3B active count. What the small active count buys is speed: inference runs closer to a 3–4B model’s pace despite the larger memory footprint. Codestral’s license is worth reading before you rely on it: Mistral’s Non-Production License permits local evaluation but not production or commercial deployment — fine for trying it out, not fine for a lab pipeline that runs unattended.

As a practical floor, treat the 24–32B tier at 4-bit quantization as the smallest size that holds up across a multi-step autonomous loop without frequent tool-call errors. Below that, a model is still useful as an assistant you supervise turn by turn (Section 23 covers exactly that setup), but it is not yet a safe choice to leave unattended.

! Parameter count is the wrong first question

Treat that floor as a statement about *sustained* multi-step loops, not as a filter to apply before anything else. Size predicts tool-calling ability poorly enough that checking it first will mislead you.

Measured on a 24 GB M2 against a single-function tool schema, `qwen2.5-coder:14b` returned an empty `tool_calls` field on four attempts out of four, with `finish_reason: stop` each time. It wrote a correct tool call as prose in the `content` field instead, which no harness will act on. The 4.2 GB `granite4:7b-a1b-h`, at half the parameter count, put a well-formed call in `tool_calls` three times out of three and completed a multi-turn round trip using the result. The smaller model was usable as an agent where the larger one was not, and no amount of context or prompting fixes a model whose calls never reach the field a harness reads.

The advertised capability list does not settle it either. `ollama show qwen2.5-coder:14b` lists `tools`, and the model still cannot be driven by a harness, so treat the tag as necessary rather than sufficient.

So order the questions this way:

1. **Does it emit well-formed tool calls?** One request answers this. A model that fails here cannot be an agent at any size.
2. **Does it fit, with context?** Weights plus key-value cache, verified with `ollama ps`.
3. **Is it big enough to sustain a long loop?** This is where the 24–32B floor applies.

A small model that clears the first two is worth measuring on your own tasks before concluding it cannot be left unattended, because the guardrails below, not the parameter count, are what actually bound the damage from a bad step.

Hardware tiers

VRAM (or unified memory)	Model tier	Autonomy
~8 GB	7–8B	Assistant only — keep a human reviewing every step
~12–16 GB	14–24B	Entry point for a bounded autonomous loop
~24 GB+	30–32B, with context headroom	Comfortable autonomy at the practical floor above
Apple-silicon unified memory (32 GB+)	Same tiers as above, generally slower per token	Well suited to an overnight batch job where wall-clock time matters less

These are rough guides, not guarantees: VRAM headroom for context length matters as much as VRAM for the weights themselves, and a long-running agentic loop accumulates a long conversation history that eats into that headroom as it runs. Check the current requirements on the model’s own listing (the [Ollama model library](#) states them per tag) rather than a rule of thumb, since quantization schemes change.

⚠ Apple unified memory does not map onto the VRAM column

Read the unified-memory row as its own scale rather than as the VRAM figures with a speed penalty attached. Two deductions come off the headline number before any model loads:

- **Only about 75% of unified memory is addressable by the GPU** by default, so a 24 GB machine has roughly 18 GB to work with, not 24.
- **Context is charged on top of the weights.** Measured on a 24 GB M2, raising a 14B model from 4k to 32k context moved it from 9.5 GB resident to 15 GB.

Together those rule out the 30–32B tier on a 24 GB Mac, even though the headline number matches the VRAM column. The smallest **qwen3-coder** tag is 19 GB, which exceeds the addressable ceiling on its own, leaving nothing for context. There is no smaller variant of it to fall back to.

The practical ceiling on 24 GB of unified memory is a **12–14B dense model at 32k context**, or a mixture-of-experts model of similar footprint. Confirm with `ollama ps` after loading: `PROCESSOR` reading 100% GPU means it fits, and anything less means part of the model is on the CPU and the loop will be far slower than the tier table suggests.

Action latency and memory bandwidth

When designing an interactive agent or subagent workflow, **turn turnaround time** dictates whether the tool feels responsive or unusable. In an autonomous or semi-autonomous loop,

latency per action is governed by two phases:

Action Latency = Time to First Token (Prompt Ingestion) + Tool-Call Generation (Decode)

On consumer unified memory hardware (such as Apple M-series chips with ~ 100 GB/s memory bandwidth), the memory bus sets a hard theoretical ceiling on token generation speeds:

- **14B Dense Models (4-bit, ~ 9 GB weights):** Decode is physically capped at ~ 10 – 11 tokens/s. Generating a modest 60-token tool call takes 6 seconds on decode alone. Combined with multi-turn prompt evaluation (3–8 seconds on long conversation histories), the total turn latency exceeds 10–16 + **seconds per action**, which is too sluggish for tight iterative tool loops.
- **7B–8B Models (4-bit, ~ 4.5 GB weights):** Decode reaches ~ 20 – 25 tokens/s, producing action turnarounds in the 4–7 **second** range.
- **1.5B–4B Small Language Models (4-bit, ~ 1 – 2.5 GB weights):** Decode runs at 40–90 + **tokens/s**, and prompt ingestion finishes in < 500 ms with prefix cache reuse. Total action turnaround drops to 1–3 **seconds**, making fast, real-time subagent action loops practical on laptop hardware.

i Prompt caching preserves latency across turns

In a multi-turn agent loop, the system prompt and accumulated history are resent on every iteration. Ensuring the inference server keeps the model resident in memory (`keep_alive: -1` in Ollama) and maintains prompt KV-cache reuse drops Time-To-First-Token on subsequent turns from several seconds to under 50 ms.

An interactive chat REPL is not an agent harness

A common stumbling block when running local models is typing agent instructions directly into `ollama run`:

```
ollama run llama3.2:3b
>>> grab an issue from github and write a PR to fix it
```

In plain `ollama run`, the model is running in an isolated conversational REPL with **no tool schemas, no file access, and no shell or Git access**. Because it cannot actually query GitHub or inspect your repository, it will fabricate a fictional issue (e.g. `cpython/issues/1234`), write fictional code in prose, and describe a non-existent commit. When asked “*did you push the PR?*”, it will correctly admit that it is a text-only assistant without execution capabilities.

An agent requires a **harness** (such as `aider` or a programmatic tool runner) that:

1. Translates available capabilities into structured tool schemas (`tools` parameter).
2. Intercepts the model’s structured `tool_calls` payloads.
3. Executes the corresponding commands or file edits on the local machine.
4. Feeds the command outputs back into the conversation context as tool messages.

Baking deterministic agent presets with Modelfiles

Default local model tags are configured for open-ended conversation (temperature 0.8, small 4k context). For agentic tool use, bake a dedicated model tag via a `Modelfile` to enforce deterministic schema compliance:

```
# Modelfile.llama3.2-agent
FROM llama3.2:3b
PARAMETER temperature 0.0
PARAMETER num_ctx 16384
SYSTEM You are a fast, concise autonomous coding agent. Always execute tasks directly using a
```

Create the derived model:

```
ollama create llama3.2-agent -f Modelfile.llama3.2-agent
```

This ensures:

- **Zero temperature (0.0):** Prevents hallucinated JSON keys or invalid tool parameters.
- **Expanded context (16384):** Accommodates multi-turn tool outputs and file snippets without silent truncation.
- **Direct system persona:** Suppresses conversational preamble (“*Sure, I’d be happy to help with that...*”) in favor of immediate tool invocation.

Routed architectures: a planner and an executor

Section 23 already shows the mechanics of splitting a task between two local models with `aider --architect`: a larger model plans the change, and a smaller one applies the edits. The same split has a name in the research literature and a stronger motivating argument than “it’s faster”: Belcak and NVIDIA’s small-language-model research group argue that most of what an agent does in a loop is “a small number of specialized tasks repetitively and with little variation” — reading a diff, running a test, formatting a commit message — and that a small model is “sufficiently powerful, inherently more suitable, and necessarily more economical” for that work (Beltak et al. 2025). A large model earns its cost only on the steps that genuinely need broad, general reasoning: deciding *what* to change and why.

Two shapes of this pattern are worth knowing:

- **All-local:** a single strong local model (30–32B) does both planning and execution, which is simplest to set up and is the right default for a laptop or workstation with one GPU.
- **Local planner, local executor:** a 30–32B planner drafts each step and a 7–8B executor applies it, trading some plan quality for throughput — worthwhile mainly on hardware that cannot comfortably hold two copies of a 32B model at once.
- **Cloud planner, local executor:** a frontier cloud model plans and a local model executes, which keeps the bulk of file contents on your own machine while still using strong reasoning for the decisions that matter most. This is a hybrid rather than a fully local setup — see the LiteLLM fallback pattern in Section 23 for one way to wire a cloud-with-local-fallback endpoint, which composes with this split.

The same split is the main cost lever in a public field report from a non-programmer (“[Vibe coded this game in four months](#)”, r/ClaudeCode, 2026-08-22; summarized in [issue #98](#)), who built a browser racing game over four months with coding agents. The report names three levers:

- plan with the strongest model available
- implement with cheaper ones
- keep the scope to what can realistically ship

The reported total was roughly \$200 in project-specific subscriptions over the four months, on top of a general-purpose subscription the author already held. The lab’s machine-facing configuration, [Morrison-Lab/ai-config](#), states the same routing rule for agents.

None of these routing choices substitutes for the guardrails below. A well-chosen planner still hands off to an executor that can make a per-step mistake, and the loop still needs a way to catch that.

Guardrails for autonomy

! Errors compound in an unattended loop

A frontier cloud model makes fewer per-step mistakes than a small local one, but the risk that matters here is not the per-step error rate on its own — it is that **autonomous** mode removes the human who would otherwise catch a mistake before it becomes the premise for the next ten steps. Structure the loop so that a mistake is caught by something other than a human watching in real time, or do not leave a small model fully unattended.

The mitigation for a higher per-step error rate is not a better model; it is a loop that cannot silently drift far from a known-good state. Five patterns do most of the work, and each is deliberately mechanical rather than judgment-based — the whole point is that they do not depend on the model noticing its own mistake:

1. **Verification gates after every step.** Run something that actually exercises the change — the test suite, `quarto render`, a linter, `R CMD check` — after each edit, and treat a non-zero exit as a hard stop for that step, not a suggestion. The environment’s pass/fail signal is the judge, never the model’s own claim that it “should work now.”
2. **Capped blast radius.** Run the loop in an isolated Git worktree (see the [worktree](#) feature), and commit after every step that passes its gate — never after a batch of several. A commit-per-green-step history means the worst outcome of a bad step is one commit to roll back, not an unreviewable pile of changes.
3. **Bounded loops.** Cap the total number of iterations and, separately, the number of *consecutive* failed gates. Hitting either cap should stop the loop and report what it tried, not retry indefinitely — a model that fails the same gate three times in a row is not going to succeed on the fourth attempt without a different approach, and a different approach is a decision for a human to make.
4. **Decomposition into specified, testable units, done up front.** Break the work into steps that each have a checkable definition of done before the loop starts, rather than handing the model one large, open-ended goal. A step with a clear pass/fail test is exactly the shape a small model handles well; an open-ended goal is exactly the shape that invites drift.
5. **Full logging.** Keep every prompt, tool call, and gate result the loop produced, so a run that stopped (or that a human later distrusts) can be reviewed after the fact rather than re-run blind.
6. **A tool surface small enough for the model.** Every tool the harness exposes costs context before the task starts, because its schema is sent with the prompt, and a configured Model Context Protocol (MCP) server can add thousands of tokens of definitions on its own. Against a 32k local context that overhead competes directly with the work, and a small model loses: asked to read one file, a 7B model with a full MCP surface loaded produced a paragraph about missing credentials instead, and with MCP disabled invented a task list copied from the description of a tool it had been shown. Expose the smallest set of tools the step actually needs, and prefer a light harness over a heavyweight one — the same model that failed those attempts fixed a real bug and committed it in 17 seconds through `aider`.

These are the guardrails as a reader-facing rationale. The concrete gate wiring — an `ai-config` skill that launches a capped, worktree-isolated local loop, and a `gha` reusable workflow that runs one against a pull request using this repository’s own lint, spellcheck, and render checks as its gates — is tracked separately; see [Companion work](#) below.

Stack-specific notes

This site’s own stack — R, Python, Quarto, Julia, GitHub Actions YAML, and Markdown — is largely about producing *verifiable* artifacts: a script that runs, a document that renders, a workflow that passes. That is exactly the property that makes a narrow, checkable sub-task safe for an autonomous small-model loop, but the safety margin is not the same across languages:

Language / format	Autonomy dial	Notes
Python, Markdown, GitHub Actions YAML	Loosest leash	Well represented in training data; a syntax or lint check is a strong gate on its own
R	Tighter gates	Watch for non-tidyverse idioms and unfamiliar use of S4 or Reference (R5) classes; a model trained mostly on Python code can default to non-idiomatic R
Quarto (.qmd)	<code>quarto render</code> as the pass/fail judge	Have the model edit a known-good <code>_quarto.yml</code> rather than authoring one from scratch — a render failure is a strong, cheap gate
Julia	Shortest leash, strongest model, tightest test gate	The weakest training coverage of this stack’s languages, so treat any unattended Julia change as higher risk by default

None of this changes the guardrails above; it changes how tightly you set them — a smaller step size, a lower consecutive-failure cap, or simply keeping a human in the loop for Julia while letting a Markdown fix run unattended.

Fine-tuning: closing the idiom gap

A local model’s non-idiomatic R or Julia, noted in the stack-specific table above, is a training-data problem rather than a capability problem: the model has seen far less R and Julia than Python, not that it is incapable of writing either. Two lighter options are worth trying before fine-tuning anything:

- **Retrieval**, giving the model your own package’s existing R or Julia code as context so it has concrete idiom to imitate.
- **Prompting**, stating the conventions directly — this repository’s own `CLAUDE.md` and `.github/copilot-instructions.md` are examples of exactly that.

When those are not enough, **LoRA** (Low-Rank Adaptation) and its 4-bit variant **QLoRA** are the standard way to close an idiom gap without retraining a whole model. Both freeze the pretrained weights and train a small set of additional low-rank matrices on top, which cuts the trainable parameter count by orders of magnitude compared to full fine-tuning (Hu et al. 2021). QLoRA adds 4-bit quantization of the frozen weights on top of that, which is what actually

shrinks the memory footprint enough to fine-tune a mid-sized model on a single consumer GPU (Dettmers et al. 2023). [Hugging Face’s PEFT library](#) is the common tooling entry point; the specifics of running it against this lab’s own repositories belong in `ai-config`, not here.

Whatever you fine-tune on, keep a held-out evaluation set of real tasks from your own codebase that the training data never touched, and re-check it after every fine-tuning run — a model that has memorized its training examples will look better on paper than it performs on the next genuinely new task.

Companion work

This page explains the reasoning; it does not implement a launcher or a CI gate. Two companion issues carry the runnable parts, each linking back here for rationale:

- [ai-config #1292](#): a skill that configures and launches a local autonomous loop — model choice, an Ollama or `llama.cpp` endpoint, and the guardrail caps above.
- [gha #436](#): a reusable workflow, a sibling to this repository’s own `claude.yml`, that runs a small/self-hosted-model agent against a pull request, wiring this site’s existing checks as the loop’s verification gates:
 - spellcheck
 - link check
 - non-standard-characters
 - bibliography DOIs

29 VoltAgent: TypeScript Agent Framework

[VoltAgent](#) is an open-source TypeScript framework for building LLM agents, paired with a commercial console called VoltOps that adds observability, deployment, evals, and prompt management (VoltAgent 2026g). The GitHub repository reports about 10,600 stars, 1,100 forks, and an MIT license, with the latest package releases published on 2026-08-27 (measured 2026-09-09). The project started in April 2025, so it is young but actively maintained.

How agents are expressed

An agent is a TypeScript object built from a name, a set of instructions, and a model (VoltAgent 2026a):

```
const agent = new Agent({
  name: "my-agent",
  instructions: "A helpful assistant that can check weather",
  model: openai("gpt-4o-mini"),
  tools: [weatherTool],
  memory,
});
```

The `model` field takes either a model object from the [Vercel AI SDK](#) (the `ai-sdk` packages) or a `provider/model` string that `VoltAgent` resolves for you (VoltAgent 2026a). Because the model layer is the AI SDK, swapping between OpenAI, Anthropic, Google, and other providers is a configuration change rather than a rewrite (VoltAgent 2026g). The agent exposes `generateText()` and `streamText()` methods, and both accept an `output` schema for structured results. The default step limit is five model calls per turn (VoltAgent 2026a).

Other building blocks follow the same pattern:

- **Tools** are functions with a [Zod](#) parameter schema, plus lifecycle hooks and cancellation support. MCP servers can be attached as tool sources “without extra glue code” (VoltAgent 2026g).
- **Workflows** are chains built with `createWorkflowChain()`. Steps are added with `andThen()` (plain code), `andAgent()` (an LLM call), `andWhen()` (a conditional), `andAll()` (parallel, wait for all), and `andRace()` (parallel, first to finish wins). Input, result, and suspend/resume payloads are validated with Zod schemas, and the TypeScript type of the data flowing through the chain is inferred at each step (VoltAgent 2026f).
- **Memory** is a `Memory` object wrapping a storage adapter. Adapters exist for:
 - in-process memory
 - LibSQL/SQLite
 - PostgreSQL
 - Supabase
 - Cloudflare D1
 - a hosted “Managed Memory” on VoltOps

Optional embedding and vector adapters add semantic recall, and a “working memory” slot holds a compact summary the agent can read and update through built-in tools (VoltAgent 2026b).

- **Supervisors and sub-agents** are ordinary agents passed in a `subAgents` array. The supervisor gains a `delegate_task` tool that takes a task description and the names of the sub-agents to call, and returns each sub-agent’s text plus its token usage. A `supervisorConfig` option controls the delegation guidelines, event forwarding, error handling, and the system prompt (VoltAgent 2026d).

A **VoltAgent** instance registers the agents and workflows and serves them over HTTP through a [Hono](#) server, which is how the console and other clients reach them (VoltAgent 2026g).

Observability and hosted parts

The framework emits traces through an OpenTelemetry-based module. In development, the VoltOps console at `console.voltagent.dev` connects from the browser to the local agent process, and the documentation states that in this mode no data is sent to or stored on external servers (VoltAgent 2026e). In production, export to the hosted platform switches on when VoltOps API keys are present in the environment; Langfuse and MLflow are also listed as export targets (VoltAgent 2026c).

VoltOps itself is not open source. The README labels it “Cloud” and “Self-Hosted” (VoltAgent 2026g), but self-hosting is listed only on the Enterprise plan (VoltAgent 2026h). The plans as of 2026-09-09:

- **Developer**, free: one seat, one project, 250 traces per month, seven days of retention.
- **Core**, USD 50 per month: three seats, 50,000 traces per month, 90 days of retention.
- **Pro**, USD 250 per month: 20 seats, 250,000 traces per month, 90 days of retention, SSO.
- **Enterprise**, custom pricing: unlimited tracing and self-hosted deployment.

Overage is USD 10 per 5,000 additional traces (VoltAgent 2026h). The managed RAG “Knowledge Base” (VoltAgent 2026g) and “Managed Memory” (VoltAgent 2026b) adapters are also VoltOps features, so an agent that uses them depends on the hosted service even though the framework around them is MIT-licensed.

Useful to us?

Probably not as a replacement for what we already run. Our orchestration baseline (`?@sec-orch-baseline`) is Claude Code subagents, the **Workflow** fan-out tool, and git worktrees, driven from R and Python repositories. VoltAgent’s supervisor and sub-agent pattern is the same idea expressed as a TypeScript library rather than as a coding harness: the sub-agents are LLM calls with tools, not sessions with a shell and a checkout. Adopting it would mean writing our agent logic in TypeScript and hosting a Node server, which adds a language and a runtime the lab does not otherwise use.

Where it could earn a place:

- **A long-running service** rather than a coding session. A lab chatbot over the manual, or a triage bot for issue queues, is the shape VoltAgent is built for: persistent memory keyed by user and conversation, typed tools, and an HTTP server out of the box.
- **Typed, resumable workflows.** The `createWorkflowChain()` builder with Zod-validated suspend and resume payloads is more structured than what our **Workflow** tool offers for human-in-the-loop pauses.

- **Local trace inspection.** The console's localhost mode gives a trace viewer without sending data off the machine, which matters for anything touching unpublished data. The same OpenTelemetry traces can go to Langfuse or MLflow instead, so the viewer does not lock us in.

The comparison table in [?@sec-orch-comparison](#) covers three other orchestrators and does not include VoltAgent. On that table's dimensions, VoltAgent would read as:

- **Primary domain:** general agent infrastructure, in TypeScript.
- **License:** MIT framework, paid and closed console.
- **Maturity:** young but active.
- **Execution model:** supervisor agents delegating to sub-agents in one Node process.

30 Configuring GitHub Copilot Settings

GitHub Copilot offers numerous configuration options that control how the AI assistant integrates into your development workflow. This section explains the key settings visible in your GitHub account preferences and provides guidance on which options to enable based on your use case.

Model Selection Options

GitHub Copilot provides access to multiple AI models, each with different capabilities and performance characteristics. The available models as of early 2026 include:

Anthropic Claude Models:

- **Claude Opus 4.1:** Most capable model for complex reasoning and analysis
 - *Pros:* Excellent at understanding nuanced requirements, handling complex codebases, superior code quality
 - *Cons:* Slower response times, may be overkill for simple tasks, limited availability (select option required)
 - *When to use:* Complex refactoring, architectural decisions, thorough code reviews
- **Claude Opus 4.5:** Latest version with enhanced capabilities
 - *Pros:* State-of-the-art performance, improved reasoning over 4.1
 - *Cons:* Similar trade-offs to Opus 4.1, requires selection
 - *When to use:* Most demanding tasks requiring cutting-edge capabilities
- **Claude Sonnet 4:** Balanced model optimizing capability and speed
 - *Pros:* Fast responses, strong performance, good default choice
 - *Cons:* Slightly less capable than Opus models for very complex tasks

- *When to use*: General development work, most coding tasks
- **Claude Sonnet 4.5**: Enhanced version of Sonnet
 - *Pros*: Improved over Sonnet 4 while maintaining speed
 - *Cons*: Still not as powerful as Opus for extremely complex scenarios
 - *When to use*: Most daily development tasks
- **Claude Haiku 4.5**: Fast, efficient model for simpler tasks
 - *Pros*: Very fast responses, cost-effective, good for quick questions
 - *Cons*: Less capable for complex reasoning or large codebases
 - *When to use*: Simple completions, quick questions, repetitive tasks

OpenAI GPT Models:

As of 2026-05-08, OpenAI points users to ChatGPT Codex at chatgpt.com/codex. The OpenAI quickstart describes Codex as an AI coding assistant available through the Codex IDE extension that can read files, run commands, and write changes. This quickstart guide also links to a dedicated Codex app for working with local projects: [OpenAI Codex quickstart guide](#). For additional background and platform context, Wikipedia describes Codex as an AI coding agent by OpenAI with desktop app availability on Windows and macOS as an additional access path: [Codex \(AI agent\)](#). In the GitHub Copilot model names shown below, the **-Codex** suffix identifies code-specialized variants (for example, **GPT-5.2-Codex** and **GPT-5-Codex**).

- **GPT-5.2-Codex**: Specialized for code generation
 - *Pros*: Strong code completion, good at common patterns
 - *Cons*: May hallucinate package names or APIs
 - *When to use*: Code completion, common coding patterns
- **GPT-5**: Latest general-purpose model
 - *Pros*: Broad knowledge, good general performance
 - *Cons*: Not specifically optimized for code
 - *When to use*: Mixed tasks involving code and documentation
- **GPT-5-Codex** (various versions including Mini and Max):
 - *Pros*: Specialized variants for different use cases
 - *Cons*: Fragmented options can be confusing
 - *When to use*: Specific scenarios where variant optimizations matter

Connecting Positron Assistant to OpenAI

If you are using Positron Assistant with OpenAI models, set up an OpenAI API key first.

Follow these steps:

1. Go to the [OpenAI API keys page](#).
2. Sign in, choose or create the OpenAI project you want to use, and select **Create new secret key**.
3. Copy the key immediately and store it in a secure password manager.
4. In Positron, open the Command Palette with **Cmd+Shift+P** (or **Ctrl+Shift+P** on Windows/Linux).
5. Run **Positron Assistant: Configure Language Model Providers**.
6. Select **OpenAI**, paste your API key, and complete sign-in.

The [Positron Assistant getting started guide](#) states that OpenAI is enabled by default. If OpenAI does not appear as a provider, update Positron and confirm `positron.assistant.provider.openAI.enable` is not disabled.

Sources: [Positron Assistant setup](#), [OpenAI API key help](#), [OpenAI quickstart](#).

Google Gemini Models:

- **Gemini 2.5 Pro**: High-capability model
 - *Pros*: Strong multimodal capabilities, good at understanding context
 - *Cons*: Less proven in coding scenarios than Claude or GPT
 - *When to use*: Tasks involving images or complex context
- **Gemini 3 Pro/Flash** (Preview): Latest generation
 - *Pros*: Cutting-edge capabilities, flash variant offers speed
 - *Cons*: Preview status means less stable, limited track record
 - *When to use*: Experimental workflows, evaluation of new capabilities

Lab Recommendation: For most lab work, enable **Claude Sonnet 4.5** as your default model. It provides excellent balance of capability and speed. Consider switching to **Claude Opus 4.5** for complex architectural decisions or difficult debugging sessions. Keep **Claude Haiku 4.5** enabled for quick inline completions.

Feature Settings

These settings control where and how Copilot integrates into your development environment:

Editor preview features:

- *What it does*: Enables previews of experimental features in your editor
- *Pros*: Access to latest capabilities before general release
- *Cons*: May have bugs or unstable behavior
- *Recommendation*: **Enable** if you're comfortable troubleshooting issues and want cutting-edge features

Copilot Chat in GitHub.com:

- *What it does:* Enables Copilot chat interface on GitHub.com
- *Pros:* Quick access to Copilot without opening an editor, useful for reviewing PRs
- *Cons:* Only available with paid license
- *Recommendation:* **Enable** (included in GitHub Copilot subscription)

Copilot CLI:

- *What it does:* GitHub Copilot for assistance in terminal
- *Pros:* AI help for command-line operations, shell commands, and git operations
- *Cons:* Requires separate installation and setup
- *Recommendation:* **Enable** and install via `gh extension install github/gh-copilot`

Copilot in GitHub Desktop:

- *What it does:* Enables Copilot in GitHub Desktop app
- *Pros:* AI assistance in GUI git client
- *Cons:* Limited compared to editor integration
- *Recommendation:* **Enable** if you use GitHub Desktop

Copilot Chat in the IDE:

- *What it does:* Enables chat interface in your code editor
- *Pros:* Context-aware help, refactoring assistance, code explanation
- *Cons:* Can be distracting if overused
- *Recommendation:* **Enable** (essential feature)

Copilot Chat in GitHub Mobile:

- *What it does:* Enables Copilot chat in mobile app
- *Pros:* Quick access on mobile devices
- *Cons:* Limited by mobile interface
- *Recommendation:* **Enable** for convenience

Copilot can search the web:

- *What it does:* Allows Copilot to search internet for up-to-date information
- *Pros:* Access to current documentation, recent library changes, latest best practices
- *Cons:* May introduce latency, results depend on search quality
- *Recommendation:* **Enable** for access to current information

Advanced Settings

Dashboard Entry Point:

- *What it does:* Allows instant chatting when landing on GitHub.com
- *Pros:* Quick access to Copilot without navigating menus
- *Cons:* None significant
- *Recommendation:* **Enable** for convenience

Copilot code review:

- *What it does:* Use Copilot to review your code and generate pull request summaries
- *Pros:* Automated code review suggestions, PR summary generation, catches common issues
- *Cons:* May generate false positives, shouldn't replace human review
- *Recommendation:* **Enable** (major productivity boost)

Automatic Copilot code review:

- *What it does:* Automatically reviews all pull requests you create
- *Pros:* Catches issues early without manual triggering
- *Cons:* May be noisy on simple PRs, uses API quota
- *Recommendation:* **Disable** initially; enable only after you're comfortable with code review quality

Copilot coding agent:

- *What it does:* Delegate tasks to Copilot coding agent in repositories where it is enabled
- *Pros:* Autonomous multi-file edits, can execute complex refactoring, runs tests and fixes issues
- *Cons:* Requires careful oversight, can make unwanted changes if instructions unclear
- *Recommendation:* **Enable** (see Section 21 for safe usage guidelines)

Copilot Memory (Preview):

- *What it does:* Remember repository context across Copilot agent interactions
- *Pros:* Better context awareness, learns repository patterns and conventions
- *Cons:* Preview feature governed by pre-release terms, potential privacy implications
- *Recommendation:* **Enable** to help Copilot learn your codebase patterns

MCP servers in Copilot:

- *What it does:* Connect MCP servers to Copilot in all editors and Coding Agent
- *Pros:* Extend Copilot with custom tools and integrations
- *Cons:* Requires MCP server setup and maintenance

- *Recommendation:* **Enable** if you have MCP servers configured; otherwise this setting has no effect

Copilot-generated commit messages:

- *What it does:* Allow Copilot to suggest commit messages when you make changes
- *Pros:* Saves time, generates descriptive messages based on code changes
- *Cons:* May miss important context, still requires review
- *Recommendation:* **Enable** but always review and edit suggested messages

Copilot Spaces:

- *What it does:* View and create Copilot Spaces (collaborative AI environments)
- *Pros:* Share AI context with team members
- *Cons:* Additional complexity for individual work
- *Recommendation:* **Enable** for team collaboration features

Copilot Spaces Individual Access:

- *What it does:* Create individually owned Copilot Spaces
- *Pros:* Personal AI workspaces for complex projects
- *Cons:* May fragment your workflow
- *Recommendation:* **Enable** for flexibility

Copilot Spaces Individual Sharing:

- *What it does:* Share individually owned Copilot Spaces
- *Pros:* Collaborate while maintaining ownership
- *Cons:* None significant
- *Recommendation:* **Enable** for sharing capability

Summary of Recommended Settings

For lab members, we recommend the following configuration:

Enable these features:

- All Copilot Chat options (GitHub.com, CLI, IDE, Mobile)
- Web search capability
- Dashboard Entry Point
- Copilot code review (but not automatic review initially)
- Copilot coding agent
- Copilot Memory
- MCP servers (if configured)
- Copilot-generated commit messages
- All Copilot Spaces options

Model selection:

- Default: Claude Sonnet 4.5
- Complex tasks: Claude Opus 4.5
- Quick completions: Claude Haiku 4.5

Enable with caution:

- Editor preview features (only if comfortable with potential instability)
- Automatic Copilot code review (wait until familiar with review quality)

Following these guidelines will help establish an effective Copilot configuration. The key is to enable features that add value to your workflow while maintaining awareness that AI assistance requires validation (see Section 21).

31 Connecting VS Code to a Custom Model Endpoint (BYOK)

VS Code’s built-in Chat usually talks to GitHub’s hosted models. It can also route requests to a model provider of your own; GitHub calls this “bring your own key” (BYOK). The lab uses BYOK to reach Databricks model serving endpoints, which expose an OpenAI-compatible API, through the community extension [oai-compatible-copilot](#).

This section describes the wiring, four errors that report themselves in the chat panel, and three more that do not.

Wiring the extension to Databricks

Databricks serves models over an OpenAI-compatible endpoint at `https://<workspace>.cloud.databricks.com/serving-endpoints`. Point the extension at it in VS Code `settings.json`:

```
"oaiCopilot.baseUrl": "https://<workspace>.cloud.databricks.com/serving-endpoints",
"oaiCopilot.models": [
  {
    "id": "databricks-claude-opus-5",
    "owned_by": "databricks",
    "family": "claude",
    "context_length": 64000,
    "max_tokens": 16000,
    "delay": 15000,
    "vision": true,
    "apiMode": "openai"
  },
  {
```

```

    "id": "databricks-gpt-5-4",
    "owned_by": "databricks",
    "family": "gpt-5.4",
    "context_length": 64000,
    "max_tokens": 16000,
    "delay": 15000,
    "reasoning_effort": "medium",
    "vision": true,
    "apiMode": "openai"
  },
  {
    "id": "databricks-gpt-5-3-codex",
    "owned_by": "databricks",
    "family": "gpt-5.3-codex",
    "context_length": 64000,
    "max_tokens": 16000,
    "delay": 15000,
    "reasoning_effort": "high",
    "vision": true,
    "apiMode": "openai-responses"
  }
]

```

The `id` of each model must exactly match the name of a deployed serving endpoint in your workspace. The extension sends `id` as the OpenAI `model` field, and Databricks routes the request to the endpoint of that name. Most entries use `POST /serving-endpoints/chat/completions`. Models marked as Responses-API-only in the [Databricks model catalog](#) instead require `apiMode: "openai-responses"`. That set is GPT-5.5 Pro, GPT-5.5, and GPT-5.3 Codex (measured 2026-09-09). It is not a Codex-family property: GPT-5.5 and GPT-5.5 Pro are in the set and are not Codex models. The older `gpt-5-2-codex`, `gpt-5-1-codex-max`, and `gpt-5-1-codex-mini` endpoints that earlier configurations also set to `openai-responses` no longer appear in the catalog (measured 2026-09-09). An `id` that names no real endpoint fails (see the 404 below).

To store your token, run **Set OAI Compatible Multi-Provider Apikey** from the Command Palette (Cmd+Shift+P / Ctrl+Shift+P), choose the `databricks` provider, and paste a Databricks personal access token. The extension keeps it in VS Code's encrypted secret storage (under `oaicopilot.apiKey.databricks`) and sends it as an `Authorization: Bearer` header.

Sizing context and output limits

The model metadata controls both the request and the amount of conversation history Copilot sends:

- `context_length` is the total context window that the extension advertises to Copilot. It may deliberately be smaller than the provider’s maximum window.
- `max_tokens` is the output cap. The extension maps it to `max_output_tokens` in Responses mode.
- `family` selects the closest Copilot system-prompt family.
- `vision` tells Copilot whether it may send images.

The extension advertises input capacity as `context_length` minus `max_tokens`. For example, `context_length: 64000` with `max_tokens: 16000` allows Copilot to send about 48,000 input tokens. Use the full provider window only when the workspace quota can sustain repeated agent turns at that size.

The underlying model’s maximum output is not always a good value for `max_tokens`. [Databricks reserves the requested output allowance before admitting a request](#). Under the standard pay-per-token quota, Claude, GPT-5, and Gemini models generally have a 20,000 output-token-per-minute limit. A 64,000-token request can therefore receive an immediate 429 response even when the underlying model supports that output length.

Use the quota-aware lab defaults in 1. The context column is the operational value to put in `settings.json`, not the model’s maximum capability.

Table 13: Lab defaults for Databricks-hosted models

Model group	Workspace ITPM / OTPM	OAI Copilot context	Output cap	Delay
GPT-5.6	2,000,000 /	400,000	16,000	0 ms
Sol/Terra/Luna	200,000			
Claude	200,000 / 20,000	64,000	16,000	15,000 ms
Opus/Sonnet/Haiku				
GPT-6 Astra	200,000 / 20,000	64,000	16,000	15,000 ms
and GPT-5.5				
through				
GPT-5				
Gemini	200,000 / 20,000	64,000	16,000	15,000 ms
GLM 5.3/5.3	200,000 / 20,000	64,000	16,000	15,000 ms
Flash/5.2				
Grok 4.6	200,000 / 20,000	64,000	16,000	15,000 ms
Inkling	200,000 / 10,000	56,000	8,192	15,000 ms
Kimi K3	200,000 / 10,000	56,000	8,192	15,000 ms
DeepSeek V4	200,000 / 10,000	56,000	8,192	15,000 ms
Flash				
DeepSeek V4	200,000 / 4,000	64,000	2,000	30,000 ms
Pro				

Model group	Workspace ITPM / OTPM	OAICopilot context	Output cap	Delay
GPT OSS 120B/20B	1,000,000 / 100,000	131,072	25,000	0 ms
Qwen3.5 122B	1,000,000 / 100,000	131,072	25,000	0 ms
Llama 4 Maverick	1,000,000 / 100,000	128,000	8,192	0 ms
Llama 3.3/3.1 and Gemma 3	1,000,000 / 100,000	128,000	8,192	0 ms

The ITPM / OTPM column is the Enterprise-tier pay-per-token limit from the Databricks limits page (Databricks 2026a), and the endpoint names are from the model catalog (Databricks 2026b), both measured 2026-09-09. The endpoint ids for the rows added on that date are `databricks-gpt-6-astra`, `databricks-glm-5-3`, `databricks-glm-5-3-flash`, `databricks-glm-5-2`, `databricks-grok-4-6`, `databricks-kimi-k3`, `databricks-deepseek-v4-flash-0731`, `databricks-deepseek-v4-pro-0813`, and `databricks-qwen35-122b-a10b`. The context column is a quota-derived working value, not a provider window. The catalog gives 1 million tokens of context for GLM, Kimi K3, and Inkling, and no window at all for either DeepSeek endpoint (measured 2026-09-09); on a 200,000 ITPM tier a 1 million-token `context_length` overruns every budget in Table 14. Qwen3.5’s 256K window and 25K output cap (measured 2026-09-09) sit on the 1,000,000 tier, so its row copies the GPT OSS values, and its 25,000 output cap is the provider figure, which that tier’s 100,000 OTPM admits. `databricks-inkling` is a real catalog endpoint, listed under Thinking Machine Labs as a Public Preview model that always reasons (measured 2026-09-09), so its row documents a live deployment rather than a local alias. The output cap is a per-request ceiling that most turns never reach, which is why the 20,000 OTPM rows carry a 16,000 cap at four requests per minute. A 4,000 OTPM tier has no such slack: DeepSeek V4 Pro’s 2,000 cap and 30,000 ms delay keep two requests inside the allowance, and 2,000 tokens is too little for an agent-mode edit, so treat that row as a chat-only default and prefer DeepSeek V4 Flash for agent work.

Use `openai-responses` for the endpoints named in the wiring section above, and `openai` for the rest. Read the catalog rather than that list, since an endpoint can change API mode without any model being added. Add the `delay` value to each affected model entry. The extension applies this model-specific pause between requests; models without it fall back to the global `oaicopilot.delay` value. Every context and output cap in the table is a working default, not the underlying model’s maximum capability. Workspaces with higher provisioned or priority limits can raise these after checking their actual quota.

Why the context and delay columns move together

The context and delay columns are one setting expressed twice, so changing either alone breaks the pairing.

A workspace admits ITPM input tokens per minute. The extension advertises `context_length` minus `max_tokens` as the model’s input budget, and the delay sets how often a request can be sent. Sustained work therefore needs

$$\text{input budget} \times \text{requests per minute} \leq \text{ITPM}$$

The Claude row is tuned to sit just under that ceiling. A 15,000 ms delay allows four requests per minute, and each request carries the 48,000-token input budget derived above, so a busy client draws about 192,000 against a 200,000 ITPM tier. That is roughly 96 percent of the allowance, which is why the 64,000 value looks conservative and is not.

This is what makes a larger window expensive. Raising `context_length` while leaving the delay alone multiplies straight through the inequality above. Table 14 gives the largest input budget each tier sustains at a given pace.

Table 14: Largest sustainable input budget by tier and pacing

Workspace ITPM	15 s (4/min)	30 s (2/min)	60 s (1/min)
200,000	50,000	100,000	200,000
1,000,000	250,000	500,000	1,000,000
2,000,000	500,000	1,000,000	2,000,000

A model’s own maximum window is a separate quantity from either column, and it is usually far larger. Registering it directly is the common mistake. Advertising a 1,000,000-token window on a 200,000 ITPM tier offers a single prompt of 984,000 input tokens, which is 4.9 times the entire per-minute allowance, so two full prompts would need about 295 seconds between them. The window is a real capability of the model and it is not available at that quota.

Choosing a model for agent mode

Agent mode spends the input budget faster than chat does, because tool definitions and file contents are sent before any conversation. Two configurations are possible, and the tier decides which one is open.

On a 200,000 ITPM tier the budget can be spent on a large window or on frequent turns, and not on both. Doubling the window means halving the pace, so an agent that reads several files per turn slows to a crawl exactly when it is doing the most work.

The higher tiers change the answer rather than easing it. GPT-5.6 Sol, Terra, and Luna sit on a 2,000,000 ITPM tier, ten times the Claude and Gemini families, and carry a 400,000 context

with no delay. That is 384,000 input tokens at roughly 5 requests per minute: 8 times the Claude window and faster turns at the same time. Prefer that family for agent work on a standard pay-per-token workspace, and keep the 200,000 ITPM families for chat.

Note the direction of the trade, which is the opposite of the intuition. Within one tier a smaller window buys more turns per minute, so the fastest agent configuration is rarely the widest one.

Measure before tuning either column. VS Code's status bar reports tokens used against the advertised window for the current chat, which is the only reading that reflects what a client actually sends. Read it in a **new** chat, since an existing one keeps the context metadata it was created with.

The 64,000/16,000 combination limits one prompt to about 48,000 input tokens. Together with 15-second pacing, that keeps one busy client near rather than far above a 200,000 ITPM tier. It is not a guarantee: the quota is shared across the workspace, prompt sizes vary, and concurrent users or clients consume the same allowance. Increase the delay or reduce `context_length` further when 429s continue.

Use longer retry spacing than the extension's one-second default:

```
"oaicopilot.retry": {
  "enabled": true,
  "max_attempts": 3,
  "interval_ms": 15000,
  "status_codes": []
}
```

The extension already retries 429 responses and doubles this base interval on successive attempts. The longer starting interval gives Databricks' sliding token window time to recover.

For GPT-5 and GPT OSS models, `reasoning_effort` adds a selector to the Copilot model configuration and is forwarded to Databricks. Start with `medium` for general work and `high` for Codex or difficult agentic tasks. Claude and Gemini 2.5 use provider-specific thinking controls; do not copy `reasoning_effort` onto those entries.

Databricks retires model endpoints over time. Before sharing or troubleshooting a configuration, compare every `id` with the current model catalog and remove entries that are no longer listed. An accurate local entry cannot make a retired endpoint work.

An individual endpoint page may also carry a banner reading "This serving endpoint is deprecated. Foundation models are now managed in Unity AI Gateway." That is a statement about where Databricks intends to manage these models, not an outage: the `/serving-endpoints` path keeps serving while the banner is up. Unity AI Gateway's **LLMs** and **Providers** tabs are in beta, and they are inactive in a workspace that has not been enabled for them. In that state there is no gateway base URL to move to, and the configuration

in this section is still the working one. Re-check when those tabs become active. Observed 2026-08-20.

Four errors that report themselves, and why they stack

These failures sit on top of each other: fixing one uncovers the next, so work through them top-down.

1. “No utility model is configured for ‘copilot-utility-small’”

When your main Chat model is a BYOK model, VS Code still needs a small “utility” model for background chores such as generating the conversation title and naming git branches. If none is configured, Chat fails before it ever reaches your provider:

```
No utility model is configured for 'copilot-utility-small'
while the selected main agent model is BYOK.
```

Set `chat.byokUtilityModelDefault` in `settings.json`:

- `"mainAgent"`: reuse your BYOK main model for these chores. This keeps all traffic on your provider and needs no extra endpoint, so it is the simplest choice.
- `"copilot"`: use GitHub’s hosted utility model. This needs an active Copilot subscription.
- `"none"`: the default, which errors on purpose.

This requirement arrived in a mid-2026 VS Code update. Before that, BYOK chat worked without the setting, so an editor update can make a working setup start failing here.

2. [404] ENDPOINT_NOT_FOUND

```
[404] Not Found
{"error_code":"ENDPOINT_NOT_FOUND",
 "message":"The given endpoint does not exist, please retry after
           checking the specified model and version deployment exists."}
```

The `model` name in the request is not a serving endpoint that exists in the workspace. Check that every `id` in `oaicopilot.models` matches a real, deployed endpoint (Databricks workspace → **Serving**), and remove or rename any entry that points at a name with no deployment. A stray placeholder entry, such as a leftover `copilot-utility-small`, is a common cause.

3. [403] Invalid access token

```
[403] Forbidden
{"error_code":403,"message":"Invalid access token."}
```

The stored token is expired or revoked. Databricks OAuth tokens are short-lived and can expire within the day, so a session that worked in the morning can start returning 403 by afternoon; personal access tokens last until their configured expiry. Generate a fresh token (Databricks → **Settings** → **Developer** → **Access tokens**) and re-run **Set OAI Compatible Multi-Provider Apikey**. Prefer a long-lived personal access token to avoid frequent re-authentication. No window reload is needed; the extension reads the token on each request.

4. [429] REQUEST_LIMIT_EXCEEDED

[429] Too Many Requests

REQUEST_LIMIT_EXCEEDED: Exceeded workspace input tokens per minute rate limit for databricks-claude-sonnet-5.

This example is an input-tokens-per-minute (ITPM) failure. Lowering only `max_tokens` does not fix it: Databricks counts the actual prompt and conversation history against ITPM, while `max_tokens` reserves output-tokens-per-minute (OTPM) capacity.

For an ITPM error:

1. lower the model's operational `context_length`;
2. add or increase its model-specific `delay`;
3. lengthen `oaicopilot.retry.interval_ms`;
4. start a new conversation when accumulated history is no longer useful;
5. use GitHub's utility model for background chores when your Copilot plan allows it; and
6. ask the Databricks account team for a higher tier, or use provisioned throughput for sustained workloads.

For an OTPM error, lower `max_tokens` first. For either type, the error can persist until the sliding rate-limit window recovers.

Tip

A quick way to tell 404 from 403: a 404 means the request authenticated but named a missing endpoint (a model-name or configuration problem), while a 403 usually means the token itself was rejected (an authentication problem). The IP access-list 403 below is the exception — there the token is valid and the failure is at the network boundary.

Check the workspace host before minting another token

Both the 404 and the 403 above assume the request reached the workspace you think it did. A `baseUrl` whose `dbc-` host belongs to a *different* workspace produces those same two errors and survives every remedy listed for them. The endpoint name is real and the token is valid; neither one is in the workspace being asked. Issuing a fresh token then fails in exactly the same way, indefinitely.

Compare the host in `oaicopilot.baseUrl` against the invocations URL shown at the top of the endpoint's page in the Databricks console (**Serving**, then the endpoint). Check every per-model `baseUrl` as well: each model entry may carry its own copy of the host, so a single corrected setting can leave dozens of stale ones behind it.

Observed 2026-08-20, where a stale host appeared 42 times in one `settings.json`: once at the top level and once in each of 41 model entries. A second VS Code installation on the same machine carried the same stale host in its own copy of the setting.

! A 403 that no new token can fix — IP access list

There is a third 403 with the same status line but a different message body and a different remedy:

```
[403] Forbidden
{"error_code":403,"message":"Source IP address: <ip> is blocked by Databricks IP ACL for workspace id: [ReqId: ...]"}
```

The request left from an address outside the workspace's IP access list, typically because a VPN dropped or was never connected. The token is valid, the host is correct, and the endpoint exists, so the request succeeds in intent and fails at the network boundary. No credential change resolves it — minting a fresh token repeats the same failure indefinitely, the same shape as the stale-host case one layer further out.

Connect to the network that the workspace allows (restore the VPN or move to an allowed address) and retry; the existing token will then succeed without replacement.

The IP-ACL variant is distinguishable by its message body, which names your source IP and a workspace id. The other two 403 situations are not distinguishable by body alone — both the expired-token case and the wrong-workspace case described in the preceding callout surface as `Invalid access token` (and, for a missing endpoint, as `ENDPOINT_NOT_FOUND`):

Message body	Cause	Remedy
<code>Invalid access token</code>	expired or revoked token, or <code>baseUrl</code> names the wrong workspace	mint a new token or correct the host (see preceding callout)
<code>Source IP address: ... blocked by Databricks IP ACL for workspace: <workspace-id></code>	off-network / VPN down	connect to the allowed network; no credential change

Observed 2026-08-26 on `databricks-gpt-5-6-sol` against `dbc-440c7148-9ff6`, three consecutive requests while a VPN connection was down.

Three failures that name no error in the chat panel

The four errors above print their own text. These three do not name an error. Failures 5 and 6 leave an ordinary-looking reply in the chat panel, and the only record is in VS Code's **GitHub Copilot Chat** output channel (**View**, then **Output**, then pick that channel). Failure 7 is visible in the panel as a `[object Object]` prefix on the reply; it is still a display bug rather than a named error. Open the output channel first whenever a BYOK reply is wrong in a way that names no error.

A reply that begins `[object Object]` and then answers as though you had asked nothing is the symptom that produced all three of these at once. Observed 2026-08-20 with VS Code 1.135.0-insider, Copilot Chat 0.63.2026082004, `oai-compatible-copilot` 0.4.2, and `databricks-claude-sonnet-5`.

5. OAI Compatible API key not found

```
OAI Compatible API key not found  
at ...provideLanguageModelChatResponse (out/provider.js:173)
```

The extension keeps the token in VS Code's encrypted secret storage, which is per install, not per profile. `settings.json` travels through Settings Sync; the secret does not. So a second install, such as Insiders beside stable, shows a complete-looking `oai-compatible-copilot` configuration with no token behind it. A new profile in the same install still sees the existing token. Re-run **Set OAI Compatible Multi-Provider Apikey** in the install that is failing.

This is not the same as the 403 above. There, a token was sent and the provider rejected it; here, no request reaches the provider at all.

6. No lowest priority node found

```
Error: No lowest priority node found (path: ...)  
... [ConversationHistorySummarizer] summarization failed
```

This one comes from Copilot Chat's prompt renderer, not from Databricks. The renderer drops prompt elements in priority order until the prompt fits the input budget the extension advertises, which is `context_length` minus `max_tokens`, or 48,000 tokens at the Claude defaults in 1. The error is what it raises when it has nothing left to drop and the prompt is still over budget. A prompt pruned that far need not still contain your own message, which fits a model that replies it cannot see a request.

Agent mode reaches this sooner than ordinary chat, because tool definitions and instruction files consume the budget before any conversation does. Raise the model's `context_length` and leave `max_tokens` alone: that widens the input allowance without reserving more output tokens per minute. Reducing the number of active tools and instruction files works too. Weigh both against 1, whose context values are chosen to keep one client inside an ITPM tier, so buying

prompt headroom this way costs more 429s. Raising the window without lengthening the delay breaks the pairing those two columns encode, and on a 200,000 ITPM tier the headroom is not there to buy. Switching to a higher-tier family is the move that gets both, as Table 14 sets out.

7. [object Object] in the reply text

Version 0.4.2 renders streamed content with `String(deltaObj.content)` in both of its streaming paths, in `out/openai/openaiApi.js`. A chunk whose `content` is a plain string renders normally. A chunk that carries structured content instead prints as the literal text `[object Object]`, because that is what JavaScript's `String()` returns for an object. Later chunks that carry a plain string render normally. In the 2026-08-20 session the prefix appeared once, at the start of the reply. That is an observation from that session, not a guarantee that later chunks cannot also be structured.

This is a display bug in the extension rather than a configuration error, so no setting turns it off. Report it upstream and read past the prefix.

Note

Copilot's own hosted quota still applies to some background chores even when the main chat model is BYOK. A log line reading `quotaExceeded | gpt-4o-mini-2024-07-18 | [title]` is conversation-title generation failing against GitHub's models, and it says nothing about whether your provider is working. In the 2026-08-20 session above, `chat.byokUtilityModelDefault` was already set to `"mainAgent"` and the title request still went to `gpt-4o-mini`, so that setting did not cover conversation titles in this version.

32 Configuring the Agent Environment

The `.github/workflows/copilot-setup-steps.yml` file allows you to customize the development environment in which the GitHub Copilot coding agent operates. This file preinstalls tools and dependencies so that Copilot can build, test, and lint your code more reliably.

Why Configure the Environment?

While Copilot can discover and install dependencies through trial and error, this can be slow and unreliable. Additionally, Copilot may be unable to access private dependencies. Preconfiguring the environment ensures:

- Faster agent startup and execution
- More reliable builds and tests
- Access to private or authenticated dependencies
- Consistent development environment across all agent sessions

File Location and Structure

The workflow file must be located at `.github/workflows/copilot-setup-steps.yml` in your repository's **default branch**. It follows GitHub Actions workflow syntax but must contain a single job named `copilot-setup-steps`.

Basic Configuration Example

See this repository's own [.github/workflows/copilot-setup-steps.yml](#) for a configuration adapted for R and Quarto projects.

Using `actions/checkout`

The `actions/checkout` action is used to check out your repository code so that the workflow can access it. While Copilot will automatically check out your repository if you don't include this step, **explicitly including it is necessary** when your setup steps need to access repository files.

Why explicitly include checkout?

Many dependency installation steps require access to repository files:

- `r-lib/actions/setup-renv@v2` needs `renv.lock` to install R package dependencies
- `r-lib/actions/setup-r-dependencies@v2` needs `DESCRIPTION` to install R package dependencies
- `npm ci` needs `package-lock.json` to install Node.js dependencies
- `pip install -r requirements.txt` needs the `requirements` file

Without an explicit checkout step, these dependency installation commands will fail because the necessary files won't be available yet.

Basic checkout:

```
- name: Checkout code
  uses: actions/checkout@v4
```

Important: The Copilot coding agent overrides any `fetch-depth` value you set in the checkout step. According to [GitHub's official documentation](#), this override happens “to allow the agent to rollback commits upon request, while mitigating security risks.” The agent dynamically determines the appropriate fetch depth based on the pull request context.

While you cannot control the fetch depth used by Copilot, the agent still has access to sufficient git history to perform its work effectively, including comparing changes and understanding the context of your pull request.

Configurable Options

You can customize only these specific settings in the `copilot-setup-steps` job:

- `steps`: Setup commands and actions to run
- `permissions`: Access permissions (typically `contents: read`)
- `runs-on`: Runner type (Ubuntu x64 Linux only)
- `services`: Database or service containers
- `snapshot`: Save environment state
- `timeout-minutes`: Maximum 59 minutes

All other workflow settings are ignored by Copilot.

Common Setup Tasks

For Node.js/TypeScript projects:

```
- name: Set up Node.js
  uses: actions/setup-node@v4
  with:
    node-version: "20"
    cache: "npm"

- name: Install dependencies
  run: npm ci
```

For Python projects:

```
- name: Set up Python
  uses: actions/setup-python@v5
  with:
    python-version: "3.11"

- name: Install dependencies
  run: pip install -r requirements.txt
```

For R projects:

```
- name: Set up R
  uses: r-lib/actions/setup-r@v2
  with:
    r-version: 'release'
```

```
- name: Install R dependencies
  uses: r-lib/actions/setup-renv@v2
```

Environment Variables and Secrets

To set environment variables for Copilot:

1. Navigate to your repository's **Settings**
2. Go to **Environments**
3. Select or create the `copilot` environment
4. Add environment variables or secrets as needed

Use secrets for sensitive values like API keys or passwords.

Testing Your Configuration

The workflow runs automatically when you modify `copilot-setup-steps.yml`, allowing you to validate changes in pull requests. You can also manually trigger the workflow from the repository's **Actions** tab.

Setup logs appear in the agent session logs when Copilot starts working. If a step fails, Copilot will skip remaining steps and begin working with the current environment state.

Advanced Configuration

Larger runners: For projects requiring more resources, you can use larger GitHub-hosted runners:

```
jobs:
  copilot-setup-steps:
    runs-on: ubuntu-4-core
```

Self-hosted runners (ARC): For access to internal resources or private registries, use Actions Runner Controller (ARC) self-hosted runners:

```
jobs:
  copilot-setup-steps:
    runs-on: arc-scale-set-name
```

Note: When using self-hosted runners, you must disable Copilot's integrated firewall in repository settings and configure appropriate network security controls.

Git Large File Storage (LFS): If your repository uses Git LFS:

```
- uses: actions/checkout@v4
  with:
    lfs: true
```

Further Reading

For complete details, see [Customizing the development environment for GitHub Copilot coding agent](#).

33 Agent Skills

[Agent Skills](#) are a lightweight, open standard for extending AI agent capabilities with specialized knowledge and workflows. The [specification](#) defines a portable, tool-agnostic format that any compatible agent can load.

What is a Skill?

At its core, a skill is a folder containing a `SKILL.md` file. This file includes the `name` and `description` metadata [required by the specification](#), along with instructions that tell an agent how to perform a specific task. Skills can also bundle supporting resources:

- Scripts
- Reference documentation
- Templates and assets

```
my-skill/
  SKILL.md           # Required: metadata + instructions
  scripts/           # Optional: executable code
  references/         # Optional: documentation
  assets/             # Optional: templates, resources
```

How Agents Load Skills

The specification describes a *progressive disclosure* model, in which an agent typically loads a skill in three stages:

1. **Discovery:** At startup, the agent loads only the name and description of each available skill — just enough to know when it might be relevant.
2. **Activation:** When a task matches a skill's description, the agent reads the full `SKILL.md` instructions into context.

3. **Execution:** The agent follows the instructions, optionally executing bundled scripts or loading referenced files as needed.

This means full instructions load only when needed, so agents can maintain many skills with a small context footprint.

Why Use Skills?

Skills package procedural knowledge and team-specific context into portable, version-controlled folders. This gives agents:

- **Domain expertise:** Capture specialized knowledge as reusable instructions
- **Repeatable workflows:** Turn multi-step tasks into consistent, auditable procedures
- **Cross-tool reuse:** Build a skill once and use it across any skills-compatible agent

Further Reading

For the complete specification and more details, see agentskills.io.

A skill is one of several ways to customize an agent, and not always the right one. Section 39 compares it against:

- instruction files
- subagents
- hooks
- permissions

That section also explains why Claude Code’s custom slash commands are now skills themselves.

The [Morrison-Lab/ai-config](https://github.com/Morrison-Lab/ai-config) repository contains an example of personal Claude Code configuration, including user-level skills, hooks, and subagents, synced across machines via Git.

34 Tessl: Skills Registry and CLI

Tessl is a commercial platform for managing [Agent Skills](#) (see Section 33 for what a skill is) the way `npm` or `pip` manage code dependencies: a searchable registry of skills and plugins, a `tessl` command-line tool that installs them into a project, and a hosted service that scores each skill for quality and security and measures whether it changes what an agent produces (Tessl 2026k). The notes below reflect the Tessl website and documentation as read on 2026-09-09.

What it is

Tessl’s home page describes it as an “Agent Enablement Platform” and pitches it at enterprise teams whose developers have accumulated many skills with no inventory, no version control, and no security review (Tessl 2026i). The documentation lists six components (Tessl 2026k):

- a **registry and package manager** for discovering, installing, versioning, and rolling back skills and plugins
- **governance**: security scanning powered by Snyk, role-based access control, install and publish policies, and an audit trail
- **evals**, which run an agent on a task with and without a skill and compare the results
- **observability** of where skills activate across agent sessions
- an **inventory** that scans a GitHub organization and maps every `SKILL.md` across every repository
- the **Tessl Agent**, a conversational agent for driving the platform (in open beta)

The registry page counts more than 3,000 searchable skills, including Anthropic’s own `docx` and `frontend-design` skills and skills published by OpenAI, Google, GitHub, and HashiCorp, each shown with a quality score, an “agent success vs baseline” multiplier from the evals, and a security-scan result (measured 2026-09-09) (Tessl 2026h).

The pages fetched for this review do not describe spec-driven development: apart from one blog article linked from the home page, the current site is about managing skills, not about generating code from specifications.

Relation to the Agent Skills standard

Tessl does not define its own skill format. A Tessl skill is the same `SKILL.md` folder the Agent Skills specification defines, and Tessl’s conformance review checks a published skill against that specification (Tessl 2026f). What Tessl adds is a lifecycle around the standard: a manifest (`tessl.json`) that records which skills a project depends on, versioned publishing with `--bump patch|minor|major`, and `tessl outdated` and `tessl update` for keeping dependencies current (Tessl 2026j).

Tessl distinguishes two kinds of context (Tessl 2026a):

- **rules**, which apply to every task (the role instruction files play in Section 39)
- **skills**, which load only when the task matches their description

A **plugin** bundles skills and rules together, and can also carry MCP servers and hooks, so it is roughly the same unit as a Claude Code plugin.

The registry and the Posit skill collection in Section 35 are two different distribution routes for the same kind of artifact. Posit publishes its skills as a Git repository that the `skills` CLI or Claude Code’s plugin marketplace installs from; Tessl installs from a GitHub repository URL as well as from its own registry (the documentation’s example is `tessl install`

<https://github.com/anthropics/skills>), and either way writes the skill into the agent's skills directory.

Installing and using it

The CLI installs natively (`curl -fsSL https://get.tessl.io | sh`), through Homebrew, or on Windows through `winget install tessl.tessl`; the `npm` package is deprecated (Tessl 2026b). If your organization's policy requires it, download and inspect the script before running it rather than piping it straight to `sh`. It needs Node.js 22.17 or later (Tessl 2026e), and by default it checks for updates every three hours and installs them silently (Tessl 2026b). Logging in (`tessl login`) uses a GitHub or Google account and is optional for searching and installing.

In a project, `tessl init` detects the coding agents present and configures each of them. The supported list is Claude Code, Cursor, Codex, Gemini CLI, Antigravity, GitHub Copilot CLI, and GitHub Copilot for VS Code, with manual setup documented for other MCP-capable agents such as OpenCode and OpenClaw (Tessl 2026e). `tessl search "code review"` searches the registry by meaning rather than by keyword, and `tessl install <workspace>/<skill>` installs a skill into `.tessl/plugins/` and links it into each agent's directory, so for Claude Code the skill appears under `.claude/skills/` (Tessl 2026f). Adding `--global` installs into `~/.tessl/` for every project instead. On Windows the links are directory junctions, so moving the project folder requires running `tessl install` again.

Before installing a third-party skill, `tessl review run security ./path-to-skill` scans it and returns a Snyk severity from LOW to CRITICAL; `--fail-on high` makes the command exit non-zero for use in CI (Tessl 2026c). Installing from the registry warns on critical and high findings but leaves the decision to the user, unless a workspace administrator has set an install policy that blocks them.

Pricing

Tessl charges by usage credits rather than by seat. Publishing and installing skills and plugins is always free; credits pay for reviews, evals, and Tessl Agent sessions, and frontier models consume credits faster than the defaults. The tiers as of 2026-09-09 (Tessl 2026g):

- **Free:** 1,000 credits per month, one workspace, free best-practice reviews on publicly published plugins
- **Team:** \$100 per month for 5,000 credits, top-ups, spending limits, and role management within a workspace
- **Enterprise:** custom pricing with multiple workspaces, SAML single sign-on, install and publish controls, mandatory skills, and bring-your-own-key models

The CLI collects telemetry by default, and the documentation says this may include conversations, code snippets, and file contents; it can be turned off in the CLI configuration (Tessl 2026d). Tessl also reserves the right to train on data from free-tier usage, though it states it has not done so (Tessl 2026d).

Useful to us? Marginally, for vetting skills; not as a package manager

The lab’s skills already live in [Morrison-Lab/ai-config](#), where they are version-controlled, reviewed in pull requests, and installed as a Claude Code plugin, and the Posit skills the lab uses are copied into that repository (see Section 35). Tessl’s package-manager features solve a problem the lab does not have: one repository is the single source of skills, so there is no sprawl to inventory, and the governance, role, and policy features are built for organizations with many teams rather than one. Adopting the CLI would add a Node.js dependency, a `tessl.json` manifest and `.tessl/` directory to each project, and telemetry that includes code by default.

Two parts are worth using without adopting the rest. The registry is a convenient place to browse public skills and see a security-scan result and an eval score before copying one, and the free `tessl review run security` scan is a quick check on a skill from an unfamiliar source before it goes into `ai-config`. The eval model, running the same task with and without a skill and comparing the results, is the only part of Tessl the lab has no equivalent for, and it is worth keeping in mind if the lab ever wants evidence that a skill helps rather than an impression that it does.

35 The Posit Skill Collection

[posit-dev/skills](#) is the collection of [Agent Skills](#) published by Posit (see Section 33 for what a skill is) for R, Python, Quarto, and Shiny work. It is MIT-licensed, and its skills load in Claude Code, Claude.ai, or through the Claude API, or in any other skills-compatible agent. The notes below reflect the repository’s README as read on 2026-09-01.

What it contains

The skills are grouped into eight categories, each of which can be installed on its own:

- **posit-dev**: general developer skills, including an adversarial `critical-code-reviewer`, `describe-design` for architecture documentation with Mermaid diagrams, and `review-testing` for auditing test code after a change
- **github**: pull-request workflows, such as `pr-create`, which opens a PR and then watches CI and debugs failures until it passes, and `pr-threads-address` / `pr-threads-resolve` for review threads
- **open-source**: `create-release-checklist` and `release-post` for R and Python package releases

- **r-lib**: `testing-r-packages` (testthat 3+), `cli`, `cran-extrachecks`, `lifecycle`, `r-package-development`, `mirai` (asynchronous and parallel R), and `alt-text`
- **ggsq1**: writing queries in `ggsq1`, a grammar of graphics for SQL
- **shiny**: `brand-yml`, `shiny-bslib`, and `shiny-bslib-theming`
- **quarto**: `brand-yml`, `authoring` (including migration from R Markdown and `bookdown`), and `alt-text`
- **connect**: `deploy-to-connect` for Posit Connect

Each skill carries an `author`, `version`, and `license` in its `SKILL.md` frontmatter, so a copied skill stays attributable.

How to install it

The README documents these routes:

- the cross-agent skills CLI (`npx skills add posit-dev/skills --list`, `--all`, or `--skill <name>`), which targets Claude Code, Codex, Cursor, Cline, and others
- in Claude Code, adding the repository as a plugin marketplace (`/plugin marketplace add posit-dev/skills`) and then choosing categories in the plugin UI
- in Claude Code, installing a category directly (`/plugin install <category>@posit-dev-skills`)
- copying skill folders into the agent’s skills directory by hand
- uploading a skill to Claude.ai, or loading it through the Skills API

Once installed, the agent activates a skill from its description rather than on an explicit command, which is the progressive-disclosure model Section 33 describes.

Useful to us? Yes, and largely already adopted

The R and Quarto skills match this lab’s daily work, and nine of them are already copied into [Morrison-Lab/ai-config](#), adapted for that repository’s conventions with the upstream MIT license text kept alongside them:

- `brand-yml`
- `cli`
- `cran-extrachecks`
- `create-release-checklist`
- `lifecycle`
- `quarto-authoring`
- `r-package-development`
- `release-post`
- `testing-r-packages`

Copying the skills in rather than installing the marketplace plugin lets them follow the lab’s own review and hook conventions, at the cost of tracking upstream changes by hand. Four categories are not adopted. Three of them have the least overlap with lab work: Shiny app development, `ggsql`, and Posit Connect deployment. The fourth, `posit-dev`, overlaps with practices the lab already has: its `critical-code-reviewer` parallels the adversarial self-review subagent that the lab’s agent configuration dispatches before every push, and `new-work` / `working-on` keep a per-task tracking document where the lab uses the issue tracker and a session notebook. The `github` skills likewise overlap with the lab’s own [pull-request workflow](#) and with the review setup in Section 44, so both categories are worth reading for ideas rather than installing alongside that tooling. The contribution guidance in the repository recommends Anthropic’s `skill-creator` skill for authoring new skills.

36 Posit AI Tooling on GitHub

The [posit-dev](#) GitHub organization holds the open-source work of Posit PBC that does not live under the `rstudio`, `tidyverse`, `r-lib`, or `quarto-dev` organizations. Much of the LLM tooling from Posit started there, so this section reviews the organization for repositories relevant to working with AI. The Posit skill collection, `posit-dev/skills`, is covered separately in Section 35 and is not repeated here.

The listing below was derived from the GitHub API (`gh repo list posit-dev --limit 300`) rather than from browsing the website. The organization had 226 public repositories, of which the 16 below are about building with or evaluating LLMs (measured 2026-09-09). Star counts and last-push dates are as of the same date. The rest of the organization is Positron and its extensions, Shiny for Python, `great-tables`, `pointblank`, `air`, `ggsql`, Posit Team deployment tooling, container images, and workshop material.

AI-related repositories

Repository	Purpose	Language	Stars	Last push	Verdict for this lab
querychat (Posit PBC 2026m)	Natural-language exploration of tabular data, translated to SQL, for R and Python Shiny apps	Python, R	211	2026-09-08	Maybe: a safe pattern for letting an LLM query a data frame

Repository	Purpose	Language	Stars	Last push	Verdict for this lab
mcptools (Posit PBC 2026i)	Model Context Protocol for R: expose a running R session as an MCP server, or register MCP servers with ellmer chats	R	194	2026-08-22	Yes: the MCP layer btw uses, and the way to expose a lab function as a tool
chatlas (Posit PBC 2026e)	Python client for LLM chat and tool calling, the Python counterpart of ellmer	Python	175	2026-09-08	No: the lab builds in R
shinychat (Posit PBC 2026p)	Chat UI component for Shiny, in R and Python	TypeScript, R, Python	137	2026-09-09	Maybe: only if a lab app needs a chat panel
btw (Posit PBC 2026d)	Toolkit for giving LLMs context about an R session: copy context to the clipboard, chat in the IDE, or connect R to coding agents through mcptools	R	135	2026-09-09	Yes: the most direct fit for an R-centric workflow

Repository	Purpose	Language	Stars	Last push	Verdict for this lab
mcp-repl (Posit PBC 2026h)	Sandboxed MCP server that keeps a persistent R or Python session alive across an agent's tool calls, with plots returned as images	Rust	65	2026-07-27	Yes, untried: replaces <code>Rscript -e</code> round trips in Claude Code
shiny-assistant (Posit PBC 2026o)	Source of the hosted Shiny Assistant chat app	Python	58	2026-01-17	No: an application, not a library
raghilda (Posit PBC 2026n)	Retrieval-augmented generation pipeline in Python, with DuckDB or ChromaDB storage	Python	36	2026-09-02	No: the R counterpart is <code>tidyverse/ragnar</code>
shinyrealtime (Posit PBC 2026q)	Voice chat in Shiny apps via the OpenAI Realtime API	Python, R, TypeScript	25	2026-07-31	No
commons (Posit PBC 2026g)	Experimental framework for data agents built on trusted R or Python calculations and data dictionaries	R, Python	20	2026-09-09	Watch: experimental, but aimed at the “agent over our own analyses” problem

Repository	Purpose	Language	Stars	Last push	Verdict for this lab
code-index (Posit PBC 2026f)	Semantic code search over a repository, exposed to coding agents through MCP; parses R, Python, Quarto, and Markdown	Go	6	2026-04-29	Maybe: for large codebases only
ai-lib (Posit PBC 2026a)	TypeScript LLM-provider infrastructure behind Posit Assistant, successor to the archived ai-provider-bridge	TypeScript	5	2026-09-08	No: internal infrastructure
bluffbench2 (Posit PBC 2026c)	Evaluation of whether agents notice subtle data-quality artifacts in plots, implemented with vitals	R	4	2026-07-21	Yes, as a reference: an eval built in R about data-analysis judgment
positron-copilot-chat (Posit PBC 2026l)	Fork of Microsoft's Copilot Chat extension bundled into Positron	TypeScript	4	2026-05-06	No: consumed through Positron itself

Repository	Purpose	Language	Stars	Last push	Verdict for this lab
assistant-feedback (Posit PBC 2026b)	Issue tracker for feedback on Posit Assistant	none	4	2026-05-28	Reference: where to report Positron Assistant problems
nesevals (Posit PBC 2026j)	Evaluation tooling for next-edit-suggestion prompts and models	R	2	2026-04-07	No: relevant to IDE builders

Two tools the lab might expect to find here live elsewhere. [ellmer](#), the R package for LLM chat and tool calling that `btw` and `querychat` build on and that `mcptools` and `shinychat` integrate with, and [ragnar](#), its retrieval companion, are in the `tidyverse` organization. Positron Assistant has no public repository of its own. Going by the repository descriptions, its provider layer is `ai-lib`, the Copilot fork above supplies its chat surface, and the rest is part of the [positron](#) repository (Posit PBC 2026k). The organization also holds workshop and demo repositories (`positron-ai-workshop`, `ai-powered-app-workshop`, `shiny-querychat-workshop`, `posit-conf-chat`) that are worked examples of the packages above rather than tools.

Useful to us? Yes, three tools and one reference

The lab’s workflow is R and Quarto driven by Claude Code and Copilot, so the question is which of these repositories connect that workflow to an R session.

- `btw` is the direct fit. Its clipboard mode pastes the structure of the objects in an R session into whatever chat is open, and its MCP server route (through `mcptools`) lets Claude Code read package documentation and inspect the global environment instead of guessing at column names. The MCP route follows the setup pattern in Section 59.
- `mcptools` is the layer underneath `btw`, and the package to use directly when a lab package wants to expose one of its own functions to an agent as a tool.
- `mcp-repl` addresses a cost the lab already pays: each `Rscript -e` call an agent makes starts a fresh process, so data and packages are reloaded on every check. A persistent, sandboxed session with inline plots is a better fit for iterating on an analysis with an agent, at the cost of a Rust binary to install. It has not yet been tried in the lab.

- **bluffbench2** is a reference rather than a dependency. It is an evaluation written in R with **vitals** that measures whether an agent notices data-quality problems, which is the kind of judgment the lab expects a human reviewer to keep. Anyone building an eval of the lab's own agent configuration can start from its harness.

querychat and **shinychat** matter only if the lab ships Shiny apps with a chat interface. **commons** is a framework for data agents rather than a chat component, and it is still marked experimental. The Python packages (**chatlas**, **raghilda**) duplicate R tools the lab already prefers. Positron Assistant is configured through Positron rather than through these repositories; see Section 30 for its OpenAI setup and Section 23 for running it against Ollama.

37 Useful plugins

This site's Quarto sources already use **Semantic Line Breaks** (SemBr): a line break after each substantial unit of thought, so the source is easier to edit while the rendered HTML still reads as ordinary paragraphs.

sembr/skills packages that convention as Agent Skills for any skills-compatible tool. Install it one of these ways:

- Claude Code: `/plugin marketplace add sembr/skills`
- Cursor: from the **Cursor Marketplace**, or **Settings > Rules > Add Rule > Remote Rule (Github)** with **sembr/skills**
- skills CLI: `npx skills add https://sembr.org`
- Pi: `pi install git:github.com/sembr/skills`

Ask the agent to apply SemBr on new or revised prose (the **sembr-reformat** skill); there is no need to reformat an entire document in one pass.

ponytail makes the agent think like the laziest senior dev in the room: the best code is the one you never write. Before writing, the agent walks a seven-rung ladder:

1. does this need to exist (YAGNI);
2. is it already in the codebase;
3. is it in the stdlib;
4. is it a native platform feature;
5. is it an installed dependency;
6. can it be one line;
7. only then write the minimum that works.

The agent reads the touched code first and is lazy about the solution, never about reading, and never cuts validation, error handling, security, or accessibility. Measured on twelve real Claude Code tasks in a FastAPI + React repo (Haiku 4.5, n=4), it averages about 54 percent less code (up to 94 percent where the agent would otherwise overbuild, for example a date picker)

with about 20 percent lower cost and 27 percent faster, while staying fully safe. It installs as a plugin or skill for more than twenty agents (Claude Code, Codex, Copilot, Cursor, OpenCode, Gemini, and others) and works from a checkout via `AGENTS.md` where a plugin is not needed.

[Contextify](#) keeps your Claude Code and Codex history forever in a private, searchable timeline. Claude Code deletes history after 30 days; Contextify watches both tools, summarizes each message (on-device via Apple Intelligence on macOS 26, or Lite Mode on macOS 15), and lets you search every conversation you ever had. It runs local-first with no account required, and optionally syncs across devices via Cloud Sync or a self-hosted instance you operate. The ambient timeline lets you follow sessions in real time or skim what happened while you were away.

The lab’s portable agent config lives in [Morrison-Lab/ai-config](#). It is a plugin-or-symlink install of skills, hooks, and memories — not a third marketplace next to SemBr. How that config actually reaches a machine, and how a doubled plugin install fails, is Section 40. Section 39 is the worked example of what the corpus contains. Section 38 maps the official and community marketplaces these four sit in and sets each against `ai-config`.

38 The plugin catalog

Section 37 names four plugins the lab already uses. This section is the map they sit on: what Anthropic’s official marketplace contains, which community plugins are worth knowing about, and how each of them relates to the lab’s own [Morrison-Lab/ai-config](#) (Morrison Lab 2026a). It closes with the question that prompted it: what the `ralph-loop` plugin is, and whether it is the same thing as `ai-config`’s `ardi`.

Plugin counts and names change weekly, so every figure below is stamped with the date it was read. The anatomy of a plugin bundle (skills, hooks, MCP servers, agents, commands) is Section 63; this section assumes it.

The three Anthropic marketplaces

Anthropic runs three plugin catalogs for Claude Code, and they differ in who curates them (Anthropic 2026w):

- `claude-plugins-official` ([anthropics/claude-plugins-official](#) (Anthropic 2026q)). Curated by Anthropic at its discretion; there is no application process. Claude Code registers it automatically the first time it starts interactively, so `/plugin install <name>@claude-plugins-official` works with no setup. Browse it with `/plugin` (the **Discover** tab) or at [claude.com/plugins](#).

- **claude-community** ([anthropics/claude-plugins-community](#) (Anthropic 2026s)). Third-party submissions that passed Anthropic’s automated validation and safety screening, each pinned to a commit SHA and synced nightly from the review pipeline. Added by hand: `/plugin marketplace add anthropics/claude-plugins-community`, then `/plugin install <name>@claude-community`. Pull requests against the mirror are closed automatically; submissions go through a form.
- **claude-code-plugins** ([anthropics/claude-code-plugins/](#) (Anthropic 2026p)). The demo marketplace: thirteen example plugins (`commit-commands`, `code-review`, `feature-dev`, `hookify`, `ralph-wiggum`, and others) that show what the plugin system can do. Most of them also ship in the official marketplace under the same or a renamed entry (`ralph-wiggum` there is `ralph-loop` here).

The official README carries the warning that governs all three: Anthropic does not control what MCP servers, files, or other software a plugin includes, and cannot verify that it works as intended or will not change (Anthropic 2026q). A plugin runs arbitrary code with your user privileges, so the trust decision is yours, per plugin, every time.

What the official marketplace contains

The official catalog’s `marketplace.json` lists 287 plugins (measured 2026-09-09 from the copy Claude Code caches locally; (Anthropic 2026q)). Only 38 of them live in the repository’s own `plugins/` directory, which the README describes as the plugins Anthropic develops and maintains. Another 15 sit in `external_plugins/`, and the remaining 234 are pointers to third-party repositories (Amazon, Microsoft, Google’s `gemini-cli-extensions`, Databricks, Hugging Face, Sentry, and many vendors), each pinned to a commit SHA. So “official” mostly means “listed and pinned by Anthropic”, not “written by Anthropic”. By declared category the catalog is:

- development (119)
- productivity (49)
- database (38)
- monitoring (20)
- security (18)
- a long tail of deployment, design, automation, learning, location, testing, migration, and math

The 38 Anthropic-maintained plugins fall into five groups. The verdicts are for this lab’s work (R packages, Quarto sites, Python and shell tooling, GitHub pull requests), not for software teams in general.

Code intelligence

Twelve `*-lsp` plugins (`clangd`, `csharp`, `gopls`, `jdtls`, `kotlin`, `lua`, `php`, `pyright`, `ruby`, `rust-analyzer`, `swift`, `typescript`) connect a Language Server Protocol server so that Claude sees type errors and missing imports after every edit and can jump to definitions instead of grepping (Anthropic 2026w). The plugin does not install the language server binary; you do, and cloud sessions never start it.

Verdict: `pyright-lsp` is worth installing wherever the lab writes Python. There is no R entry in the catalog (measured 2026-09-09), so R work gets no diagnostics from this route; the docs describe an `.lsp.json` for writing your own LSP plugin, which is the path if anyone wants to wire up R's `languageserver`.

Development workflows

- `commit-commands`: commit, push, and open a PR from slash commands.
- `code-review`: multi-agent PR review with confidence-based scoring.
- `pr-review-toolkit`: reviewer agents specialized by concern (comments, tests, error handling, type design).
- `feature-dev`: a phased explore-design-implement workflow with dedicated agents.
- `code-simplifier`: an agent that refines code for clarity without changing behaviour.
- `code-modernization`: a preflight-assess-transform workflow for legacy codebases.
- `security-guidance`: pattern warnings on every edit plus an LLM diff review at **Stop**, with fixes applied in the same session.
- `claude-security`: deeper vulnerability scanning of your own code, at a chosen effort level.
- `ralph-loop`: the self-referential iteration loop, treated on its own below.

Verdict: `security-guidance` fills a gap `ai-config` does not cover and costs little, so install it. `commit-commands`, `code-review`, and `pr-review-toolkit` overlap the lab's forge workflow (claim, PR-on-claim, ardi, the `@claude` review action) and would run beside it rather than replace it; skip them unless you are working outside a lab repository. `feature-dev` and `code-modernization` are aimed at application codebases and have not earned a place here.

Building your own extensions

- `plugin-dev`: seven skills covering hooks, MCP integration, and plugin structure.
- `skill-creator`: create, improve, and benchmark skills.
- `hookify`: write a hook from a conversation pattern or an explicit instruction.
- `mcp-server-dev` and `agent-sdk-dev`: building MCP servers and Claude Agent SDK programs.
- `mcp-tunnels`: reach a private MCP server through an Anthropic tunnel.
- `claude-md-management`: audit `CLAUDE.md`, capture session learnings, keep project memory tidy.

- `claude-code-setup`: analyze a codebase and recommend hooks, skills, and MCP servers for it.

Verdict: `skill-creator` and `plugin-dev` are the ones to reach for when adding to `ai-config`, since the corpus is itself a plugin. `claude-md-management` duplicates `ai-config`'s `ums` and `memorize` skills and its `MEMORY.md` conventions; skip it in lab repositories. `hookify` generates a hook from a description; `ai-config`'s hooks are hand-written Python with a test per hook, so use `hookify` for a personal one-off and the corpus for anything shared.

Output styles and reports

`explanatory-output-style` recreates the deprecated built-in Explanatory style (commentary on implementation choices), and `learning-output-style` implements a Learning style that never shipped (prompts for you to write key pieces yourself). `receipts`, `session-report`, and `project-artifact` generate HTML reports of what you shipped, what a session cost, and a project status page. `playground` builds single-file interactive HTML explorers; `frontend-design` targets distinctive web interfaces; `math-olympiad` solves competition mathematics with adversarial verification; `cwc-makers` sets up a hardware kit.

Verdict: `learning-output-style` is a reasonable choice for a student who wants to learn a codebase rather than delegate it, per Section 49. `session-report` is useful when a quota question comes up. The rest are not relevant to lab work.

External integrations

The `external_plugins/` directory and many third-party entries bundle a pre-configured MCP server: `github`, `gitlab`, `atlassian`, `asana`, `linear`, `notion`, `figma`, `slack`, `sentry`, `vercel`, `firebase`, `supabase`, `context7`, `playwright`, `serena`, `terraform`, and messaging bridges for Discord, Telegram, and iMessage (Anthropic 2026w).

Verdict: `github` is the one that matters here; Section 59 covers configuring it, and Section 48 what it adds. `context7` (live library documentation lookup) is worth a try for Python and JavaScript work. Everything else depends on whether the lab uses the service.

Community plugins worth knowing

Beyond the three Anthropic catalogs, any GitHub repository with a `.claude-plugin/marketplace.json` is a marketplace (`/plugin marketplace add owner/repo`), and the four plugins in Section 37 (`sembr/skills`, `ponytail`, `Contextify`, and `ai-config` itself) all reach you that way. One more is widely enough used to describe here.

Superpowers ([obra/superpowers](#) (Vincent 2026a)) calls itself “a complete software development methodology for your coding agents, built on top of a set of composable skills”. The 6.3.0 build installed on one lab machine carries fourteen skills (measured 2026-09-09):

- brainstorming
- writing-plans and executing-plans
- test-driven-development
- systematic-debugging
- subagent-driven-development and dispatching-parallel-agents
- requesting-code-review and receiving-code-review
- using-git-worktrees
- finishing-a-development-branch
- verification-before-completion
- using-superpowers and writing-skills, the two meta-skills

The `using-superpowers` skill loads at session start so the others trigger on their own. It is MIT-licensed and installs into more than a dozen agents, among them:

- Claude Code, via `/plugin install superpowers@claude-plugins-official`
- Cursor, Codex, and Copilot CLI
- Gemini CLI, OpenCode, and Antigravity
- Pi and Hermes

Verdict: install it if you want an opinionated end-to-end process and are not already running `ai-config`. Running both means two session-start bootstraps competing to set the workflow, and `ai-config` already covers the same ground in its own vocabulary (`brainstorm`, `st`, `ardi`, `adversarial-reviewer`, `clean-worktrees`, `wrap-up`). The skills that do not overlap (`systematic-debugging`, `test-driven-development`) are worth reading even if you do not install the plugin.

The community marketplace itself is small so far (four plugin directories visible on 2026-09-09: `eli5`, `quickdesign`, `testdino`, `tres-finance-plugin`; (Anthropic 2026s)), so “branching out” today means the official catalog’s third-party entries and independent repositories, not that mirror.

Compared with `ai-config`

[Morrison-Lab/ai-config](#) (Morrison Lab 2026a) is the lab’s own plugin. The copy Claude Code caches on one lab machine holds 198 skills and 90 hook scripts (measured 2026-09-09), plus memories and the `shared/` fragments this site vendors into `?@sec-fully-clean` and its neighbours. It installs as a Claude Code plugin (`/plugin marketplace add Morrison-Lab/ai-config`), as a Cursor plugin, and via `bootstrap.sh` for Codex, Gemini CLI and Antigravity, VS Code Copilot, and OpenCode (Section 40).

Set beside the marketplaces above, four differences decide what to install:

- **Scope.** A marketplace plugin does one thing: one language server, one MCP server, one workflow. `ai-config` is a whole working style: forge etiquette (`claim`, `PR-on-claim`, `ardi`, `mwc`), lab coding and writing conventions, quota management, and the memory

that carries lessons between sessions. Nothing in the official catalog is shaped like that; **superpowers** is the closest, and it stops at the software-methodology layer.

- **Enforcement.** The official workflow plugins are mostly skills and agents, which advise. **ai-config** pairs its skills with hooks that block: an unauthorized merge, a force push, a reply that promises without a mechanism. The one official plugin with a comparable **Stop-hook** design is **security-guidance**, which is why it is the clearest addition.
- **Cross-agent reach.** Both **ai-config** and **superpowers** install into several agents from one repository; the official plugins are Claude Code only. If a lab member works in Codex or Antigravity as well, that decides it.
- **Ownership.** A finding in an official plugin is a bug report to Anthropic; a finding in **ai-config** is a PR you can open today, and the corpus’s UMS discipline expects exactly that.

The practical composition, as of 2026-09-09: **ai-config** as the base, **pyright-lsp** and **security-guidance** from the official marketplace, **github** where the MCP server is not already configured by hand, and **sembr/skills** for prose repositories. Everything else is a per-person choice.

What is ralph-loop, and is it ardi?

ralph-loop (official marketplace; **ralph-wiggum** in the demo marketplace) packages Geoffrey Huntley’s “Ralph” technique (Anthropic 2026ad; Huntley 2025). Ralph, in Huntley’s words, “is a Bash loop”:

```
while ;; do cat PROMPT.md | claude-code ; done
```

The same prompt file is fed to a fresh agent run, over and over. Nothing changes between iterations except the repository: the previous run’s files, commits, and test results are what the next run reads. Tests supply the backpressure that keeps each pass honest, and the operator “tunes Ralph by adding a sign” to the prompt when a failure pattern shows up (Huntley 2025). Huntley says it works best on greenfield projects, one task per loop (Huntley 2025).

The plugin moves that loop inside a single Claude Code session. `/ralph-loop "<prompt>" --max-iterations <n> --completion-promise "<text>"` installs a **Stop** hook that intercepts the agent’s attempt to end the turn and feeds the same prompt back, until the agent’s output contains the completion-promise string exactly, or the iteration cap is hit, or you run `/cancel-ralph` (Anthropic 2026ad). The caveats in the README are the important part: the promise is an exact-string match, so it cannot distinguish “done” from “blocked”, and `--max-iterations` (unlimited by default) is the only real safety net. It lists tasks needing human judgment, one-shot operations, unclear success criteria, and production debugging as cases where not to use it.

So: is it like **ardi**? Both are loops that refuse to let the agent stop early, and both rely on a **Stop**-time mechanism (**ai-config**'s own **Stop** hooks are what block a placeholder reply or an empty promise). Past that they answer different questions.

	ralph-loop	ardi
What it loops over	one prompt, re-fed to the same session	one pull request, across review rounds and sessions
Who decides “done”	the agent, by emitting an agreed string	an independent reviewer’s verdict plus green CI on the current head
What changes each round	nothing but the repository state	the reviewer’s new findings, each Addressed, Rebutted, or Deferred
Safety valve	an iteration cap you set	a human merge gate; the loop reports ready and never merges
Where it fits	before a PR exists: grind an implementation against tests	after a PR exists: drive it to fully clean

ardi is not self-referential. Each iteration is triggered by something outside the agent (a review landing, a check turning red), it re-arms itself with a timer rather than a **Stop** hook, and its terminal condition is **?@sec-fully-clean**, which an agent cannot declare about itself. **ralph-loop** has no reviewer in the loop at all; its judge is whichever tests the prompt tells it to run.

The two compose rather than compete. A Ralph loop is a reasonable way to get a well-specified, well-tested change to the point of opening a PR; **ardi** takes over from there. What the lab should not do is use **ralph-loop** as a substitute for review: a loop whose exit is a string the agent writes will exit whether or not the work is right.

39 Customizing an Agent

Section 33 describes one way to extend an agent. It is not the only one, and a lab that knows only that one tends to write every customization as a skill, including the ones that should have been something else.

This section maps the whole surface. The mechanisms differ less in what you can write in them — most are Markdown with a **YAML front matter** header, as Section 8 describes — than in **when they fire and who decides**.

The Question That Picks the Mechanism

Ask two things about the behavior you want:

- **Who triggers it?** You, by typing something; the model, by judging it relevant; or the harness, on a fixed event.
- **What happens if the model disagrees?** Some mechanisms are advice the model may ignore. Others are enforced by the harness whatever the model decides.

That second question is the one people get wrong, and Claude Code’s own documentation is blunt about it. Instruction files are [described](#) as “context, not enforced configuration”, delivered “as a user message after the system prompt”, so “there’s no guarantee of strict compliance.” The same page names the remedy:

To block an action regardless of what Claude decides, use a `PreToolUse` hook instead.

A rule you cannot afford to have ignored does not belong in a `CLAUDE.md`.

Instruction Files: Always-On Context

`CLAUDE.md` and [AGENTS.md](#) are prose the harness loads at the start of a session, with no front matter and no schema (Section 8).

Three properties matter when you write one:

- **Files concatenate; they do not override.** Claude Code [walks up the directory tree](#) from the working directory, and “all discovered files are concatenated into context rather than overriding each other”, ordered from the filesystem root down. A project file does not replace your personal one.
- **Imports exist, and they skip code spans.** The `@path/to/import` syntax pulls in another file, recursively, to a maximum depth of four hops. Paths resolve relative to the importing file. To *mention* a path without importing it, wrap it in backticks — import parsing skips fenced code blocks and code spans.
- **The two filenames are not interchangeable.** Claude Code [reads CLAUDE.md](#), [not AGENTS.md](#), and recommends a `CLAUDE.md` whose first line is `@AGENTS.md` so both tools read one source. A symlink works on macOS and Linux; on Windows it needs Administrator privileges or Developer Mode, so the import is the portable choice. GitHub Copilot, by contrast, [reads all of AGENTS.md](#), `CLAUDE.md`, and `GEMINI.md`.

`AGENTS.md` is stewarded by the Agentic AI Foundation under the Linux Foundation, and it is worth being precise about what it standardizes: a filename, a location, and a nearest-file-wins precedence rule. Asked whether there are any required fields, its own FAQ answers that there are none — “AGENTS.md is just standard Markdown.” So two agents reading the same file are guaranteed to *see* the same text and guaranteed nothing about acting on it alike.

Workspace Rules and Activation Modes

Instruction files and rules can be scoped globally or to a specific workspace. Google Antigravity and Cursor extend basic instruction files by supporting explicit rule activation modes and workspace rules (`.agents/rules/` or `.cursor/rules/*.mdc`):

- **Always On:** Included unconditionally in every prompt context for the workspace (e.g. `GEMINI.md`, `AGENTS.md`, or `.mdc` files with `alwaysApply: true`).
- **Glob Scoped:** Activated automatically only when matching specific file patterns or paths in the workspace (e.g. `globs: ["src/ui/**/*"]` or Copilot's `applyTo`).
- **Model Decision:** Dynamically injected into context when the model judges the rule relevant to the current task or user prompt.
- **Manual:** Explicitly invoked by name or `@mention` during a session.

In Antigravity, workspace rules live under `.agents/rules/` (project-level) with global rules in `~/.gemini/GEMINI.md`, and workspace discovery operates via directory structures (`.agents/skills/` and `.agents/plugins/`).

Skills, and the Commands That Became Them

The distinction most people still draw here is out of date. Claude Code's documentation [states](#):

Custom commands have been merged into skills. A file at `.claude/commands/deploy.md` and a skill at `.claude/skills/deploy/SKILL.md` both create `/deploy` and work the same way.

Existing `.claude/commands/` files keep working, and if a command and a skill share a name, the skill wins. So “slash command versus skill” is now a question about **an older file layout versus a newer one**, not about two different capabilities.

What did *not* collapse is the invocation question. It moved into front matter, where it is now set per skill rather than implied by which directory the file sits in:

Front matter	You can invoke	The model can invoke
<i>(default)</i>	yes	yes
<code>disable-model-invocation: true</code>	yes	no
<code>user-invocable: false</code>	no	yes

This is the setting to think hardest about. A skill the model cannot invoke will never fire unless someone remembers it exists; a skill the model *can* invoke costs context on every turn, because its description sits in the listing whether or not it is ever used. Note also that `user-invocable` controls menu visibility rather than access: to block programmatic invocation, use `disable-model-invocation`.

The portable core is small and worth knowing exactly. The [Agent Skills specification](#) requires precisely two front matter fields — **name** and **description** — and says of the body that there are no format restrictions. Everything past that is a vendor extension: Claude Code’s own docs describe invocation control, subagent execution, and dynamic context injection as extensions to the standard. Portability is therefore real but shallow: the folder and its metadata travel, and how much of the *behavior* travels depends on how alike two agents happen to be.

One concrete sign that the format genuinely crosses vendors: GitHub Copilot [looks for skills](#) in `.github/skills`, `.agents/skills`, and `.claude/skills` — a competitor’s directory name.

Subagents: Delegating to a Fresh Context

Section 7 and Section 10 cover what a subagent *is* and how the harness runs one. The authoring question is narrower: a subagent is a single Markdown file with front matter in `.claude/agents/` (project) or `~/.claude/agents/` (personal), whose body becomes that agent’s entire system prompt.

Two details are easy to get wrong.

Precedence runs the opposite way from skills. For subagents, a project definition [outranks](#) a personal one. For skills, personal [outranks](#) project. If you keep a personal copy of something the repository also defines, which one wins depends on which mechanism you chose.

An @-mention picks the worker, not the words. Naming a subagent guarantees which one runs. It does not hand that subagent your sentence:

Your full message still goes to Claude, which writes the subagent’s task prompt based on what you asked. The @-mention controls which subagent Claude invokes, not what prompt it receives.

Hooks: The Part the Model Cannot Talk Its Way Around

Hooks are the mechanism this manual has not previously covered, and the one that changes what a customization is *worth*. They are [defined in JSON settings files](#) rather than in Markdown, and they run as shell commands, HTTP calls, or LLM prompts at fixed points in the harness’s lifecycle. The [page on instruction files](#) draws the contrast plainly:

Hooks execute as shell commands at fixed lifecycle events and apply regardless of what Claude decides to do.

An event fires, a matcher selects which handlers apply (by tool name, for instance), and the harness passes the handler JSON describing the event. Documented events include **SessionStart**, **UserPromptSubmit**, **PreToolUse**, **PostToolUse**, **Stop**, and **SessionEnd**, among a longer list; consult the reference rather than this page for the current set.

The practical rule: anything stated as “always” or “never” in a `CLAUDE.md` is a candidate to become a hook. Prose asks; a `PreToolUse` hook decides.

Settings and Permissions: The Boundary Around All of It

Settings live in `~/.claude/settings.json` (user), `.claude/settings.json` (project, shared), and `.claude/settings.local.json` (project, private), with a managed-policy layer above them for organizations. The [precedence](#) runs managed, then command-line arguments, then local, then project, then user.

One exception deserves emphasis, because it is the opposite of what the ladder implies:

Permission rules behave differently because they merge across scopes rather than override.

Rules are written as `Tool(specifier)` — for example `Bash(npm run test *)`, `Read(./env)`, or `Skill(commit)` — and sorted into `allow`, `ask`, and `deny`. Because a project’s rules merge with yours rather than replacing them, a repository can tighten what you allow, and cannot quietly loosen it.

MCP Servers Are a Different Axis Entirely

Everything above changes what the agent *knows or must do*. An [MCP](#) server changes what it *can reach*: typed tools, data resources, and reusable templates exposed over a standard protocol. The specification is explicit that it “does not dictate how AI applications use LLMs or manage the provided context.”

So MCP is never the answer to “how do I make the agent follow our convention”, and always a candidate answer to “how do I let the agent query our issue tracker”. [Section 59](#) covers configuration and its failure modes.

Choosing

If you want to...	Use	Fires when
State a convention that should color everything	<code>CLAUDE.md</code> / <code>AGENTS.md</code>	every session, as context
Package a procedure the model should notice on its own	a skill	the model judges it relevant
Package a procedure <i>you</i> will invoke by name	a skill with <code>disable-model-invocation</code>	you type <code>/name</code>
Hand off self-contained work to a fresh context	a subagent	the model delegates, or you <code>@-mention</code>

If you want to...	Use	Fires when
Enforce something regardless of the model	a hook	a lifecycle event
Constrain what may run at all	permissions in settings	every tool call
Give the agent access to an external system	an MCP server	the model calls the tool

What Travels Between Tools

Mechanism	Portable?	Evidence
Agent Skills (<code>SKILL.md</code>)	yes, in format	open standard with a published specification, adopted across Claude Code, Codex, Copilot, Cursor, and Google Antigravity (and legacy Gemini CLI; see Section 11 on Gemini CLI's sunset and folding into Antigravity CLI)
<code>AGENTS.md</code>	yes, as an open specification	standardizes filename, location, and precedence across Codex, Google Antigravity (and legacy Gemini CLI; see Section 11), Cursor, Aider, and Copilot (Claude Code reads <code>CLAUDE.md</code> by default, or imports <code>@AGENTS.md</code>)
MCP servers	yes	open protocol with multiple independent clients
<code>CLAUDE.md</code> / <code>GEMINI.md</code>	by courtesy	vendor instruction files read by default in their respective environments (Claude Code and Google Antigravity; see Section 11 on the legacy Gemini CLI product folding into Antigravity CLI), and supported by courtesy in GitHub Copilot

Mechanism	Portable?	Evidence
Cursor Rules (<code>.cursor/rules/*.mdc</code>)	no	Markdown Cursor (<code>.mdc</code>) files with <code>alwaysApply</code> and <code>globs</code> frontmatter, scoped to Cursor
<code>.github/copilot-instructions.md</code> <code>.github/instructions/*.instructions.md</code>	no	GitHub Copilot only
<code>*.prompt.md</code> prompt files	no	GitHub Copilot only, and not on every Copilot surface Copilot only , and “only available in VS Code, Visual Studio, and JetBrains IDEs”
Hooks, settings, permissions	no	each harness defines its own

The lesson for a lab is to keep the portable layer carrying the meaning. Conventions belong in `AGENTS.md` and skills, which survive a change of tool; hooks and permissions are worth writing, and are worth writing as enforcement of rules that are also stated somewhere portable.

A Worked Example

The [Morrison-Lab/ai-config](#) repository is one lab member’s configuration, versioned and synced across machines. Counting its `main` branch on 4 August 2026 — `git ls-tree -d --name-only origin/main skills/ | wc -l`, and the equivalent for the other directories — it carries:

- 177 skill directories
- 1 file under `commands/`, from before the merge described above
- 21 hook scripts, registered across `UserPromptSubmit`, `PreToolUse`, and `Stop`
- 7 subagent definitions

That distribution is itself the argument. Nearly everything is a skill, because a skill is the mechanism the model can reach for unprompted. The hooks are few and specific, because each one exists to make a rule that was being forgotten impossible to forget.

This repository is a smaller example of the same idea: it carries a `.github/copilot-instructions.md` for conventions that apply everywhere, plus path-scoped files under `.github/instructions/` whose `applyTo` globs attach them only when you edit a matching file.

40 How the Config Reaches a Machine

Section 39 describes *which* mechanism a customization should use. This section is about the step after that decision: how a config like a shared instruction repository actually reaches a machine, and how a broken install fails.

Two agents can load the identical instruction corpus and still behave differently, because behavior depends not only on what the config says but on how it is installed where the agent runs. An install problem is quiet by construction — nothing errors, the work still gets done, and a capability simply goes missing with no message that it existed.

Two Ways the Same Config Reaches a Machine

A repository of skills, hooks, and instruction files can be delivered two ways, and they are alternatives rather than layers.

- **As a plugin.** You enable the repository as a plugin from a marketplace, and the harness loads its skills, commands, and hooks directly from the plugin's own checkout at session start. It refreshes when the marketplace updates.
- **As a per-machine install.** You symlink (or copy) the repository's children into `~/.claude/` — `skills/`, `shared/`, `memories/`, `hooks/`, and `CLAUDE.md` — and register the hooks into `~/.claude/settings.json` by hand or with a script.

The trap is running both at once. The plugin path and the settings-file path carry different command strings, so the harness keeps both, and every hook fires twice. This fails in the safe direction — a doubled guard warns or blocks twice, it never goes missing — which is exactly why it is easy to leave in place unnoticed. Pick one path.

The `~/.claude` Layout

On a symlink-capable system, the children of `~/.claude` are symlinks into a working checkout of the repository, so a `git pull` in that checkout refreshes every skill and rule for free. Windows Git Bash is the common exception: without symlink privileges configured (Developer Mode, or `MSYS=winsymlinks:nativestrict`), its `ln -s` falls back to real copies, so a pull does not propagate and the copies must be re-synced.

Two health checks answer two different questions, and a clean answer to one says nothing about the other.

- **Files.** Does `~/.claude/hooks/<script>` still track the checkout, or has it drifted, gone missing, or become a dangling link?
- **Bindings.** Does `settings.json` actually invoke that script on an event?

A guard can be a perfectly linked file that is registered to nothing, and an unregistered guard and a guard with nothing to block produce the same output: none. So verify both, and read the two results separately.

One Plugin Per Capability

Enabling the *same* repository as a plugin from two marketplaces — a personal fork and a lab org, say — loads its whole skill set into every session twice. That is not just untidy. A large instruction corpus is already a substantial share of a session’s context, and a duplicated one can push a session far enough over the model’s context limit to break subagent delegation, because a subagent independently re-loads that same duplicated skill set at start, and that alone can exceed the limit before any work begins. Keep one copy enabled.

Failure Modes and Their Symptoms

A broken install rarely announces itself; you read it backward from a symptom.

- **A stale hook reference blocks the tool it guards.** A `settings.json` entry pointing at a hook script that no longer exists errors before the guarded tool runs, so every call to that tool fails with the hook’s error rather than the tool’s output. A `PreToolUse` hook on `Bash` that references a missing script, for instance, blocks *all* shell commands until the entry is removed.
- **A dangling symlink silently drops a file.** A `~/.claude` child symlinked into a temporary worktree or a since-deleted clone becomes a dead link when that directory is removed, and the instructions or agent definition it provided quietly stop loading.
- **A check run against the wrong checkout cries wolf.** A verification that compares the installed copies against a stale or unintended checkout can report every entry as misdirected. That is a false alarm from a check aimed at the wrong target, not a fleet of real problems; re-run it against the checkout the install is actually anchored to before acting.

The common thread is that the install layer is a real surface, distinct from the content of the config, with its own failure modes and its own checks. When an agent behaves as though a rule or skill you wrote does not exist, suspect the install before you suspect the rule.

41 Anthropic’s Public Repositories

Anthropic publishes its code under the [anthropics](#) GitHub organization. The organization held 108 repositories when surveyed (measured 2026-09-09). Claude Code itself is not open source, but its public repository is still useful, and so are several others. This section inventories the repositories that matter to us, says what the two headline repositories actually contain, and ends with a verdict on each.

All counts below come from the GitHub API on the survey date (measured 2026-09-09); stars and push dates drift daily, and the license column reports what the API detected. `none` means the API found no license; for `claude-code` and `claude-agent-sdk-typescript` that is because `LICENSE.md` is a proprietary notice rather than a recognized open-source license, marked `none (proprietary)`, and for the other `none` rows there is no license file at all. `not detected` means a license file exists that the API could not classify (it reports `NOASSERTION`). A single row can also hide a split: `skills` has no root license file, so the API reports none, while its README says four of its skills are source-available.

Which repositories matter to us

Forks and archived repositories together are 38 of the 108 (measured 2026-09-09): forks of infrastructure projects that Anthropic contributes patches to (`tokio`, `rayon`, `argo-cd`, `httpcore`, `orjson`, `beam`), and archived companions to research papers (`hh-rlhf`, `toy-models-of-superposition`, `sleeper-agents-paper`). Neither group concerns a lab that consumes Claude as a product. The rest sorts into six groups.

Repository	Purpose	Language	Stars	Last push	License	Verdict
<code>claude-code</code>	Issue tracker, changelog, and bundled plugins for the Claude Code CLI	Python (plugin scripts)	144.5k	2026-09-08	none (proprietary)	Track issues and read the plugins
<code>claude-code-action</code>	GitHub Action that runs Claude Code on PRs and issues	TypeScript	8.8k	2026-09-08	MIT	Already in use; read its examples/
<code>claude-code-base-action</code>	Read-only mirror of <code>base-action/</code> from the repository above	TypeScript	974	2026-09-08	MIT	Reference only; do not pin to it

Repository	Purpose	Language	Stars	Last push	License	Verdict
claude-code-security-review	Security-focused PR review action	Python	6.2k	2026-02-11	MIT	Low priority; superseded by a review prompt
claude-agent-sdk-python	Python SDK wrapping the Claude Code agent loop	Python	8.1k	2026-09-08	MIT	Useful for scripted agents
claude-agent-sdk-typescript	Issue tracker and changelog for the TypeScript agent SDK	Shell	1.7k	2026-09-08	none (proprietary)	Track issues only
anthropic-sdk-python	Raw Messages API client	Python	3.9k	2026-09-04	MIT	Useful for direct API calls
sandbox-runtime	OS-level sandbox (<code>srt</code>) for processes, MCP servers, and agents	TypeScript	5.2k	2026-09-07	Apache-2.0	Trial on Linux and macOS; alpha on Windows
skills	Reference Agent Skills plus the <code>spec/</code> for the skill format	Python	175.4k	2026-09-03	none	Read the spec; borrow document skills
claude-plugins-official	Anthropic-curated plugin marketplace	Python	36.1k	2026-09-09	Apache-2.0	Install from it; see Section 37

Repository	Purpose	Language	Stars	Last push	License	Verdict
claude-plugins-community	Read-only mirror of the community marketplace	Python	3.7k	2026-08-25	Apache-2.0	Browse before writing our own
knowledge-work-plugins	Role-specific plugins built for Claude Cowork	Python	23.9k	2026-09-09	Apache-2.0	Templates for a lab-role plugin
claude-cookbooks	Notebook recipes for the Claude API	Jupyter Notebook	52.6k	2026-09-03	MIT	Reference when scripting the API
courses	Five API and prompting courses	Jupyter Notebook	22.8k	2026-08-28	not detected	Introductory material
prompt-eng-interactive-tutorial	Nine-chapter prompting tutorial	Jupyter Notebook	38.1k	2026-08-28	none	Introductory material; dated model names
claude-quickstarts	Starter apps you can deploy (support agent, computer use)	TypeScript	17.6k	2026-09-04	MIT	Skim only
devcontainer-features	Dev Container feature that installs the CLI	Shell	302	2025-12-16	MIT	Useful for a GitHub Codespaces setup

Repository	Purpose	Language	Stars	Last push	License	Verdict
<code>claude-code-monitoring-guide</code>	Telemetry and cost-tracking guide	Markdown	369	2025-07-29	none	Read once if we meter usage
<code>claude-ai-mcp</code>	Issue tracker for MCP inside <code>claude.ai</code>	Markdown	467	2026-06-08	not detected	Search before filing an MCP bug

The six groups, and what each is for:

- **Product trackers.** `claude-code`, `claude-agent-sdk-typescript`, and `claude-ai-mcp` exist mainly so users can file issues against a closed product.
- **Automation.** `claude-code-action`, its mirror, and `claude-code-security-review` run Claude inside GitHub Actions.
- **Programmatic access.** The `anthropic-sdk-*` clients (Python, TypeScript, Go, Java, Ruby, C#, PHP) speak the raw API; `claude-agent-sdk-python` wraps the whole Claude Code agent loop and bundles the CLI, and the TypeScript package does the same from npm, though its repository here is only a tracker.
- **Extensions.** `skills`, the two plugin marketplaces, and `knowledge-work-plugins` hold the skills and plugins the harness loads.
- **Infrastructure.** `sandbox-runtime`, `devcontainer-features`, and `claude-code-monitoring-guide` are about where the CLI runs and what it costs.
- **Learning material.** `claude-cookbooks`, `courses`, `prompt-eng-interactive-tutorial`, and `claude-quickstarts` teach the API rather than the CLI.

Two repositories one might look for are not in the organization. There is no `mcp` repository, because the Model Context Protocol lives under its own `modelcontextprotocol` organization, and the former `dxt` repository (Desktop Extensions) now redirects to `modelcontextprotocol/mcpb`, the renamed MCP Bundles format (measured 2026-09-09).

What the public `claude-code` repository actually contains

The repository named after Claude Code, second in the table only to `skills` by stars, does not contain Claude Code. The tree at `main` has 333 paths and no `src/` directory (measured 2026-09-09) (Anthropic 2026e). `LICENSE.md` is a one-line proprietary notice pointing at Anthropic’s commercial terms, which is why the API reports no license. The CLI itself ships as a compiled npm package (and, since the README marked npm installation deprecated, as a native installer, a Homebrew cask, and a `winget` package; read 2026-09-09) (Anthropic 2026e), and this repository is the public face of that closed product.

The repository holds five things the lab can use:

- **CHANGELOG.md** is the only authoritative per-release changelog. It is the file to read when a behavior changes between versions, and its entries are detailed enough to diagnose regressions (the entry for 2.1.266, for example, names the gateway environment variable, `CLAUDE_CODE_USE_GATEWAY`, that 2.1.265 had started honoring on its own, and says no configuration change is needed; read 2026-09-09) (Anthropic 2026f).
- **plugins/** holds thirteen first-party plugins and a `.claude-plugin/marketplace.json` that publishes them as the `claude-code-plugins` marketplace (Anthropic 2026o). Several map directly onto lab workflows: `code-review` (five parallel review agents with confidence scoring), `pr-review-toolkit` (six specialist review agents, including a `silent-failure-hunter`), `commit-commands`, `hookify` (generates hooks from observed misbehavior), `ralph-wiggum` (a `Stop-hook` loop that keeps re-running one task), and `security-guidance` (a `PreToolUse` hook watching nine patterns). `plugin-dev` is the toolkit for writing more.
- **examples/** holds the reference configurations that the documentation describes in prose: `settings/` (strict, lax, and bash-sandbox profiles), `hooks/` (a Bash command validation hook), `mdm/` (managed settings for macOS and Windows fleets), and `gateway/` (AWS and GCP gateway setups).
- **.devcontainer/** is the container Anthropic uses for its own sandboxed sessions, including `init-firewall.sh`, the allowlist firewall that Section 22 discusses.
- **.github/workflows/** is a live example of running the action against a very large issue tracker: `claude.yml` (the `@claude` agent), `claude-issue-triage.yml`, `claude-dedupe-issues.yml`, and `auto-close-duplicates.yml`, with the TypeScript behind them in `scripts/`. The repository had 12,563 open issues (measured 2026-09-09), which is why that automation exists.

So the honest description is “issue tracker plus plugins plus example configs”. The harness internals are described from the outside in Section 8; nothing in this repository lets you read them.

claude-code-action: modes and what ships with it

`claude-code-action` is genuinely open source (MIT), and the whole action is readable: `src/` holds the entry points, the GitHub client, an in-process MCP server for file operations, and the two execution modes; `test/` has a test file per concern (comment sanitizing, branch validation, permissions, public-comment redaction, SSH signing) (Anthropic 2026g).

Section 44 already explains that review is a prompt, not a separate action. The general form of that observation is the mode detector in `src/modes/detector.ts`, documented in `docs/experimental.md` (Anthropic 2026y):

1. If the workflow supplies a **prompt** input, the action runs in **agent mode**: it executes the prompt directly on whatever event fired, which is how scheduled maintenance, issue triage, and one-shot review work.
2. If there is no **prompt** but the event carries an **@claude** mention, an assignment, or the trigger label, it runs in **tag mode**: it posts a tracking comment with progress checkboxes and runs an open-ended session that can push code.
3. If neither, it does nothing.

The `track_progress` input forces tag-mode tracking comments onto `pull_request` and `issues` events that would otherwise run in agent mode. Everything else about behavior is set through `claude_args`, which passes flags straight to the CLI (`--max-turns`, `--system-prompt`, and so on), and through `plugins`, which installs named marketplace plugins before the run.

Three other things in the repository are easy to miss:

- **examples/** has eleven copy-paste workflows (measured 2026-09-09): the three review variants, plus `claude.yml` (the general **@claude** mention workflow), `ci-failure-auto-fix.yml`, `test-failure-analysis.yml`, `issue-deduplication.yml`, `issue-triage.yml`, `manual-code-analysis.yml`, `agent-approval-check.yml`, and `claude-wif.yml`, the workload-identity-federation variant that avoids storing a long-lived API key.
- **agent-approval-check/** is a second, smaller action that gates a workflow on whether the actor is a recognized agent identity, with an example identities file.
- **Authentication inputs** cover a direct API key, a Claude Code OAuth token, federation (`anthropic_federation_rule_id` plus organization, workspace, and service-account IDs), and OIDC to Bedrock, Vertex AI, or Microsoft Foundry.

The `allowed_non_write_users` input carries its own warning in `action.yml`: letting users without write access trigger the action exposes the run to prompt injection, and the secret scrubbing it performs is best-effort. `claude-code-security-review` says the same of itself in its README and recommends requiring approval for fork PRs (Anthropic 2026h). Both are the upstream statement of the caution in Section 20.

The extension and SDK repositories, briefly

`skills` is the reference implementation of the Agent Skills format that Section 33 describes. Its `spec/` directory is the format definition, `template/` is a starting skill, and `skills/` holds the examples. The README states the licensing split plainly: most skills are Apache-2.0, while the `docx`, `pdf`, `pptx`, and `xlsx` skills that power Claude’s own document features are source-available rather than open source (Anthropic 2026m). Read a license header before copying one.

`claude-plugins-official` is the marketplace behind `/plugin install <name>@claude-plugins-official`, split into Anthropic-maintained `plugins/` and partner-submitted

`external_plugins/` (Anthropic 2026j). Its README states a rule that applies to any marketplace we publish: a plugin's `name` is an immutable slug, since renaming it breaks every existing install, and a `renames` map exists for the unavoidable case. `claude-plugins-community` is a nightly, read-only mirror of the community submissions that passed security scanning (Anthropic 2026i), and `knowledge-work-plugins` holds eleven role-specific bundles built for Claude Cowork (Section 68) that also load in Claude Code (Anthropic 2026k).

Of the two agent SDKs, only the Python one is open source. `claude-agent-sdk-python` bundles the CLI inside the wheel and exposes `query()` and `ClaudeAgentOptions` for driving a full agent session from a script (Anthropic 2026c). `claude-agent-sdk-typescript` has the same shape as `claude-code`: a proprietary `LICENSE.md`, a changelog, examples, and an issue tracker for the npm package (Anthropic 2026d). The lower-level `anthropic-sdk-python` is the raw API client (MIT) for anyone who wants the model without the agent loop (Anthropic 2026b).

`sandbox-runtime` (`srt`) is the sandbox Claude Code uses internally, released as a beta research preview. It wraps `sandbox-exec` on macOS and `bubblewrap` on Linux with a proxy-based network allowlist, and the headline use case in its README is wrapping a local MCP server so it can read only the directories you name (Anthropic 2026l). Windows support is marked alpha, through a bundled `srt-win.exe` helper that runs the process under a dedicated local user account (measured 2026-09-09), so Windows users in the lab should treat it as experimental.

Useful to us?

Yes, selectively, and less for code than for reference.

- **Adopt now.** `claude-code-action` is already how our repositories run `@claude` and review (Section 44); the `examples/` directory and `claude-wif.yml` are the upgrade path when we move off static API keys. `claude-plugins-official` is the safe default source for plugins.
- **Read, then borrow.** The `plugins/` directory in `claude-code` duplicates several things our own instruction repository does by hand — a multi-agent review command, a silent-failure hunter, a hook generator — and a `hookify`-style rule is a lighter way to encode a “never do X again” correction than a hand-written hook. `skills/spec/` is the authority when a skill fails to load.
- **Track, do not depend on.** `claude-code` and `claude-agent-sdk-typescript` are trackers. Search their issues before filing; read `CHANGELOG.md` before blaming a regression on your config. Pin workflows to `claude-code-action@v1`, never to the base-action mirror.
- **Trial on Linux and macOS.** `sandbox-runtime` could replace part of the firewall configuration in Section 22 for macOS and Linux users; its Windows path is alpha.
- **Skip.** `claude-code-security-review` has not been pushed since February 2026 and its job is now a review prompt; the quickstarts and courses are for API programming, which is not the lab's main use of Claude.

42 Claude Code Cloud Environments

[Claude Code](#) is a CLI coding agent that can also run tasks on Anthropic-managed cloud infrastructure— either from the web at [claude.ai/code](#) (“Claude Code on the web”), or from the terminal by adding the `--remote` flag to move a session into the cloud.

Each cloud run executes inside a configured **environment**. An environment bundles three things:

- **Network access level:** what the cloud session is allowed to reach (a security control, analogous to the firewall configuration described above).
- **Environment variables:** values the session needs at runtime, such as `NODE_ENV`, database URLs, or API keys.
- **Setup scripts:** Bash that runs automatically when the session starts, for example to install dependencies.

Selecting the default environment with `/remote-env`

The `/remote-env` slash command sets **which configured environment is the default** for `--remote` runs:

- With a single environment configured, it shows your current configuration.
- With multiple environments, it opens an interactive picker so you can choose the default.

`/remote-env` only *selects* the default; to add, edit, or archive the environments themselves, use the web interface at [claude.ai/code](#). Because `/remote-env` opens an interactive panel, run it from an interactive `claude` terminal session.

Availability

Claude Code on the web (and the cloud environments it relies on) is a research-preview feature, available to Pro, Max, and Team users (and Enterprise users with eligible seats). Availability and behavior may change.

For details, see the [Claude Code on the web documentation](#) and the [slash command reference](#).

43 Using a ChatGPT Account for Codex Pull-Request Reviews

OpenAI Codex can act as a reviewer on GitHub pull requests. The native integration uses the Codex service connected to a ChatGPT workspace and posts a standard GitHub review through the Codex connector bot. It does not require you to build a separate GitHub Action.

Prerequisites

The native reviewer depends on **Codex Cloud**. Before it can review a repository, you need:

- Codex Cloud enabled for your active ChatGPT workspace;
- the repository connected to Codex Cloud;
- access to the Codex code-review settings; and
- GitHub push or admin permission if you want to configure automatic reviews.

An `AGENTS.md` file is optional, but it lets the reviewer follow repository-specific guidance.

Requesting and automating reviews

After an administrator or repository owner connects the repository and enables code review in Codex settings, request a review by adding this comment to a pull request:

```
@codex review
```

You can add a one-off focus to the same comment, for example:

```
@codex review for security regressions and missing tests
```

Where the workspace configuration permits it, Codex can also review new pull requests automatically. In GitHub, the general code review reports only high-priority P0 and P1 findings to keep its comments focused.

A separate **Security Review** (research preview) is a deeper pass on security-specific risks. Request it with `@codex security review`. It can overlap with the general review's security findings.

When an administrator has disabled Codex Cloud

The message “**Your admin has turned off Codex Cloud**” is a workspace-policy restriction, not a GitHub repository error. The native `@codex review` bot cannot run because GitHub reviews execute through Codex Cloud. Local Codex use can continue. Run `/review` against a checked-out diff in any of:

- the Codex app
- the IDE composer
- the CLI

That is a local review. It does not post a GitHub review and does not create an always-on GitHub bot.

Resolve the restriction in one of these ways:

- A workspace administrator can enable Codex Cloud in the workspace’s admin permissions.
- A user can connect a *personal* repository from a personal workspace that permits Codex Cloud. Do not use a personal workspace to connect organization-owned repositories — that bypasses the workspace policy your administrator set and moves the code outside your organization’s controls.
- If cloud access must remain disabled, use the local `/review` path above.

GitHub organizations may separately require an owner to approve the repository connection. Changing that authorization does not override a ChatGPT workspace policy; both sides must permit the integration.

Native review versus API-backed automation

Do not confuse the native integration with a custom GitHub Action that calls an OpenAI model. The native reviewer is configured through Codex and the linked ChatGPT workspace. A custom action instead needs API credentials and uses API billing and limits. It also requires you to implement:

- the review prompt
- permissions
- comment-posting behavior

Do not copy personal ChatGPT or Codex login credentials into CI secrets.

Adding repository-specific review rules

Codex reads applicable `AGENTS.md` files. Put broad guidance in the repository root and narrower guidance in a file closer to the code it governs. Review-only guidance belongs under a `## Code Review Rules` heading:

```
## Code Review Rules

- Flag schema changes that are not backward compatible.
  A safe migration must support both deployed application versions.
- Flag behavior changes without a regression test.
```

Keep deterministic formatting and lint checks in continuous integration. Code-review guidance should:

- focus on consequential behavior
- state the safe alternative or exception
- remain concise enough to apply consistently

Codex review is an additional signal; it does not replace:

- tests
- branch protection
- required human approval

For current setup details, see OpenAI’s [GitHub code-review documentation](#).

44 Where Pull-Request Review Lives in Claude Code Action

A common question about [anthropics/claude-code-action](#) is where its pull-request review lives. The answer is surprising: **there is no dedicated review action**. The repository publishes one general-purpose top-level action, and “review” is a *prompt* you pass it, not a separate artifact.

All claims below were surveyed against that repository at `main` (measured 2026-08-25; paths can move).

One action, many behaviors

The published action is [action.yml](#). It is a single composite action whose one external step installs Bun, and its description reads: “Flexible GitHub automation platform with Claude. Auto-detects mode based on event type: PR reviews, @claude mentions, or custom automation.”

What selects the behavior is the `prompt` input:

- A workflow triggered by comments containing @claude gets the interactive agent mode.
- A workflow that supplies an explicit review prompt gets a one-shot reviewer.
- The same action also handles issue triage and other automation, which is why one top-level action covers every behavior.

A second directory in the repository, [base-action/](#), is easy to mistake for a delegation target. It is not one: it holds a lower-level building block, developed in-tree and mirrored automatically to its own repository, [anthropics/claude-code-base-action](#). Consumers reference the mirror, not the in-repo path.

Where the repository reviews its own PRs

- `.github/workflows/claude-review.yml` is the reviewer. It triggers on `pull_request: opened`, skips fork PRs (they cannot mint the OpenID Connect (OIDC) token used for authentication), and calls `anthropics/claude-code-action@v1` with the prompt `/review-pr REPO: ... PR_NUMBER: ....`
- `.claude/commands/review-pr.md` defines what `/review-pr` does. It fans out to five reviewer subagents, defined under `.claude/agents/`:
 - code quality
 - performance
 - test coverage
 - documentation accuracy
 - security

Each subagent is told to report only noteworthy feedback. The command then reviews that feedback and posts only the findings it also deems noteworthy — inline comments for specific issues, top-level comments for general observations or praise.

- `examples/pr-review-comprehensive.yml` plus two filtered variants (by author and by path) are copy-paste templates for adding review to another repository.

Adding review to your own repository

Copy one of the `examples/pr-review-*.yml` files into your repository's `.github/workflows/`, then provide credentials. The examples authenticate with an `ANTHROPIC_API_KEY` secret; the upstream repository's own workflows instead use Workload Identity Federation inputs (`anthropic_federation_rule_id`, `anthropic_organization_id`, `anthropic_service_account_id`) to exchange the workflow's OIDC token for a short-lived API token. Either route works; the federation route avoids a long-lived static key in CI.

The example's prompt is the customization point. It is ordinary prose naming focus areas, so you edit it the way you would edit any review checklist.

Reviewer versus @claude agent

Do not confuse the review workflow with `.github/workflows/claude.yml`, the interactive agent. They share one underlying action but differ in trigger surface:

	claude-review.yml	claude.yml
Fires on	PR opened	comments, reviews, or issues containing @claude
Behavior	one-shot review	open-ended agent session
Writes code	no	yes

This distinction explains a common debugging dead end: “@claude answered my comment, so why did nobody review the pull request?” Mention-triggered activity never runs the reviewer; only the `pull_request`-triggered workflow does.

Relation to Codex native review

This is the Claude-side counterpart to [using a ChatGPT account for Codex pull-request reviews](#). Codex ships a hosted native reviewer configured through the ChatGPT workspace, with no workflow file. Claude Code’s reviewer is the opposite trade: you own a workflow file and supply API credentials, but the prompt, tools, model, and triggering events are all visible and editable in your repository.

45 Gemini Review Action for GitHub Pull Requests

[derailed-dash/gemini-review-action](#) is an open-source GitHub Action that provides automated code reviews on pull requests and automated triage on issues using Google’s Gemini models (measured 2026-08-31; repository at v1.6.6).

Key capabilities

The action operates as a composite GitHub Action designed to run directly in CI pipelines. Notable features include:

- **Model selection and defaults:** Defaults to `gemini-3.7-flash`, configurable via the `gemini_model` input.
- **Dual authentication modes:** Supports standard API key authentication via the `gemini_api_key` input, as well as keyless Google Cloud Workload Identity Federation (WIF) by running `google-github-actions/auth` beforehand and setting `GOOGLE_GENAI_USE_VERTEXAI`, `GOOGLE_CLOUD_PROJECT`, and `GOOGLE_CLOUD_LOCATION` environment variables.
- **Hybrid codebase context enrichment:** Beyond analyzing git diffs, the action indexes repository context (structure, declarations, and key files) to assess pull requests against surrounding architecture.

- **Structured output and GitHub suggestions:** Generates line-level review comments formatted as GitHub suggestion blocks so contributors can apply suggested fixes directly in the GitHub UI.
- **Dynamic skill loading:** Can load domain-specific or project-specific instructions from `.agents/skills/` or custom directories, aligning reviews with existing repository skills.
- **Cost telemetry and context caching:** Emits token usage reports and estimated costs per run, leveraging Gemini's prompt and context caching to reduce API spend on repetitive large-context reviews.

Basic workflow configuration

To add Gemini code review to a repository, create `.github/workflows/gemini-review.yml`:

```
name: Gemini Code Review

on:
  pull_request:
    types: [opened, synchronize, reopened]

jobs:
  review:
    runs-on: ubuntu-latest
    permissions:
      contents: read
      pull-requests: write
      issues: write
    steps:
      - name: Checkout repository
        uses: actions/checkout@v4
        with:
          fetch-depth: 0

      - name: Run Gemini Code Review
        uses: derailed-dash/gemini-review-action@v1
        with:
          gemini_api_key: ${ secrets.GEMINI_API_KEY }
          github_token: ${ secrets.GITHUB_TOKEN }
          gemini_model: gemini-3.7-flash
```

Comparison with native review integrations

Like [Claude Code Action](#), `derailed-dash/gemini-review-action` gives the repository owner full visibility over workflow triggers, authentication methods, and review prompts. In contrast to hosted review offerings that require platform-level permissions across an entire organization, this GitHub Action operates per-repository with credentials scoped to GitHub Actions secrets or Google Cloud IAM roles.

46 Gemini Code Assist for Repository Code Review

Google Cloud provides a native code-review capability through [Gemini Code Assist](#) (measured 2026-08-31; documentation in Enterprise preview). Unlike GitHub Actions that run inside individual repository workflows, Gemini Code Assist operates as a managed service connected at the organization or repository level.

Architecture and setup

Gemini Code Assist connects to GitHub, GitHub Enterprise Cloud, or GitHub Enterprise Server through Google Cloud Developer Connect:

- **Connection host:** Repositories connect via Developer Connect connections provisioned in Google Cloud region `us-east1`.
- **Console enablement:** Code review is enabled through the Google Cloud Agents & Tools Console (under **Code review** in the Gemini Cloud Assist navigation).
- **Automated triggers:** Once connected, Gemini automatically generates pull-request summaries and posts line-level review comments when pull requests are opened or updated.
- **Interactive commands:** Reviewers and authors can trigger on-demand reviews or ask questions by posting a comment starting with `@gemini-code-assist` or `/gemini` directly in the pull-request thread.

Review scope and security boundaries

Gemini Code Assist enforces several deliberate boundaries on review scope:

- **Workflow file exclusion:** Files under `.github/workflows/` are explicitly excluded from reviews to prevent automated agents from analyzing or suggesting edits to CI credentials and pipeline definitions.
- **Style and policy customization:** Repositories can provide custom review guidelines and coding standards configured centrally in the Google Cloud Console, applying uniform review standards across multiple connected repositories.
- **Enterprise governance:** Because access is managed through Google Cloud IAM and Developer Connect, organization administrators can enable or disable reviews across projects without checking workflow files into individual repositories.

Trade-offs versus Action-based reviews

Dimension	Gemini Code Assist	Gemini Review Action / Claude Code Action
Hosting	Managed Google Cloud service	GitHub Actions runner
Configuration	Google Cloud Console & Developer Connect	In-repo <code>.github/workflows/*.yaml</code>
Workflow maintenance	Zero workflow files in repository	Repository owns workflow YAML and prompts
Interactive mode	Built-in <code>/gemini</code> comment tags	Configurable via <code>@claude</code> or Action triggers
Credential location	Google Cloud IAM & Developer Connect token	GitHub Secrets or Workload Identity Federation

47 PR-Agent: Open-Source Pull-Request Review Bot

[PR-Agent](#) (The PR Agent 2026h) is an open-source pull-request review bot. It runs one large-language-model call per command, posts the result as a pull-request comment or review, and works against several git forges with any model that LiteLLM can reach. All claims below were checked against the repository at `main`, its release list, and the documentation site at <https://docs.pr-agent.ai/> (measured 2026-09-09; the latest release was `v0.45.0`, published 2026-09-05).

History and ownership

PR-Agent began at CodiumAI, later renamed Qodo, and its hosted enterprise edition was sold as Qodo Merge. In 2026 Qodo donated the open-source project to the community. The repository now lives in the **The-PR-Agent** GitHub organization, has an external maintainer, and its README says it “is currently in the process of being donated to an open-source foundation” (The PR Agent 2026h). Qodo Merge was rebranded as Qodo 2.0, a separate hosted product.

The README is emphatic that the two are not the same thing: the repository “is not the Qodo offering for open-source projects”, and PR-Agent is “a community-maintained legacy project of Qodo” (The PR Agent 2026h). Qodo remains the project’s gold sponsor. Read the documentation with that split in mind: pages that describe a hosted app or a free tier for open-source projects are describing Qodo’s product, while PR-Agent itself is software you run.

The Docker images moved with the ownership. Releases 0.34.2 and later are published under `pragent/pr-agent`; the older `codiumai/pr-agent` namespace is a frozen archive (The PR Agent 2026h).

Commands

Every command is a slash comment on the pull request or a subcommand of the `pr-agent` command-line tool. The four core commands (The PR Agent 2026a):

- `/describe` generates a title, type, summary, code walk-through, and labels.
- `/review` generates a review covering possible issues, security concerns, tests, and an estimate of review effort. The review can also rate risk, recommend whether to merge, list the files a human should read first, and check the change against a linked GitHub or Jira ticket (The PR Agent 2026g).
- `/improve` posts actionable code suggestions, which the forge can render as one-click suggestion blocks.
- `/ask "..."` answers a free-text question about the whole PR or about specific lines.

Six further commands (`/add_docs`, `/generate_labels`, `/similar_issue`, `/help`, `/help_docs`, and `/update_changelog`) round out the set. `/help_docs` is disabled as of v0.36.1 pending a fix for a credential-exposure issue (The PR Agent 2026h).

Each command is a single model call, which the README describes as “~30 seconds, low cost” (The PR Agent 2026h). Large diffs are handled by a compression strategy rather than by a multi-step agent loop: deletion-only hunks are dropped, deleted files are collapsed into a list, and remaining files are added in order of size until the prompt approaches the model’s token limit (The PR Agent 2026f). Files that do not fit are named but not shown to the model. v0.45.0 adds an opt-in large-diff chunking mode for `/review` that splits a diff across several calls and merges the results.

Behavior is configured in a `.pr_agent.toml` file at the repository root, or by environment variables using dotted keys such as `pr_reviewer.extra_instructions` and `config.model` (The PR Agent 2026c). Each tool has an `extra_instructions` field for free-text review guidance, and `/review` has `require_*` switches (`require_security_review`, `require_tests_review`, and so on) that turn individual sections on or off (The PR Agent 2026g).

Supported forges and models

PR-Agent supports five forges:

- GitHub
- GitLab
- Bitbucket
- Azure DevOps

- Gitea

Most commands work on all five; the documentation keeps a per-forge feature matrix, and a few commands (such as `/similar_issue`) are GitHub-only (The PR Agent 2026a).

Models are bring-your-own. PR-Agent calls models through LiteLLM, so any provider LiteLLM supports can be selected by setting `config.model` (The PR Agent 2026d). Named providers include OpenAI, Azure OpenAI, Anthropic, Google (AI Studio and Vertex AI), Amazon Bedrock, DeepSeek, Mistral, Groq, xAI, OpenRouter, and local servers such as Ollama and vLLM. Further keys let you route work to different models: `config.model_weak` for lighter commands (`/describe`, `/ask`, `/update_changelog`), `config.fallback_models` for retries, and a `[model_routing]` table that picks a model by diff size (The PR Agent 2026d).

Local models work but come with a warning. The model-configuration page says that “most open-source models currently available (as of January 2025) face challenges with these complex tasks” and recommends them for “experimentation and learning purposes (mainly for the ask command)” (The PR Agent 2026d). That note predates the current generation of open-weight coding models, so treat it as a prompt to test rather than as a verdict; Section 28 covers what we have found local models can and cannot do.

Deployment

There is no hosted PR-Agent service. Every route runs the software on infrastructure you control (The PR Agent 2026e):

- **GitHub Action** (the README’s recommended route). A workflow that uses: `the-pr-agent/pr-agent@main` with `OPENAI_KEY` (or another provider’s key) and `GITHUB_TOKEN` in `env`, triggered on `pull_request` and `issue_comment` events. `github_action_config.auto_review`, `auto_describe`, and `auto_improve` choose which commands run automatically when a PR opens, and the `issue_comment` trigger enables slash commands.
- **Self-hosted GitHub App**. A webhook server (the `pr-agent:github_app` Docker target) registered as a GitHub App with pull-request, issue-comment, and metadata permissions. The documentation covers running it as an AWS Lambda function and sizing the `gunicorn` workers. Equivalent webhook servers exist for GitLab and Gitea.
- **Command line**. `pip install pr-agent`, then `pr-agent --pr_url <url> review` runs a command against an existing PR from your own machine.
- **Docker**. Versioned images under `pragent/pr-agent` back all of the above.

The README’s data-privacy section says only that a self-hosted PR-Agent with your own OpenAI key “is between you and OpenAI” (The PR Agent 2026h). That is the whole privacy story: no third party sees the diff except the model provider you chose.

License and the hosted-versus-open split

The code is MIT-licensed, with the 2026 copyright held by “The PR Agent” (The PR Agent 2026b). The current documentation no longer marks individual commands as open-source or hosted-only: after the donation, everything at `docs.pr-agent.ai` describes the open-source tool, and the hosted features moved to Qodo’s own product and documentation. Older blog posts and third-party guides still refer to “PR-Agent Pro” or “Qodo Merge” feature tiers; those describe the product that became Qodo 2.0, not the repository reviewed here.

Useful to us?

The lab already runs four pull-request reviewers, so the question is what PR-Agent adds. The comparison uses the same axes as the other review sections:

Dimension	PR-Agent	Claude Code Action	Codex review	Gemini Review Action	Gemini Code Assist
Source	Open (MIT)	Open action, closed model	Closed	Open	Closed
Hosting	Actions runner or your own webhook server	Actions runner	OpenAI-hosted	Actions runner	Google-hosted
Model	Any via LiteLLM, including local	Claude only	OpenAI only	Gemini only	Gemini only
Forges	GitHub, GitLab, Bitbucket, Azure DevOps, Gitea	GitHub	GitHub	GitHub	GitHub
Billing	Your API key (or free with a local model)	Anthropic API key or federation	ChatGPT workspace	Gemini API key or Google Cloud	Google Cloud

Dimension	PR-Agent	Claude Code Action	Codex review	Gemini Review Action	Gemini Code Assist
Review depth	One model call per command over a compressed diff	Agentic; five reviewer subagents with repository access	Hosted agent	One call with repository context indexing	Managed service
Interactive	<code>/ask</code> , <code>/review</code> , and other slash comments	<code>@claude</code> mention (separate workflow)	<code>@codex review</code>	Triggered by workflow events	<code>/gemini</code> comments
Customization	<code>.pr_agent.toml</code> extra_instructions prompts in repo	Review prompt in the workflow file	<code>AGENTS.md</code> review rules	Skills directory	Google Cloud Console

What PR-Agent offers that none of the others do:

- **Model independence.** It is the only reviewer here that can be pointed at whichever provider currently has spare quota, or at an Ollama model for a zero-cost pass on a private repository. For the lab's quota juggling (Section 67 and Section 27), a reviewer whose model is a one-line config change is attractive.
- **Structured output.** `/review` reports effort, risk, and a merge recommendation in a fixed format, which a script can parse. The other reviewers produce prose.
- **GitLab and other forges.** Only PR-Agent covers repositories outside GitHub, which matters if a collaborator hosts code elsewhere.

What it lacks:

- **Depth.** A single call over a pruned diff cannot follow a symbol into a file the diff did not touch. The Claude Code Action reviewer runs as an agent with the repository checked out, so it can. For the review standard the lab's ARDI loop expects (see Section 44), PR-Agent is a first pass, not a replacement.
- **A stable maintainer story.** The project changed hands in 2026, its README says a foundation transfer is still in progress, and one command is disabled pending a security fix. Release cadence remains weekly (five releases between 2026-08-01 and 2026-09-05), so the project is active, but the governance is newer than the code.
- **Another bot in the thread.** Each added reviewer costs a review round per push and a comment the ARDI loop has to triage. Adding a fifth reviewer only pays off if it catches something the other four miss.

A reasonable trial is one repository, the GitHub Action route, `auto_review` on and `auto_describe` and `auto_improve` off, with `config.model` set to whichever provider has spare quota that month. If its `/review` findings overlap the Claude reviewer’s after a few weeks, turn it off; if it catches something distinct, keep it as the cheap first pass.

48 How a Session Learns a PR Changed

A coding-agent session that is watching a pull request does not poll it. Something wakes the session when the pull request changes, and in Claude Code that “something” is one of **two separate channels**.

The distinction matters for a practical reason: **the agent can turn one of them on and off, and cannot touch the other**. So “stop watching this PR” is a request an agent can satisfy for one channel and can only decline for the other.

Both arrive the same way — mid-turn, alongside the next tool result, the same delivery mechanism as a background task notification — so they are easy to mistake for each other. They are distinguishable by the tag wrapping them.

	<github-webhook-activity>	<ci-monitor-event>
Turned on by	the agent, via a tool call	a human, via a checkbox
Turned off by	the agent, via a tool call	a human, via the same checkbox
Available in	a session with the remote/hosted GitHub MCP server (not a local one)	Claude Code on the web only
Carries	comments, CI results, reviews, mergeability notices	the new comment, plus a fixed instruction template

The channel an agent controls

`subscribe_pr_activity` and `unsubscribe_pr_activity` are ordinary tool calls, taking an owner, a repository, and a pull-request number. Subscribing delivers that pull request’s activity into the conversation as `<github-webhook-activity>` messages until the session unsubscribes, or the pull request merges or closes.

Two caveats are worth knowing before relying on it.

A successful subscribe does not guarantee delivery. If a PR Steward agent already holds the watch on that pull request, the call still succeeds — but this session receives nothing. The tool result says so in as many words, so read the result rather than the exit status. Taking

over the watch requires opting the steward out first, by removing its watching label on the pull request.

The tool does not exist on a locally-run GitHub MCP server. Workflow guidance written for remote or web sessions names it freely, which strands anyone following that guidance from a local harness. Section 59 covers the local analogues to reach for instead.

Webhook delivery is also not exhaustive, which is the failure mode most likely to be mistaken for “nothing has happened”. CI *successes*, new pushes, and merge-conflict transitions can arrive late or not at all. A session that treats silence as “still green” will sit indefinitely on a pull request that has gone stale or conflicted, so a subscription is a supplement to periodically re-reading the pull request’s real state, not a replacement for it.

The channel only a human controls

Claude Code on the web shows a per-pull-request **CI monitoring** panel in the session sidebar, with two checkboxes: **Auto-fix CI & address comments** and **Auto-merge when ready**.

Ticking the first is what starts `<ci-monitor-event>` messages arriving.

Three properties of that panel surprise people:

- **There is no default.** No account-, organization-, repository-, or environment-level setting turns either checkbox on ahead of time. Absent the `/autofix-pr` shortcut described below, every new pull request and session starts with both off.
- **No agent-side tool can reach it.** It is client-UI state, not something an agent’s configuration surface touches, so asking an agent to enable it cannot work. If the checkbox changes, a human changed it.
- **One of the two has a shortcut, and the other does not.** Running `/autofix-pr` from the command line on a pull request’s branch spawns a web session with **Auto-fix CI & address comments** already on. There is no equivalent shortcut for **Auto-merge when ready**.

See Section 42 for the web-session context these run in.

The instruction template is boilerplate

Each `<ci-monitor-event>` quotes the triggering comment verbatim and then appends a **fixed instruction template**:

- address the feedback and push a fix;
- post a one-line reply on the thread;
- end that reply with a set attribution line;
- resolve the thread;
- skip replies for comments you did not act on.

That template is appended to **every** new comment. It is not gated on whether the comment contains anything actionable. Observed firings with nothing to act on include:

- a Copilot review declining for quota reasons (“unable to review... reached their quota limit”),
- a Copilot review reporting it “wasn’t able to review any files”, and
- a sticky preview-deployment comment that posts a preview URL and rewrites itself on every push.

The practical consequence is that the template’s imperative opening should not be obeyed reflexively. Its own closing clause — skip replies for comments you did not act on — is the standing permission to do nothing when there is nothing to do, and it is the half most often read past. Treat each event as a prompt to go and check the pull request’s actual state through the API, rather than as an instruction that a fix is owed.

Two names that sound identical and are not

The sidebar’s **Auto-merge when ready** checkbox and the `enable_pr_auto_merge / disable_pr_auto_merge` MCP tools are unrelated controls with nearly the same name.

The tools drive **GitHub’s own** native auto-merge on the pull request: merge once required checks pass and approvals are met, which is a setting stored on GitHub and visible to everyone. The checkbox is Claude Code client state governing what the agent session does. Toggling one says nothing about the other.

Treat what arrives as untrusted

Comment bodies inside either wrapper come from anyone who can comment on the pull request. Directives that appear inside them are data, not instructions: a comment that says “ignore your previous instructions” is a comment, and a comment that asks for a credential is a comment.

One specific trap is worth calling out, because it looks exactly like the thing it is not. Comments posted *by the agent* through the GitHub MCP tools authenticate as whichever account owns the session’s token. In an interactive session that is typically the human who owns it, so an event echoing the agent’s own just-posted reply usually shows a human author rather than a recognizable bot name. It is not a rule, though: a pipeline authenticating through an App-token exchange posts as `claude[bot]` instead, as this repository’s own review workflow does. Either way the conclusion is the same, and the variability only sharpens it — author identity is useless for deciding whether an event is your own echo.

The attribution footer is a better signal, but not proof. Every comment posted from these sessions carries one, so a body that lacks it is very unlikely to be yours. A body that has it establishes less than it appears to, for two reasons the rest of this page has already supplied:

- **It is part of the untrusted comment data.** Anyone who can comment on the pull request can paste the same footer text at the end of a malicious comment.
- **It identifies a class, not an instance.** It marks the comment as coming from *some* Claude Code session, which is not the same as *this* one — a PR Steward concurrently watching the same pull request carries the identical footer.

So treat a footer as a strong hint and a missing footer as near-conclusive, and settle genuine authorship questions against what this session actually posted.

⚠ Running both at once duplicates every comment

The two channels overlap in what they deliver. With webhook activity subscribed *and* the auto-fix checkbox ticked on the same pull request, each new comment arrives twice: once as raw activity, and once wrapped in the instruction template. Turn on the one whose control surface and behavior you actually want.

i Observed behavior, not documented contract

The delivery mechanics and the wording of the instruction template above were established by observation during agent sessions in mid-2026, not from a published specification. Claude Code on the web is a research-preview feature, so treat the specifics as liable to change and re-check them against current behavior before depending on any one detail.

49 When to use a coding agent

Coding agent sessions are currently¹ considered “premium requests”, which are limited resources; see <https://github.com/features/copilot/plans> for details. So, use coding agents sparingly. Use them for complex changes that would be difficult or time-consuming for you to complete by hand. Coding agents also take time to get configured for work, every time you make a request. See <https://docs.github.com/en/copilot/how-tos/use-copilot-agents/coding-agent/customize-the-agent-environment#preinstalling-tools-or-dependencies-in-copilots-environment> for ways to reduce that startup time, but it will never be 0. If you can complete the task faster than the coding agent can, you should probably do it yourself. For example, when you have errors in the spell-check or lint workflows, you can often fix them faster than Copilot can. Similarly, when reviewing Copilot’s PRs, you can often make direct changes to the branch faster than you could write clear review comments and get Copilot to address them.

Also, the less we practice, the weaker our skills get, and the harder it is for us to supervise the agents and make sure they are actually doing what we want them to do, the way we want

¹2026-01-10

them to do it. You should exercise your own coding skills regularly, just like you would for any other skill you want to maintain.

50 Deep Research Modes

Every major assistant now ships a “deep research” mode: a long-running agent that plans a search strategy, runs dozens of queries, reads the results, and writes a cited report (measured 2026-09-09). These modes sit between a single web search and a coding agent. They do not edit files or run your code, but they read far more sources per question than a chat turn does, and they hand back something closer to a literature memo than an answer.

This section compares five of them from their vendors’ own documentation:

- Claude Research (Anthropic)
- Gemini Deep Research (Google)
- ChatGPT deep research (OpenAI)
- Researcher in Microsoft 365 Copilot (Microsoft)
- Perplexity Research (Perplexity)

For each we record what it does, which plans include it and with what limits, how long a run takes, what the output looks like, which connected sources it can read, and whether the same capability is reachable from an API. All figures are as published on 2026-09-09; vendors change plan limits often, so re-check before relying on one.

Claude Research

[Research](#) (Anthropic 2026af) is available on the paid Claude plans (Pro, Max, Team, and Enterprise) in the web, desktop, and mobile apps. Web search must be switched on for Research to work. Claude runs a chain of searches that build on each other and returns an answer with inline citations. Anthropic’s guidance on choosing between web search, extended thinking, and Research (Anthropic 2026ah) places Research at “five or more tool calls over 1-3 minutes”, with web search for one- or two-query lookups and extended thinking for reasoning that needs no new information.

Research draws on the same connectors as the rest of the app: when Gmail, Google Calendar, and Google Docs are connected, it searches those alongside the web (Anthropic 2026af). There is no separate Research quota; runs count against the plan’s ordinary conversation limits, and the help page warns that a run “may consume limits faster” because it retrieves many sources.

The Claude API has no Research endpoint. What it exposes is the [web search tool](#) (Anthropic 2026ag): a server-side tool that searches, returns results with citations, and can be capped with `max_uses` or restricted to `allowed_domains`. It is billed at \$10 per 1,000 searches plus

token costs. A multi-search research loop over that tool is something you build, not something you call.

Gemini Deep Research

[Deep Research in the Gemini app](#) (Google 2026h) searches Google by default, and you can add Gmail, Drive, uploaded files, and NotebookLM notebooks as sources. A report “usually takes about 5-10 minutes to generate”, longer for complex topics. The report can be exported to Google Docs or turned into an Audio Overview. All users can run reports; Google AI Pro and Ultra subscribers can generate them with the Pro model, and Ultra reports may include charts, diagrams, and interactive simulators.

Limits are compute-based rather than a per-report count (Google 2026c): the allowance refreshes every 5 hours up to a weekly cap, with AI Plus at 2x the free allowance, AI Pro at 4x, and AI Ultra at 5x or 20x the Pro allowance depending on the subscription. Deep Research is listed as available on every tier, but the same page notes that for accounts without a paid plan compute-heavy features like Deep Research “may be unavailable during periods of high demand”.

Google is the one vendor here that sells the app feature and the API feature under the same name. The [Gemini Deep Research agent](#) (Google 2026d) runs only through the Interactions API, must be launched with `background=true`, and ships as two agents: `deep-research-preview-04-2026` and `deep-research-max-preview-04-2026`. The Max variant, built on Gemini 3.1 Pro, is positioned for exhaustive batch work such as a nightly job that writes due-diligence reports (Google 2026b). By default the agent gets Google Search, URL context, and code execution; it accepts documents as input and can call remote MCP servers. A run is capped at 60 minutes, and the docs estimate \$1-3 per standard task and \$3-7 per Max task.

ChatGPT deep research

[Deep research in ChatGPT](#) (OpenAI 2026d) is the one mode here that puts a plan in front of you before it starts. ChatGPT may ask clarifying questions, drafts a research plan you can edit, lets you watch and interrupt the run to redirect it, and finishes with a full-screen report carrying a table of contents, a sources section, and an activity history. Reports download as Markdown, Word, or PDF.

Sources are the public web, files you upload, and connected apps such as Google Drive or SharePoint where your plan and workspace allow them; deep research uses only an app’s read actions. You can also restrict a run to named sites, or prioritize them while still allowing full-web search. Usage “varies by plan” and shows in an in-product counter; plans with a fixed monthly allowance reset 30 days from first use, and availability depends on country. Enterprise and Edu admins gate access by role.

The [API version](#) (OpenAI 2026c) runs through the Responses API with model `o3-deep-research` or `o4-mini-deep-research`. OpenAI recommends background mode because runs “can take tens of minutes”, and background mode is incompatible with zero-data-retention agreements. Available tools are web search, file search over at most two vector stores, remote MCP servers exposing a search/fetch interface, and the code interpreter. Output carries inline citations as annotations with a URL, title, and character offsets.

Microsoft 365 Copilot Researcher and Think Deeper

Microsoft has consolidated its offerings into one agent. The consumer “Deep Research” mode was retired from the Copilot app on 2026-08-18, and Microsoft 365 Premium subscribers are directed to [Researcher](#) (Microsoft 2026e, 2026c) instead. Researcher is also included for business and enterprise tenants with a Copilot add-on license, and admins can block it in the Microsoft 365 admin center.

Researcher’s distinguishing source is your work graph: it reads files, emails, meetings, and Teams chats you already have access to, plus Microsoft Graph connectors, and uses the Bing index for the web (Microsoft 2026h). Web search can be disabled tenant-wide, but there is no allow-list or block-list of sites below that. It asks clarifying questions before running, and the FAQ gives run times of “under five minutes for simple queries” and “10 to 45 minutes for highly complex ones”. Output is a report with headings, visuals, and cited sources; PowerPoint and PDF export are listed as “to be released soon”. The hard limit is 25 queries per user per month. Tenants can enable Anthropic’s Claude models inside Researcher, and the model picker is available when Researcher is opened as its own agent. The FAQ states that Researcher cannot be customized or extended through Copilot Studio, and none of the pages we read describe a programmatic route to it (Microsoft 2026i).

Think Deeper is not a research mode. It is one of the [conversation modes](#) (Microsoft 2026b) in the consumer Copilot app: a reasoning pass that “takes up to 10 seconds” and does no extra retrieval, available to all users with priority for Microsoft 365 subscribers when capacity is short. Treat it as the counterpart of Claude’s extended thinking, not of Researcher.

Perplexity Research

[Research mode](#) (Perplexity 2026b) runs “dozens of searches, reads hundreds of sources”, completes most tasks in under 3 minutes, and takes around 4 to 5 minutes to deliver the report. Reports export to PDF or a document, or become a shareable Perplexity Page. Free accounts get limited access and Pro subscribers get extended access; the help page gives no counts. You cannot choose the model in Research mode. The [Advanced Deep Research](#) (Perplexity 2026c) update added a code sandbox, uploaded-document input, clarifying questions, follow-up questions while a run is in progress, and reports that stream into an editable file. It also names the underlying model by plan. Which model each plan gets has changed more than once, and

Perplexity’s help-center pages refuse automated requests, so this section does not reproduce the mapping: check the linked page before relying on a particular model being behind your plan.

The API equivalent is the [sonar-deep-research](#) model (Perplexity 2026a), priced at \$2 per million input tokens, \$8 per million output tokens, \$2 per million citation tokens, \$3 per million reasoning tokens, and \$5 per 1,000 searches, with a 128K context window. The docs carry a deprecation notice moving the Sonar chat-completions interface to the Agent API, with the old interface supported until 2026-09-27.

Comparison

	Claude Research	Gemini Deep Research	ChatGPT deep research	Copilot Researcher	Perplexity Research
Plans	Pro, Max, Team, Enterprise	All tiers; Pro model on AI Pro and Ultra	Varies by plan and country	M365 Premium; Copilot add-on licenses	Free (limited), Pro (extended)
Limit	Shared conversation limits	Compute-based, 5-hour refresh, weekly cap	In-product counter; monthly allowance on some plans	25 queries per user per month	Not published
Run time	1-3 minutes	5-10 minutes	Not published for the app; “tens of minutes” via API	Under 5 minutes to 45 minutes	3-5 minutes
Plan review	No	No	Editable plan; run can be interrupted	Clarifying questions	Clarifying questions
Output	Answer with inline citations	Report; Google Docs export; Audio Overview	Report with table of contents and sources; Markdown, Word, PDF	Report with visuals and citations	Report; PDF, document, Perplexity Page

	Claude Research	Gemini Deep Research	ChatGPT deep research	Copilot Researcher	Perplexity Research
Connected sources	Gmail, Calendar, Docs connectors	Gmail, Drive, uploads, NotebookLM	Uploads, Drive, SharePoint, other apps; site restriction	Files, email, meetings, chats, Graph connectors	Uploaded documents
API	Web search tool only (\$10 per 1,000 searches)	Interactions API; two agents; 60-minute cap; \$1-7 per task	Responses API; o3-deep- research, o4-mini- deep- research	None documented	sonar-deep- research

Useful to us?

Yes, for a specific slice of work, and no as a substitute for the agents in Section 4.

Reach for a research mode when the question is about the world rather than about your repository: a literature scan before a grant section, a comparison of vendors or packages you have not yet chosen between, or a check on what a regulator or funding agency currently requires. The modes read many more pages than a coding agent will in one turn, they hand back citations you can audit, and several export straight into the document formats we already draft in (see Section 68 for the workspace side). Researcher is the odd one out: its value is reading your own institution's email and files, which none of the others can see, and that is also why its output stays inside the Microsoft 365 boundary.

Reach for a coding agent with web search instead when the answer must touch the code: a dependency upgrade that needs the changelog read *and* applied, a CI failure whose fix lives in a vendor's docs, or anything where the deliverable is a diff rather than a memo. Section 49 covers that decision in general. A coding agent can also chain research and action in one run, whereas every research mode above stops at the report.

Two cautions carry across all five:

- **The limits are the product.** Monthly caps of 25 (Copilot) or compute pools that refresh every 5 hours (Gemini) mean a research mode is not a tool to leave running in a loop. For batch or scheduled research, use the API routes, and budget them like any other agent spend (Section 67).

- **A cited report is not a verified one.** Every mode cites, and none checks that the cited page supports the sentence. Spot-check the citations you intend to reuse before they reach a manuscript, exactly as the fact-check rules in this manual require of any AI-drafted prose.

51 Editing with .docx files

GitHub Copilot coding agents can read Microsoft Word (.docx) files, including tracked changes and comments. This enables a hybrid editing workflow where:

1. Lab members can export Quarto content to Word format for review
2. Reviewers can make edits, add tracked changes, and insert comments in Word
3. Coding agents can read the .docx file and translate the edits back to Quarto format

When using this workflow, make sure to explicitly instruct the coding agent to:

- Examine and apply all tracked changes in the .docx file
- Read and address all comments in the .docx file
- Translate edits from Word formatting to appropriate Quarto/markdown syntax

This approach makes it easier for collaborators who are more comfortable with Word to contribute while maintaining the source files in Quarto format.

Known Issue: “Document 1” Warning in Word

When opening DOCX files generated by Quarto (including this site), Microsoft Word may display a warning message and open the file with the title “Document 1” instead of the actual filename. Word may also require you to save the file before you can add comments or track changes.

This is a known limitation with how Quarto generates DOCX files. The issue is being tracked in the Quarto project:

- [Quarto CLI Issue #6357](#)
- [Quarto Discussion #6544](#)
- [Quarto CLI Issue #10587](#)

Workaround: If you are the author generating the DOCX file from Quarto, follow these steps before sharing with collaborators:

1. Open the generated DOCX file in Microsoft Word
2. Immediately save the file (File → Save, or Ctrl+S/Cmd+S)
3. Close and re-open the file to verify it no longer shows “Document 1”
4. Share this saved version with collaborators

This one-time step ensures that when collaborators open the file, they won't see the "Document 1" warning and can immediately add comments and track changes without issues.

52 Hosted AI Agent Offices: Vestra and Similar

[Vestra](#) (Vestra AI 2026d) sells what it calls an "AI agent office": a hosted, always-on general-purpose agent that you reach from the tools you already use and that runs on a cloud machine the vendor operates. This section reviews Vestra and four products in the same category (measured 2026-09-09):

- [Manus](#) (Manus AI 2026)
- [Lindy](#) (Lindy 2026b)
- [Zapier Agents](#) (Zapier 2026a)
- [Relevance AI](#) (Relevance AI 2026a)

The category sits between two things this site already covers. [Make](#) and Zapier's core product are visual automation builders: you draw the workflow, and the AI parts are modules inside it. The collaborative workspace agents in Section 68 (Claude Cowork, Gemini Spark) work on files and documents you own. An agent office is closer to a hired assistant than to either: you describe an outcome in chat, the vendor's agent plans and executes it on the vendor's infrastructure, connects to your accounts through OAuth, and keeps a memory of your organization between tasks. [?@sec-openclaw](#) describes the self-hosted, open-source version of the same idea.

52.1 Vestra

Vestra's marketing describes two named agents. "Bash" is the executor: you describe "outcomes, not instructions", it "plans it, in the open, before anything runs", and it can split work across specialist sub-agents, running up to ten tasks in parallel. "Nami" is the memory layer, which the handbook says builds a company memory by "watching everything: every task Bash finishes, every document, every decision" plus whatever is in the connected tools (Vestra AI 2026a). Each task runs on "a real cloud machine with files, a browser, and a terminal", and the platform routes each task to a model it picks from several providers unless you pin one.

- **Target user.** The use-case pages address:
 - founders
 - sales and marketing
 - support
 - finance
 - recruiting
 - operations
 - research and strategy teams

Press coverage aims it at non-technical domain experts and small teams. The site describes a private beta.

- **Pricing** (Vestra AI 2026b). Four self-serve tiers, all “entire office, all features”:
 - Lite at USD 19/month for 2,000 credits on a 4 GB machine
 - Pro at USD 97 for 10,000 credits
 - Max at USD 197 for 20,000 credits
 - Ultra at USD 497 for 50,000 credits on a 32 GB machine

A credit is “the unit of work” and the cost per task varies. The first 1,000 credits are free, and custom engagements with forward-deployed engineers start at USD 4,000/month.

- **Data handling** (Vestra AI 2026c). The privacy policy (effective 2026-09-04) says content is not used for training by default, but task content is sent to whichever model provider handles the task, and one listed model tier (Meta’s Muse Spark 1.3) permits that provider to train on it. OAuth flows and access tokens are handled by a subprocessor, Composio, Inc. Deleting the account removes stored data within 30 days. User-created agents may be indexed by search engines by default, with opt-out by email.
- **Agent and API access.** Chat channels are Slack, Telegram, WhatsApp, iMessage, Gmail, and a desktop overlay on the Mac app. “Abilities” package skills, specialist agents, connectors, and guardrails into installable bundles. We found no public API documentation. The memory store can be exported as plain Markdown.

52.2 Manus

Manus (Manus AI 2026) is the best-known product in the category. Its home page lists a browser operator, a web app builder, “Wide Research” for parallel research tasks, slide and image generators, email (“Mail Manus”) and Slack integrations, a team plan with SSO, and an API for developers. Its pricing page renders client-side and its privacy policy returned only a title when we fetched them, so we could not verify plan prices or the training policy from the vendor (measured 2026-09-09); check manus.im/pricing and trust.manus.im directly before relying on either.

52.3 Lindy

Lindy (Lindy 2026b) is a “Slack-native teammate”: everyone on a team gets a private assistant in Slack DMs and a shared one in channels. It runs scheduled routines, connects to over 1,000 apps, supports MCP and computer use, and ships with “40+ skills: research, data analysis, decks, dashboards” that a team can extend by saving a completed task as a new skill.

- **Pricing** (Lindy 2026a). Plus at USD 29.99/month for 3,000 credits per user, Pro at USD 99.99 for 15,000, Max at USD 199.99 for 35,000, and an Enterprise tier with HIPAA, a BAA, SSO, and audit logs. Credits do not roll over.
- **Data handling.** “Your data is encrypted, never sold, and never used to train models”, with SOC 2 Type II, GDPR, HIPAA, and PIPEDA compliance claimed. “Nothing irreversible happens without your approval.”
- **Agent and API access.** MCP support is the extension route; the pricing page lists no public API.

52.4 Zapier Agents

Zapier Agents (Zapier 2026a) are “superhuman teammates” layered on Zapier’s 9,000-app integration catalog. You describe the agent, attach knowledge sources (FAQs, documentation, public links), and it acts across those apps, with real-time activity monitoring. Templates cover lead enrichment, content, support, candidate ranking, and expense classification.

- **Pricing** (Zapier 2026b). Agents bill in *activities*, separate from Zap tasks: “any billable action an agent takes: using a trigger, answering from a knowledge source, running an action, web browsing or scraping, a web search, or a message sent via the Chrome extension.” The Free plan includes 400 activities/month with a 10-activity cap per run; paid plans start at about 1,500 activities/month with a 40-activity cap. Zapier’s paid plans start at USD 19.99/month billed annually; nonprofits get 15% off.
- **Data handling.** Neither the Agents page nor the pricing page states how AI features handle data, so read Zapier’s separate privacy and security documents.
- **Agent and API access.** Agents inherit Zapier’s triggers, webhooks, and Chrome extension; this is the most “automation builder” of the five.

52.5 Relevance AI

Relevance AI (Relevance AI 2026a) calls itself “the home of the AI Workforce”. Agents are built three ways: a drag-and-drop canvas, plain-language description, or programmatically through MCP and an API. Each agent “owns one narrow task”, and a workforce nests them with parallel streams, human-in-the-loop approvals, evaluation pass-rate tracking, audit trails, and triggers from CRM events, email, calendar, webhooks, and cron.

- **Pricing.** The public pricing page lists only an Enterprise tier with custom pricing (Relevance AI 2026c); a case study on the agents page quotes USD 0.01 to 0.11 per task.
- **Data handling** (Relevance AI 2026b). The data security policy covers AES-256 encryption in transit, need-to-know staff access, and background checks, but does not address model training on customer data, data residency regions, or certifications.
- **Agent and API access.** The strongest developer surface of the five: API, MCP, webhooks, and cron.

52.6 Useful to us? Not for research work

For an epidemiology lab whose work is code, data, and manuscripts, these products solve a problem we mostly do not have. Their value is in gluing SaaS business tools together (CRM, inbox, calendar, Slack) for people who do not write code. Our work lives in Git repositories, R and Python environments, and Quarto documents, where the coding agents in Section 4 and the collaborative workspace agents in Section 68 already reach the files directly.

Three concerns weigh against adopting any of them for lab work:

- **Data.** An agent office works by connecting to your accounts and remembering what it sees. Vestra’s memory layer is designed to watch “every document, every decision”; its policy names one model tier that trains on task content; and Manus’s policy could not be read at all. Human-subjects data, unpublished results, and grant drafts should not flow through a vendor whose training and residency terms we cannot verify. Lindy’s explicit no-training statement and HIPAA/BAA tier is the exception, and it costs enterprise pricing.
- **Reproducibility.** A task that ran on a vendor’s cloud machine with an auto-selected model leaves no script we can re-run. For analysis, that is disqualifying.
- **Cost.** Credit-metered plans at USD 19 to 497 per person per month buy less than a coding-agent subscription for our use, because the expensive part (a hosted VM with a browser) is what we least need.

Where they could earn a place is in lab administration: a scheduled routine that drafts a weekly summary from a shared inbox, triages a calendar, or files reminders. Zapier Agents is the cheapest way to try that, because the free tier covers 400 activities a month and the integration catalog is the largest. If a member wants to try Vestra or Manus, use a throwaway account connected only to non-sensitive tools, and treat **?@sec-openclaw** as the option that keeps the data on our own machine.

53 Copilot Instructions for this Repository

A `.github/copilot-instructions.md` file contains repository-specific instructions and guidelines for GitHub Copilot coding agents. This file helps ensure that AI-generated contributions follow the project’s formatting standards, coding conventions, and documentation practices.

For a Quarto-based repository like this one, the copilot instructions file typically specifies:

- Markdown and Quarto formatting rules (e.g., blank lines before lists, line breaks in prose)
- R code style guidelines (e.g., using native pipe `|>`, following tidyverse style)
- File organization patterns (e.g., using Quarto includes for modular content)
- How to work with DOCX files for hybrid editing workflows

- Version control best practices (e.g., do not commit rendered artifacts that are generated only for preview/review)
- Repository-specific best practices

By having these instructions in `.github/copilot-instructions.md`, you ensure that coding agents produce consistent, high-quality contributions that align with the project’s established practices. This reduces the review burden and helps maintain consistency across all contributions, whether made by humans or AI assistants.

See this repository’s own [.github/copilot-instructions.md](#) for a working example.

54 Using Copilot Review Before Human Review

Before requesting review from other humans, **always have Copilot review your pull request first**—even if Copilot created the PR itself. AI review provides fast, thorough feedback that helps catch issues before involving human reviewers, saving everyone time and improving code quality.

Why review with Copilot first:

- **AI has more bandwidth:** Copilot can review code immediately without competing priorities
- **Catch common issues early:** Copilot excels at identifying bugs, logic errors, security vulnerabilities, and style inconsistencies
- **Improve human review quality:** When humans review cleaner code, they can focus on higher-level concerns like design and architecture rather than basic issues
- **Learn from feedback:** Even experienced developers benefit from Copilot’s perspective on best practices and potential improvements
- **Growing capabilities:** AI review capabilities continue to improve over time, making this investment increasingly valuable

Copilot review workflow:

1. **Assign Copilot as a reviewer:** On your pull request page, assign Copilot to review the PR the same way you would assign any other reviewer. Click “Reviewers” in the right sidebar and select Copilot from the list.
2. **Review Copilot’s comments:** Once Copilot completes its review, carefully examine each comment. For each comment, decide whether you agree with the suggestion:
 - **If the comment is correct:** Address it by making code changes yourself or ask Copilot to apply the fix using GitHub’s suggestion features
 - **If the comment is incorrect or not applicable:** Dismiss the comment with an explanation for why it doesn’t apply

- **If you're uncertain:** Seek a second opinion from a human reviewer or do additional research
3. **Request another Copilot review:** After addressing or dismissing all comments, request another review from Copilot. This creates an iterative improvement process.
 4. **Iterate until satisfied:** Repeat the review-and-address cycle until Copilot stops providing valuable suggestions. This typically takes 1-3 iterations depending on the complexity of the changes.
 5. **Request human review:** Only after you've addressed Copilot's feedback should you request review from human team members. At this point, the code should be in better shape, allowing human reviewers to focus on higher-level concerns.

Important considerations:

- **Copilot isn't perfect:** AI review can produce false positives or miss important issues. Always apply your own judgment when evaluating Copilot's suggestions.
- **Don't blindly accept all suggestions:** Some of Copilot's recommendations may not fit your specific context or requirements. It's perfectly appropriate to dismiss comments that don't apply.
- **Human review remains essential:** Copilot review supplements but does not replace human code review. Humans bring domain knowledge, understanding of business requirements, and judgment about trade-offs that AI cannot replicate.
- **Document dismissals:** When dismissing Copilot comments, briefly explain why. This helps human reviewers understand your reasoning and can serve as documentation for future reference.

For pull request authors:

Even if you're highly experienced, treating Copilot review as a required pre-review step helps maintain code quality and makes the best use of everyone's time. The few minutes spent on Copilot review often save hours of back-and-forth with human reviewers.

For human reviewers:

When you receive a PR for review, check whether the author has completed the Copilot review process. If Copilot hasn't reviewed the PR yet, consider asking the author to complete that step first before you invest time in review. This ensures you're reviewing code that has already been through initial automated quality checks.

55 Reviewing a Copilot PR You Didn't Create

When reviewing a pull request where someone else prompted Copilot to make changes, follow these guidelines to avoid confusion and ensure smooth collaboration:

Understanding PR roles:

The general PR roles (issue creator, author, reviewer, merger, assignee, and PR steward) are described in the [UCD-SERG Lab Manual's GitHub chapter](#). Copilot-assisted workflows add two more:

- **PR prompter:** Assigns a developer (human or AI) to start working on a PR, often by assigning an issue to Copilot. The PR prompter is sometimes the same person as the issue creator, but is often a project maintainer who reviews and triages external issue reports. When Copilot creates commits, both Copilot and the prompter are listed as co-authors. Typically the PR prompter becomes the PR manager.
- **PR manager:** Supervises and guides the PR authors, assigns reviewers, and controls the PR workflow, deciding when and how Copilot makes additional changes. The PR manager can hand off this role to someone else.

The scenario:

- One team member (the “PR prompter”) assigned Copilot to work on an issue or explicitly prompted Copilot to start working
- The prompter may or may not be the same person who originally created the issue
 - In projects with a user base, users often submit issues (bug reports, feature requests)
 - A project maintainer then steps in, adds their perspective, and assigns the issue to Copilot
 - In this case, the maintainer who assigned Copilot is the prompter, not the original issue creator
- Copilot created the PR with the prompter as co-author
- The prompter (now acting as PR manager) requested your review
- Copilot may have also automatically reviewed the PR

As a non-manager reviewer, your role is to provide feedback, not to directly initiate more work by Copilot. The PR manager should remain in control of when and how Copilot makes additional changes.

Recommended review workflow:

1. Use “**Comment**” or “**Request changes**” based on the severity of issues:
 - Use “**Comment**” for suggestions, questions, or minor issues that don't block merging

- Use “**Request changes**” for significant issues that must be addressed before merging
- Both options allow you to provide feedback without directly triggering Copilot

2. **Don’t ask Copilot to make changes directly:**

- Avoid using features that would trigger Copilot to start working immediately
- Let the PR manager decide whether to ask Copilot to address your comments or make changes themselves

3. **Write clear, actionable comments:**

- Explain what needs to change and why
- Suggest specific solutions when appropriate
- The PR manager will decide how to address your feedback

For PR managers:

After receiving reviews from other team members:

1. **Review all comments carefully:**

- Decide which comments you agree with
- Dismiss or respond to comments you don’t entirely agree with
- This ensures Copilot only addresses feedback you’ve validated

2. **Choose how to address valid feedback:**

- **Option A:** Make the changes yourself (faster for simple fixes)
- **Option B:** Ask Copilot to address the feedback (better for complex changes)
- **Option C:** Add your own review summarizing which comments Copilot should address, then ask Copilot to respond to the open comment threads

3. **Maintain clear communication:**

- Let reviewers know how you plan to address their feedback
- Mark conversations as resolved after addressing them
- Request re-review from humans after Copilot makes significant changes
- Update the PR’s “Assignees” field to reflect who is currently responsible for the PR

Transferring the PR manager role:

The original PR manager can hand over a PR to another person, who then becomes the new PR manager with control over Copilot’s work on that PR. This might be useful when:

- The original PR manager is unavailable or on leave
- Someone with different expertise needs to guide the remaining work
- Responsibilities are being redistributed within the team

To transfer the PR manager role:

1. The original PR manager should clearly communicate the handover to all reviewers
2. The new PR manager should review the PR's history and any open feedback
3. The new PR manager should take over responding to Copilot and managing future iterations
4. The team should update the PR's "Assignees" field and comments or description to reflect the current PR manager

This workflow ensures the PR manager maintains control over the development process while benefiting from collaborative human review and Copilot's implementation capabilities.

56 Agent Sessions and Handoff in Visual Studio Code

In Visual Studio Code, interactions with AI coding assistants are structured around [Agent Sessions and Handoff](#) (measured 2026-08-31). Understanding how sessions organize work and transfer state across tools is essential for managing multi-step agent workflows.

Anatomy of an agent session

An agent session represents a stateful stream of interaction between a developer and an agent. Each session maintains:

- **Chats and model selection:** One or more chats within a session, each with its own agent or model selection.
- **Interaction history:** User prompts, assistant responses, and tool calls executed during the task.
- **Accumulated context:** Referenced files, code snippets, and conversational state gathered across turns.
- **Checkpoints and branching:** Points in session history allowing developers to fork or roll back state when exploring alternative implementation paths.

Managing sessions across surfaces

VS Code provides unified session discovery and access across editor surfaces:

- **Shared surfaces:** The primary Chat view and the dedicated Agents window share the same sessions, allowing developers to switch views without losing conversational context.
- **External session discovery:** VS Code discovers sessions initiated outside the primary GUI, including CLI agent sessions (such as Copilot CLI, Claude Code, and Codex).
- **Cloud synchronization:** Synced sessions are backed up to your GitHub account, enabling developers to access active and past sessions across devices.

Session handoff types

Session handoff transfers context and intent from an active session to a specialized workflow without manual re-prompting:

- **Harness to harness:** Switch the active session between different agent harnesses to leverage distinct agent runtime capabilities on the same task.
- **Plan to implementation:** Hand off a high-level architectural plan or task specification directly to an implementation session to generate code.
- **Continue in the cloud:** Hand off a local session to run in a cloud-hosted agent environment (such as background tasks leading to pull requests), freeing local editor resources while the agent executes in the background.

57 Herdr: A Terminal Multiplexer for Coding Agents

[Herdr](#) (herdrdev 2026e) is a terminal multiplexer built for running several coding agents at once (measured 2026-09-09). It describes itself as “the runtime your coding agents live on”: a background server owns the real terminal processes, clients attach to render them, and the server keeps every agent running when the client closes or an SSH connection drops. On top of that `tmux`-like core it detects which agent is running in each pane and reports whether that agent is working, blocked, done, or idle, so the operator looks only at the pane that needs attention.

Maker and activity

Herdr is developed by [herdrdev](#) (herdrdev 2026f), a one-person company founded in 2026 by Can Celik in Ankara, Turkey, and part of the Y Combinator Fall 2026 batch (Y Combinator 2026). The project is written in Rust and licensed under Apache 2.0. The GitHub repository was created on 2026-03-27 and had about 37,000 stars, 2,700 forks, and 326 open issues on 2026-09-09, with the most recent commit the day before; the latest stable release was v0.9.0 on 2026-09-07, following v0.8.2 (2026-08-19) and v0.8.0 (2026-08-03) (herdrdev 2026f). A Show HN thread drew 166 points and 110 comments (Hacker News 2026), and a community-maintained list of plugins and integrations has grown around the tool (Konur 2026). The software is pre-1.0 and changes quickly, so expect some of what follows to be out of date.

What it does

Herdr organizes terminals into a hierarchy of sessions, workspaces, tabs, and panes (herdrdev 2026b). A pane is a real terminal (a PTY) that the server owns; a workspace groups the tabs and panes for one repository, task, or investigation. The distinctive features, beyond ordinary multiplexing, are:

- **Agent status detection:** Herdr recognizes 21 coding agents out of the box, including Claude Code, Codex, Cursor Agent CLI, OpenCode, GitHub Copilot CLI, Hermes Agent, Pi, Amp, Grok CLI, Antigravity CLI, and Kiro CLI, with Gemini CLI and Cline detected but less thoroughly tested (herdrdev 2026a). Where an agent ships lifecycle hooks, the hook reports state authoritatively; otherwise Herdr matches the bottom of the live terminal buffer against a TOML detection manifest, and marks a pane `blocked` only when that snapshot matches a known approval, question, or permission prompt.
- **Git worktree management:** the socket API exposes `worktree.list`, `worktree.create`, `worktree.open`, and `worktree.remove`, which “manage Git checkouts as Herdr workspaces” (herdrdev 2026i). Creating a worktree checks out (or creates) a branch and opens the checkout as a new workspace grouped under the source workspace; the CLI form is `herdr worktree create --branch <name>` (Copes 2026).
- **A CLI and socket API for agents to drive:** the two are “the same surface agents drive”, speaking newline-delimited JSON over a Unix socket (a named pipe on Windows) at `~/.config/herdr/herdr.sock`. Methods include `pane.read` to read a pane’s recent output, `agent.prompt` to send a prompt, and `agent.wait` to block until another agent reaches `done` or `blocked` rather than polling (herdrdev 2026i).
- **Persistence and resume:** detaching with `ctrl+b q` leaves every process running; a server restart stops processes but restores workspaces, tabs, panes, working directories, and layout from a `session.json` snapshot, and supported agents can resume their own conversation (for example `claude --resume <id>`) from a saved session reference (herdrdev 2026h).
- **Remote machines:** `herdr machine add <ssh-host>` registers a machine reachable over ordinary SSH, installs or starts the Herdr server there, and folds its agents into one combined agent list; authentication stays with OpenSSH, and Herdr stores no keys or passwords (herdrdev 2026c). Remote servers must run Linux or macOS; a native Windows server is not supported as an SSH target.

Install and use

Herdr ships stable binaries for Linux, macOS, and Windows x86_64 (herdrdev 2026g). The one-line installers are:

```
curl -fsSL https://herdr.dev/install.sh | sh
```

```
powershell -ExecutionPolicy Bypass -c "irm https://herdr.dev/install.ps1 | iex"
```

Homebrew (`brew install herdr`), `mise`, and Nix are also supported. Run `herdr` to start the server and attach a client, `herdr agent list` to see detected agents and their states, `herdr status` to summarize the runtime, and `herdr update` to upgrade. The default prefix key is `ctrl+b`, as in `tmux`, and the configuration file lives at `~/.config/herdr/config.toml` (`%APPDATA%\herdr\config.toml` on Windows) (herdrdev 2026d). The maintainers publish an

[agent guide](#) (herdrdev 2026d) meant to be handed to a coding agent so it can set Herdr up for you.

Pricing and license

The runtime is free and open source under the Apache 2.0 license. The landing page announces a paid “Herdr Cloud” as “coming soon”, described as connecting your own machines without configuring SSH, with a waitlist and no published pricing (measured 2026-09-09) (herdrdev 2026e).

Useful to us?

Herdr overlaps with two things the lab already does (?@sec-orch-baseline): running several Claude Code sessions in parallel, and isolating each one in its own git worktree. It does not replace either. Herdr runs the same agent CLIs unchanged, and its worktree commands wrap the same `git worktree` calls that Claude Code’s own worktree isolation makes. What it adds is the layer between those sessions and the person watching them:

- **An attention queue instead of a wall of terminals.** When a dozen sessions run at once, the expensive part is finding the one waiting on a permission prompt. The blocked, working, and done states Herdr aggregates across every workspace and every machine answer that question at a glance, which is the gap one early adopter reported it filling when herding parallel agents on a remote box (Coles 2026).
- **Sessions that survive a closed laptop.** Long runs of the lab’s review-and-iterate loop currently die with the terminal that started them unless they are already running on a server under `tmux`. Herdr gives the same persistence with a friendlier client, and can reattach to a Claude Code conversation by session ID after a restart.
- **A machine-readable control surface.** `agent.wait` and `pane.read` let a supervising agent spawn helpers in separate worktrees, wait until each is done or blocked, and read their output, without the polling loops our current scheduled check-ins use.

The same early adopters name the limits (Coles 2026; Copes 2026). Herdr provides no sandboxing and does not isolate file changes between agents sharing a directory; worktrees, permissions, and network egress remain the user’s problem, so our hooks and permission rules stay in place unchanged. Windows support is in beta, a Windows machine cannot serve as a remote target, and the tool is pre-1.0. It also sits beside, not inside, the editor-hosted session views described in Section 56: VS Code discovers CLI sessions and hands work between harnesses, while Herdr owns the terminals those sessions run in.

The practical recommendation is to try Herdr on a Linux workstation or server where a lab member already runs several Claude Code sessions, using one workspace per worktree. It is a small, reversible addition (no wrapper around the agent, no change to `ai-config`), and the status sidebar alone may justify it. Adopting the socket API as an orchestration layer is a larger step that we should defer until the project reaches a stable release.

58 Installing Claude Code on Windows

Claude Code is Anthropic’s command-line coding agent. Installing it on Windows works well, but a few platform-specific pitfalls can cost you hours if you don’t know about them. These notes capture a setup that works, and the gotchas to watch for.

Note

These notes were written in June 2026. As with the rest of this chapter, treat the specifics with caution— installers and behavior change quickly.

Run it from a Unix-like terminal

Claude Code expects a Unix-like shell. On Windows, run it from one of:

- **Git Bash** (ships with [Git for Windows](#));
- **MSYS2** (<https://www.msys2.org/>), which also gives you a package manager (**pacman**) and optional **zsh**;
- **WSL** (Windows Subsystem for Linux), the most Unix-faithful option.

If you use WSL, install Claude Code *inside* WSL— a Windows install will not carry over, because WSL has its own filesystem and **PATH**. Inside WSL the standard Linux install applies, and the Windows-specific **PATH** and **rehash** gotchas below don’t apply.

Install Claude Code

There are two common routes:

1. **Native installer** (recommended):

```
curl -fsSL https://claude.ai/install.sh | bash
```

This installs the binary to `~/.local/bin`. If your organization’s policy requires it, download and inspect the script before running it rather than piping it straight to **bash**.

2. **npm** (requires [Node.js](#)):

```
npm install -g @anthropic-ai/claude-code
```

Claude Code migrates itself out of npm

Even if you install via npm, current versions **migrate themselves to a native install** the first time you run **claude** (at `~/.local/bin/claude`, or `~/.local/bin/claude.exe` on Windows) and remove the npm copy. You can watch this happen in the terminal

output the first time you run `claude`; the current install methods are documented in the [Claude Code setup guide](#).

This is the single most confusing Windows gotcha: a path that worked a moment ago “disappears.” **Do not** hardcode the npm location (e.g. `.../AppData/Roaming/npm/...`) in your shell config. Point your `PATH` at the shell-appropriate directory shown in the next section instead.

Make sure your shell can find `claude`

The native binary lives in `~/.local/bin`. The installer adds this directory to `PATH` for ordinary shells, but on Windows two things commonly break that.

MSYS2 does not inherit the Windows `PATH` by default. It starts in a “minimal” path mode, so tools installed elsewhere on Windows are invisible to it. Add the binary’s directory explicitly in your `~/.zshrc` (or `~/.bashrc`). Because MSYS2’s `$HOME` is `/home/<you>` (its own home, not the Windows profile where the installer puts the binary), point `PATH` at the absolute Windows path:

```
export PATH="/c/Users/<you>/.local/bin:$PATH"      # MSYS2
```

In **Git Bash**, `$HOME` already is `/c/Users/<you>` (your Windows user profile), so the shorthand works there:

```
export PATH="$HOME/.local/bin:$PATH"      # Git Bash
```

Rehash after changing `PATH`. `zsh` and `bash` cache the locations of executables. If you add a directory to `PATH` in your shell config, the shell may still report `command not found` *even though the directory is on `PATH`*, because its command table is stale. Force a rebuild in the same shell right after editing `PATH`:

```
rehash      # zsh; use 'hash -r' in bash
```

This bites hardest with [oh-my-zsh](#), which builds `zsh`’s command table *before* your custom `PATH` line runs, so opening a fresh window may not clear the stale `command not found` on its own until you `rehash` (or move the `PATH` line before `oh-my-zsh` initializes).

Do not edit dotfiles with PowerShell redirection

⚠ PowerShell writes the wrong encoding

Never write to `~/.bashrc`, `~/.zshrc`, or other shell config files using PowerShell's `>` or `>>` redirection. Windows PowerShell writes **UTF-16 with a byte-order mark**, which `bash/zsh` read as a stray character at the start of the file:

```
bash: $'\377\376export': command not found
```

Edit dotfiles from *inside* the shell (e.g. with `nano`, `vim`, or `echo 'export PATH=...' >> ~/.zshrc` run in `bash/zsh`), or with an editor that saves UTF-8 without a BOM (e.g. VS Code).

Authenticating the GitHub CLI in Git Bash or MSYS2

If you also use the [GitHub CLI](#) (`gh`) to push code or open pull requests, `gh auth login` may fail with:

```
could not prompt: ... running in MinTTY without pseudo terminal support
```

Git Bash and MSYS2 both default to the MinTTY terminal, which can't host the interactive prompt. Wrap the command with `winpty`, or run it from PowerShell / Windows Terminal instead:

```
winpty gh auth login
```

`winpty` ships with Git Bash, but in MSYS2 it's a separate package — install it first with `pacman -S winpty` if you get `command not found`.

Verify the install

Open a **new** terminal window (so it picks up your updated config) and run:

```
claude --version      # prints the installed version number
```

If you get a version number, you're ready to run `claude` in your project directory. If you get `command not found`, re-check the two `PATH` issues above: the directory must be on `PATH`, and you must `rehash` (or open a fresh window) after changing it.

59 Setting up MCP servers

The [Model Context Protocol](#) (MCP) is how a harness gains typed access to external systems. Configuring a server is usually a one-line command. Diagnosing one that *silently* isn't working is the part worth writing down, because the common failure mode produces no error at all — only a quiet absence of tools you assumed were there.

Note

These notes were written in July 2026, from a real diagnosis on a Linux machine. As with the rest of this chapter, treat the specifics with caution — harness internals and vendor defaults change quickly.

Installed is not registered

A server binary sitting on disk is not available to your agent. The harness knows only what its configuration declares, so installing [github-mcp-server](#) and *registering* it are two separate acts. Skipping the second is easy to miss, because the first one felt like the hard part.

The check is one command:

```
claude mcp list
```

A broken server can occupy the name your working one wanted

This is the failure worth knowing about, and it is nastier than a missing entry, because `claude mcp list` shows you something that looks right.

Plugin marketplaces can register servers of their own. An official GitHub plugin may install a **remote** server under exactly the name you meant to give your **local** one. The listing then reports a `github` server that is not your setup at all, and the local binary you installed goes unregistered and unnoticed.

So read the listing for the *transport and address*, not just the name:

```
plugin:github:github: https://api.githubcopilot.com/mcp/ (HTTP) - X Failed to  
connect - HTTP 400: ... Authorization header is badly formatted
```

An (HTTP) entry pointing at a vendor URL is a remote server. A local one shows a command path instead.

400 and 401 mean different things

The status code is the whole diagnosis here, and it is easy to skim past.

The configuration a plugin ships may hardcode a credential placeholder:

```
{"headers": {"Authorization": "Bearer ${GITHUB_PERSONAL_ACCESS_TOKEN}"}}
```

If that variable has no value when the harness starts, the header goes out as a bare **Bearer** with nothing after it. That is **malformed**, not **unauthorized**:

- **401** means the server read your token and rejected it — a real credential problem: expired, wrong scopes, wrong account.
- **400** means no token was ever substituted — a *configuration* problem, and no amount of re-issuing tokens will fix it.

Chasing a 400 as though it were a 401 sends you to the token-minting page for a problem that lives in a config file.

Install the binary the platform's own way

These notes started from a Linux install, where the binary lands under `$HOME/.local/bin`. On macOS the same server is a Homebrew formula, and getting it is one command:

```
brew install github-mcp-server
```

That installs to `/opt/homebrew/bin` on Apple silicon (`/usr/local/bin` under Intel Homebrew), so no manual download is needed.

The official install guide leads with a Docker recipe instead, and it has a catch worth knowing before you follow it: **docker** being on `PATH` does not mean the daemon is running. With Docker Desktop stopped, the server fails to connect to the Docker API, and that failure surfaces at *server start*, not at registration — so `claude mcp add` succeeds, and the break only shows up later, as a silent absence of tools. That is one more reason to prefer the binary path above.

Because the binary's location differs by platform and by installer, a launch wrapper should resolve it from `PATH` rather than hardcode it, with an override for the case where it isn't on one.

Supply credentials without storing a token

The obvious registration bakes a token straight into harness config:

```
claude mcp add github -e GITHUB_PERSONAL_ACCESS_TOKEN=<pat> -- <server> stdio
```

That writes a live credential to a config file in plain text, and pins you to one token that will eventually expire.

A launch wrapper avoids both. It reads the credential at start time from a tool that already holds one, so nothing is stored and the server follows whatever account you are currently logged in as:

```
#!/bin/sh
set -eu

SERVER="${GITHUB_MCP_SERVER_BIN:-$(command -v github-mcp-server || true)}"
if [ -z "$SERVER" ]; then
    echo "github-mcp-server not on PATH; install it or set GITHUB_MCP_SERVER_BIN" >&2
    exit 1
fi

GITHUB_TOOLSETS="${GITHUB_TOOLSETS:-default,actions}"
export GITHUB_TOOLSETS

GITHUB_PERSONAL_ACCESS_TOKEN="$(gh auth token)"
if [ -z "$GITHUB_PERSONAL_ACCESS_TOKEN" ]; then
    echo "empty token; run 'gh auth login'" >&2
    exit 1
fi
export GITHUB_PERSONAL_ACCESS_TOKEN

exec "$SERVER" stdio "$@"
```

Register the wrapper rather than the binary:

```
claude mcp add --scope user github -- ~/.local/bin/github-mcp-server-stdio
```

Note the explicit failure when the token comes back empty. A wrapper that silently exports an empty string reproduces the bare-Bearer bug you just finished diagnosing.

Toolsets are opt-in, and the default may omit what you need

A server does not necessarily expose everything it can do. GitHub's server exposes a default group, and that default carries **no continuous-integration access at all** — no workflow

runs, no job logs, no re-run trigger. As of this writing (server v1.7.0) the actions toolset's four tools are `actions_get`, `actions_list`, `actions_run_trigger`, and `get_job_logs` — the exact names have moved around across releases, so treat this list as a snapshot rather than a promise.

If your workflow involves driving pull requests to a clean state, that omission matters, because reading check status is most of the job. Request the extra group explicitly:

```
GITHUB_TOOLSETS=default,actions
```

Note that the selection **replaces** the default rather than extending it, which is why the value above names `default` explicitly. Writing `actions` alone would silently trade away every default tool for the four you asked for — a net *loss* that looks like a successful configuration change.

So compare the tool list before and after, and confirm the count went up rather than sideways. On the same v1.7.0 server, the default group carries 44 tools, and `default,actions,discussions,dependabot,labels,notifications` carries 63 — the count moving the right direction, as the check above expects.

subscribe_pr_activity isn't a local-server tool

Workflow guidance written for remote or web agent sessions sometimes names `subscribe_pr_activity` as the way to watch a pull request's activity. It doesn't appear in a locally-run GitHub MCP server, under any toolset combination.

The local analogues are `manage_notification_subscription` and `manage_repository_notification_subscription`. Reach for those instead when working from a local harness.

Verify with a real call, and expect to restart

Two habits close this out.

Verify by calling, not by reading a list. A tool appearing in the registry proves the harness parsed a config file. It does not prove the server started, authenticated, or can reach the API. One cheap identity call plus one read (for GitHub: `get_me`, then listing pull requests on a repo you know) proves the whole path end to end.

Expect the tools to be missing until you restart. MCP servers connect when a session starts, so a server registered mid-session is inert for the rest of it. This is a common false alarm: the registration worked, and the tools genuinely are not there yet.

Finally, note which new tools can *write*. A re-run or dispatch tool can trigger CI, and permissive permission modes will not prompt before it does. Treat those the way you would treat a merge — something a human authorizes, not something an agent does in passing.

Copilot on GitHub uses a different config surface

The notes above are for a local harness (`claude mcp list`, a binary on `PATH`). [Copilot cloud agent](#) and Copilot code review on GitHub.com do not read that file.

Repository administrators configure those agents from **Settings > Copilot > MCP servers** using a JSON `mcpServers` object. GitHub's [Configure MCP servers](#) page is the source of truth for the schema. As of that page:

- The GitHub MCP server and the Playwright MCP server are enabled by default.
- Cloud agent and code review share the repository config; a separate toggle can disable MCP tools for code review only.
- Only MCP *tools* are supported, not resources or prompts.
- Remote servers that authenticate with OAuth are not supported.
- Secrets and variables must be named with a `COPILLOT_MCP_` prefix or they are invisible to the config.
- Once a tool is enabled, Copilot uses it without asking for approval, so allowlist specific read-only tools rather than `*`.

Do not copy a local `claude mcp` add registration into that JSON and expect it to work.

Granola MCP: your meetings as context

[Granola MCP](#) is the other side of that same pattern: it is not a code-host server but a meeting-context server. Granola is an AI notepad for back-to-back meetings; its MCP exposes your meeting notes to any MCP client.

The gap it closes is the copy-paste loop: without it, using something you discussed in a meeting while working in Claude, ChatGPT, or Cursor means finding the note, copying the relevant bit, and pasting it in. With the MCP connected, that context rides with you. Use it to turn a standup into Linear tickets, scaffold a feature from what was agreed, or draft a follow-up from what was actually said.

It connects through the standard [Model Context Protocol](#) (MCP) as a remote server at `https://mcp.granola.ai/mcp`. For Claude or ChatGPT, enable it from the app's connector/app settings and authenticate; for Cursor, Claude Code, or any other MCP client that supports a manual URL, register that URL directly (see [the announcement](#) for per-client steps). On an Enterprise plan it is an early-access beta, off by default until an admin enables it.

60 Google Antigravity Python SDK

The [google-antigravity/antigravity-sdk-python](#) repository provides the official Python SDK for building and automating agents on the Google Antigravity agent runtime (measured 2026-08-31; distributed via PyPI as `google-antigravity`).

Architecture and runtime model

The SDK embeds the compiled Antigravity runtime engine directly into Python applications:

- **Lifecycle management:** Agents are instantiated through **Agent** objects configured via **LocalAgentConfig**, managed within asynchronous Python context managers (**async with**).
- **Autonomous agentic loop:** The underlying runtime drives multi-turn reasoning, streaming model responses, subagent spawning, and tool dispatch without requiring hand-rolled state machines.
- **Binary distribution:** The Python package distributes pre-compiled native runtime binaries via wheels, ensuring consistent agent execution across Windows, macOS, and Linux environments.

Extensibility and policy enforcement

Developers can customize agent behavior and enforce safety policies:

- **Custom tools and MCP servers:** Register custom Python functions as callable agent tools or connect external Model Context Protocol (MCP) server endpoints.
- **Steering hooks:** Attach pre-tool and post-tool lifecycle hooks to inspect, steer, or veto actions (such as restricting command execution or gating repository modifications).
- **Skill discovery:** Load skill catalogs (**skills/**/SKILL.md**) dynamically, allowing agents to leverage shared procedural instructions.

Comparison with interactive Antigravity surfaces

Dimension	Antigravity Python SDK	Antigravity CLI & IDE Extensions
Primary use case	Automated pipelines, CI evaluation, custom harnesses	Interactive terminal and GUI pair programming
Control plane	Python API (async with Agent(...))	Interactive CLI prompt or editor chat panel
Tool definitions	In-process Python callables & MCP	JSON manifests, plugins, and CLI scripts
Runtime engine	Embedded native binary	Managed local service

61 Unbounded Context with Magic Context

[cortexkit/magic-context](#) is an open-source self-managing memory engine designed to provide unbounded context for AI coding agents (measured 2026-08-31). It operates as a background

memory subsystem—often described as a “hippocampus for coding agents”—that extracts, consolidates, and retrieves long-term repository state without pausing the active coding turn.

Core architecture and agent roles

Rather than requiring the primary coding agent to interrupt its execution to prune conversation buffers, **magic-context** delegates memory lifecycle operations to specialized background workers:

- **The Historian:** Runs background context compaction on completed turns, compressing verbose tool outputs and dialog history while preserving architectural decisions and code rationale.
- **The Dreamer:** Executes periodic consolidation passes to deduplicate memory records across multiple sessions, distilling recurring observations into canonical project facts and persistent guidelines.
- **The Sidekick:** Acts as an on-demand retrieval companion that augments active prompts with relevant project context, supplying historical context matched to the current file or task.

Cache-aware memory management

A key challenge with dynamic prompt injection is preserving prompt caching efficiency. **magic-context** addresses this through cache-conscious orchestration:

- **Cache-stable prompt layout:** Maintains a deterministic prompt layout and replay ordering, preserving provider prompt caching prefixes across conversational turns without invalidating cached tokens.
- **Deferred background extraction:** Memory analysis and summarization tasks are deferred to idle windows or subagent threads, preventing token churn and latency spikes during high-tempo coding loops.
- **Cross-session persistence:** Extracted knowledge persists in lightweight local stores across IDE restarts, enabling coding agents to resume work with full institutional memory of past decisions.

62 Spec-Driven Development with Conductor

[gemini-cli-extensions/conductor](#) is an open-source plugin for AI coding agents (including Google Antigravity and Claude Code) that implements **Spec-Driven Development** (measured 2026-08-31). Rather than relying on conversational chat history that degrades over extended sessions, Conductor anchors agent behavior in structured, version-controlled Markdown artifacts stored directly in the repository, providing persistent context across multi-session workflows.

Core workflow phases

Conductor structures development into four distinct, sequential phases:

- **Context establishment** (`/conductor:conductor-setup`): Interactively initializes base-line project documentation (including product goals, technical stack choices, and testing guidelines), giving coding agents persistent reference material across subsequent sessions.
- **Specification and track planning** (`/conductor:conductor-new-track`): Transforms feature requests or bug fixes into a dedicated track containing a `spec.md` (functional scope and acceptance criteria) and a `plan.md` (ordered implementation phases broken into verifiable task checklists).
- **Phased implementation** (`/conductor:conductor-implement`): Guides the agent through the active track's plan sequentially, executing file edits, running local test suites, and marking tasks complete as acceptance criteria are met.
- **Track review and plan compliance** (`/conductor:conductor-review`): Conducts an adversarial verification pass against the original track specification, ensuring that all declared acceptance criteria are satisfied and no architectural drift occurred during execution.

Architectural benefits of Spec-Driven Development

Dimension	Conversational Prompting	Spec-Driven Development (Conductor)
Context persistence	Volatile in-memory chat buffer	Version-controlled Markdown artifacts
Task boundaries	Ad-hoc user instructions per turn	Structured <code>spec.md</code> and <code>plan.md</code> checklists
Verification loop	Manual spot-checking	Milestone-level automated tests and <code>/conductor:conductor-review</code>
Handoff & resumption	Requires re-prompting or context replay	Any agent resumes from the checked-in track state

63 Anatomy of Agent Plugins

In modern AI coding assistants (such as Google Antigravity and Claude Code; see Section 11 on the sunset of legacy Gemini CLI and its folding into Antigravity CLI), **plugins** serve as the top-level packaging and distribution layer for agent capabilities (measured 2026-09-01). While individual skills or Model Context Protocol (MCP) servers extend specific tasks, a plugin aggregates multiple extensibility primitives into a unified, version-controlled bundle.

Anatomy of a plugin bundle

A plugin manifest (such as `plugin.json` or `.claude-plugin/plugin.json`) orchestrates four distinct architectural components:

- **Skills (`skills/**/SKILL.md`):** Procedural Markdown instructions that teach the agent domain-specific workflows, coding conventions, and structured checklists.
- **Model Context Protocol (MCP) servers:** External process definitions exposing executable tool functions, database connectors, and live workspace resources via standard MCP JSON-RPC endpoints.
- **Lifecycle hooks:** Deterministic executable scripts configured via hook manifests (such as `hooks.json`) attached to agent lifecycle events (such as `PreToolUse` command inspection, `Stop` review-gate verification, and `UserPromptSubmit` context injection).
- **Specialized agent roles and slash commands:** Pre-configured subagent personas (such as dedicated reviewers or researchers) and user-facing shortcut commands (`/command`).

Comparison: Skills vs. MCP Servers & Tools vs. Plugins

Dimension	Skills (<code>SKILL.md</code>)	MCP Servers & Tools	Plugins (<code>plugin.json</code>)
Primary purpose	Procedural guidance & workflows	External tool execution & data access	Unified packaging & distribution
Execution model	Progressively loaded on demand	Executed by agent harness over IPC	Discovered & loaded by agent platform
Dependencies	Plain Markdown & scripts	Language runtimes (Node.js, Python, binaries)	Bundles skills, MCP configs, and hooks
Lifecycle control	Passive context instructions	Dynamic tool calls during agent turn	Active deterministic hook gates

Managing token budget and context bloat

A common hazard when adopting large community plugin bundles is context window saturation. When multiple plugins eagerly inject verbose instructions and exhaustive tool schemas into every turn, available context for actual code and reasoning shrinks.

Effective plugin architectures mitigate this through several strategies:

- **Lazy tool discovery:** Only core system tools load eagerly; specialized plugin tools declare schemas that load lazily on demand.

- **On-demand skill activation:** Agents search skill catalogs dynamically when relevant keywords appear, rather than loading the entire skill directory into the initial system prompt.
- **Prefix caching preservation:** Static plugin definitions are placed at the root of prompt structures so provider-level prompt caching remains undisturbed during multi-turn sessions.

64 Running Codex Inside Claude Code: the `codex-plugin-cc` Plugin

[openai/codex-plugin-cc](#) is OpenAI’s official Claude Code plugin for running Codex from inside a Claude Code session (OpenAI 2026a). It adds slash commands that ask Codex to review the current work or take over a task, while Claude Code stays the harness the user is typing into. The repository is Apache-2.0 licensed, has about 33,000 GitHub stars, and its latest release is `v1.0.6` from 2026-07-08 (measured 2026-09-09).

What it does

The plugin does not ship a second Codex runtime. It wraps the `codex` binary already installed on the machine, talking to it through the Codex app server, so it uses the same login, the same `config.toml`, and the same repository checkout that the Codex CLI would use directly (OpenAI 2026a). A Node.js script (`codex-companion.mjs`) does the actual work; the Markdown command files tell Claude to invoke that script once and to return Codex’s output verbatim rather than paraphrasing it. The command files enforce this: they forbid Claude from fixing anything a Codex review reports until the user says which findings to act on.

Installation

Node.js 18.18 or later is required (OpenAI 2026a). The plugin is installed through Claude Code’s plugin marketplace mechanism (see Section 63 for how marketplaces and manifests fit together); the repository’s `marketplace.json` names the marketplace `openai-codex`, which is why the install command does not repeat the repository name:

```
/plugin marketplace add openai/codex-plugin-cc
/plugin install codex@openai-codex
/reload-plugins
/codex:setup
```

`/codex:setup` checks whether Codex is installed and logged in, and offers to run `npm install -g @openai/codex` when it is missing. Logging in is done outside the plugin with `codex login`.

What gets added

The marketplace manifest lists a single plugin, `codex`, whose bundle contains commands, one subagent, three skills, and a hook file (OpenAI 2026a).

Slash commands:

- `/codex:review` runs Codex’s built-in read-only review on the working tree or on the branch against `--base <ref>`, with `--scope` to force one or the other and `--wait` or `--background` to choose whether it blocks the session. It takes no focus text.
- `/codex:adversarial-review` is the steerable version: it questions the design and assumptions rather than only the diff, accepts free-text focus such as “look for race conditions”, and takes the same `--base`, `--scope`, `--wait`, and `--background` flags.
- `/codex:rescue` hands a task to Codex to investigate or fix, with `--model`, `--effort`, `--resume`, and `--fresh` flags. By default this run is write-capable.
- `/codex:transfer` exports the current Claude Code transcript into a Codex thread and prints the `codex resume <session-id>` command, for continuing the same conversation in Codex.
- `/codex:status`, `/codex:result`, and `/codex:cancel` manage background jobs.
- `/codex:setup` checks the install and toggles the review gate described below.

Subagent:

- `codex:codex-rescue` (pinned to the Sonnet model tier) is a thin forwarder that `/codex:rescue` invokes. Its definition tells it to make exactly one `Bash` call to the companion script and to do no repository inspection or independent reasoning of its own, beyond using the `gpt-5-4-prompting` skill to tighten the forwarded prompt.

Skills (all marked `user-invocable: false`, so Claude loads them by description and the user cannot call them by name):

- `codex-cli-runtime`, the calling contract for the companion script, attached to the subagent
- `gpt-5-4-prompting`, guidance for tightening a request into a block-structured Codex prompt, attached to the subagent
- `codex-result-handling`, rules for presenting Codex output without altering it, attached to no agent and so available to the main session

Hooks:

- `SessionStart` and `SessionEnd` hooks record the transcript path that `/codex:transfer` later reads.

- An optional **Stop** hook implements a **review gate**: when enabled with `/codex:setup --enable-review-gate`, every time Claude tries to end a turn, Codex reviews that turn and blocks the stop if it finds problems. The README warns that this can produce a long Claude-Codex loop that drains usage limits quickly, and recommends enabling it only in an actively monitored session (OpenAI 2026a).

Authentication and cost

The plugin uses whatever `codex login` set up. Two routes exist (OpenAI 2026a, 2026b):

- A **ChatGPT account** (including the Free tier) draws on the Codex usage limits included in that plan; every review or rescue run counts against them.
- An **OpenAI API key** bills at API rates, which OpenAI positions for shared or automated environments such as CI.

Either way, the Claude Code session itself still bills to the Anthropic plan. A rescue run therefore spends on both providers at once: Claude’s tokens to dispatch and read back, and Codex’s usage limits or API credits to do the work. Model and reasoning-effort defaults come from `~/.codex/config.toml` or a project-level `.codex/config.toml`, the latter only in a trusted project (OpenAI 2026a).

Comparison with the lab’s existing Codex paths

The lab already reaches Codex two ways, and the plugin overlaps with both.

delegate-to-codex in ai-config (Morrison Lab 2026b). That skill runs `codex exec` directly from a Bash call, with a read-only sandbox by default, prompts written to files, an optional JSON output schema, and a background runner that fans out several prompts at once and polls a completion marker. It exists to spend the separately billed ChatGPT plan on heavy read/draft/verify fan-out before Claude’s own quota. The plugin covers the single-task case of that skill and adds three things the skill lacks: job tracking, threads that can be resumed, and a subagent Claude can call proactively. It does not cover the fan-out case, it defaults to write-capable runs where the skill defaults to read-only, and it does not enforce structured output. The two also disagree on who orchestrates: the skill keeps Claude as the integrator that assembles Codex’s parts, while the plugin’s result-handling skill tells Claude to relay Codex’s answer and stop.

Codex as a GitHub reviewer (Section 43). That path runs through Codex Cloud and posts a review on the pull request, so it needs a connected repository and a workspace that permits Codex Cloud. `/codex:review` runs locally against the checkout instead, with no GitHub side effects. It is the same local `/review` that the GitHub-review section names as the fallback when an administrator has disabled Codex Cloud, reachable without leaving Claude Code. Its output stays in the terminal, so it does not create the durable review record that the lab’s pull-request workflow relies on.

Useful to us? Yes, for local second-opinion reviews; not a replacement

The main use is `/codex:adversarial-review --base main` as a cross-vendor second opinion before pushing, which is what `ai-config`'s adversarial self-review rule asks for and what the `delegate-to-codex` skill implements by hand. The plugin makes that one command, and the `--background` flag keeps a multi-file review from blocking the session.

Three cautions apply:

- Leave the review gate off. A `Stop` hook that reruns Codex on every turn spends on both providers at the rate of a chat conversation, and the lab's own `Stop` hooks already gate on cheaper deterministic checks.
- `/codex:rescue` writes to the working tree by default. In a shared worktree or a multi-agent session, ask for a read-only run or use the `delegate-to-codex` skill, which sandboxes by default.
- The plugin is a wrapper, so it inherits Codex's model access rules: a ChatGPT login cannot reach every `--model` value the CLI accepts, and the refusal arrives from the API after the flag is accepted.

For fan-out work, structured output, or anything a script needs to consume, the `delegate-to-codex` skill remains the right tool. For pull-request reviews that must be visible to other contributors, the GitHub integration in Section 43 remains the right tool. See Section 37 for the rest of the plugins the lab has evaluated.

65 Multi-Agent Orchestration with Oh My OpenCode / Oh My OpenAgent

[code-yeongyu/oh-my-openagent](#) (originally published as **Oh My OpenCode** or `omo`, with community forks such as [opensoft/oh-my-opencode](#)) is an open-source multi-agent orchestration framework and plugin for AI coding agent harnesses (including OpenCode and OpenAI Codex CLI) with over 65,000 GitHub stars (measured 2026-09-01). Inspired by modular terminal configuration frameworks (such as [Oh My Zsh](#)), it expands single-agent coding into a specialized multi-agent system with automated model routing and background task execution.

Hub-and-spoke agent architecture

Rather than relying on a single generalist agent to perform all research, design, coding, and review steps, the framework implements a hub-and-spoke delegation model with specialized agent personas:

- **Sisyphus (Primary Orchestrator & Task Lead)**: Acts as the lead coordinator that breaks complex user requests into discrete work packages, delegates subtasks to specialized agents, and aggregates results into a coherent solution.

- **Prometheus (Strategic Planner)**: Interviews the user to clarify scope and ambiguities, formulating detailed implementation plans before code modifications begin.
- **Metis (Pre-Planning Gap Analyzer)**: Conducts pre-planning analysis alongside Prometheus to identify hidden requirements, ambiguities, and potential failure points before plans are finalized.
- **Momus (Plan Reviewer)**: Validates proposed execution plans against clarity, verification, and context criteria before execution.
- **Atlas (Plan Executor)**: Executes approved Prometheus plans step-by-step, managing task progress and session continuity across execution sessions.
- **Hephaestus (Autonomous Deep Worker)**: Specializes in autonomous architectural refactoring and complex multi-file debugging.
- **Oracle (Architecture & Debugging Consultant)**: Serves as a read-only architecture and deep debugging consultant, analyzing system design trade-offs and diagnosing subtle runtime failures.
- **Librarian (Documentation & Code Search)**: Conducts targeted external documentation lookups and code search to supply focused context without bloating the orchestrator's context window.
- **Explore (Codebase Navigation & Fast Grep)**: Specializes in fast pattern searches, codebase navigation, and file structure discovery.

Heterogeneous model routing

A central capability of the framework is decoupling agent roles from a single model provider. Developers can route distinct tasks to the model family best suited for each workload:

- **High-reasoning tiers** (such as Anthropic Claude Sonnet or Opus) for architectural planning and deep refactoring.
- **High-throughput models** (such as OpenAI GPT models) for rapid code generation and unit testing.
- **Large-context models** (such as Google Gemini) for extensive repository research and multi-file context indexing.

Operational capabilities

Dimension	Standard OpenCode	Oh My OpenCode / Oh My OpenAgent
Agent topology	Single sequential agent loop	Hub-and-spoke multi-agent team
Model routing	Single active model per session	Dynamic per-agent model assignment
Execution monitoring	Standard terminal output	Interactive <code>tmux</code> -backed session management

Dimension	Standard OpenCode	Oh My OpenCode / Oh My OpenAgent
Extensibility	Individual plugins and MCPs	Curated bundle of tools, agents, and MCP integrations

66 The OpenCode Ecosystem

[OpenCode](#) (anomalyco 2026) is the open-source coding agent harness (MIT-licensed, about 206,000 GitHub stars, measured 2026-09-09) that the earlier OpenCode sections of this chapter build on: running it against local models (Section 25), against OpenRouter (Section 26), and under the Oh My OpenCode multi-agent framework (Section 65). Around the harness itself sits a community ecosystem of plugins, clients, and agent bundles, which the OpenCode maintainers index on a single documentation page (OpenCode 2026b). This section maps that page (as of 2026-09-09) and calls out the entries most relevant to us. Where the site already covers a project, this section points there instead of repeating it.

How the ecosystem is organized

The ecosystem page groups its entries into three lists (OpenCode 2026b):

- **Plugins** (38 entries): JavaScript or TypeScript modules that OpenCode loads at startup and that hook into its lifecycle events (tool execution, file edits, session compaction, permissions, notifications, and so on). A plugin is named in the `plugin` key of `opencode.json` as an npm package, or dropped as a file into `.opencode/plugins/` (project) or `~/.config/opencode/plugins/` (global); npm packages are fetched automatically with Bun (OpenCode 2026d).
- **Projects** (11 entries): clients and integrations built on the OpenCode server API or SDK — editor front ends, web and desktop apps, a Discord bot, an Obsidian plugin, and an extension manager.
- **Agents** (2 entries): bundles of agent definitions, prompts, and commands that reshape how OpenCode plans and executes work.

The issue that prompted this section also asked about **providers**. The ecosystem page has no provider list; model providers are configured in `opencode.json` and documented separately, and the two first-party paid options are covered under “OpenCode Zen and OpenCode Go” below.

A recurring caveat applies to almost every entry: these projects are independent, and most carry an explicit “not built by the OpenCode team and not affiliated with OpenCode” disclaimer (for example `ocx` (kdcokenny 2026a) and `opencode-worktree` (kdcokenny 2026d)). Listing on the ecosystem page is a courtesy index, not an endorsement or a security review. Every plugin

runs with the same permissions as OpenCode itself, so treat adding one the way you would treat adding any unaudited dependency (see Section 20).

Notable plugins

Activity figures below (stars, last push, latest release) were read from the GitHub API on 2026-09-09. Several repositories have moved since the ecosystem page was written; the current location is given where it differs.

Context and token management

- [opencode-dynamic-context-pruning](#) (Opencode-DCP 2026) (now under the [Opencode-DCP](#) organization; AGPL-3.0; about 4,200 stars; release v3.1.15 on 2026-08-16). Reduces token spend by deduplicating repeated tool calls, purging the inputs of failed tool calls after a configurable number of turns, and letting the model compress stale conversation content into summaries. The README reports prompt-cache hit rates of roughly 85% with the plugin versus 90% without, and says development focus has shifted to a separate tool called *Sleev*. *Useful to us?* Yes, for long ARDI sessions on paid metered providers; the AGPL license is irrelevant for a locally-run tool but worth knowing.
- [opencode-morph-plugin](#) (Morph, the vendor; MIT; 85 stars; release v2.0.17 on 2026-09-07) and its community predecessor [opencode-morph-fast-apply](#) (MIT; 170 stars). Both route edits through Morph’s hosted “Fast Apply” model to cut edit latency, which means a second paid API and a second party seeing your code. *Useful to us?* Not by default; the privacy cost outweighs the speed gain for research code.

Subscription and quota bridges

These plugins let OpenCode consume a chat subscription you already pay for instead of API credits. They are among the most-starred plugins on the page and also the most fragile, since they depend on undocumented OAuth flows that the subscription vendors can close at any time:

- [opencode-openai-codex-auth](#) (Ali 2026) (Numman Ali; about 2,200 stars; release v4.4.0 on 2026-01-09; no push since then). Uses the ChatGPT Plus/Pro OAuth flow to expose the GPT and Codex model presets, and its README restricts it to “personal development use” with your own subscription.
- [opencode-gemini-auth](#) (MIT; about 1,700 stars; release v1.4.16 on 2026-05-21). The same idea for a Gemini plan.
- [opencode-antigravity-auth](#) (Fabris 2026) (MIT; about 11,000 stars; **archived** on GitHub as of 2026-09-09). Authenticated against Google Antigravity’s model pool with a Google account, rotating across accounts to stay under quota. Its own README states that using it “violates Google’s Terms of Service” and that your account “may be suspended or permanently banned”. The similar [opencode-google-antigravity-auth](#) (375 stars) is archived as well.

Useful to us? No. Our lab policy is to use delegation budgets through their supported routes (the `codex` CLI on the ChatGPT plan, OpenCode Go or Zen, OpenRouter credits), and a bridge whose README concedes a terms-of-service violation is a liability, not a saving. The Codex and Gemini bridges are less clearly prohibited but share the fragility.

Sandboxing, isolation, and worktrees

- [opencode-daytona](#) (Daytona 2026) (Daytona, the vendor; Apache-2.0; release `opencode-plugin-v0.192.1` on 2026-09-03). Runs each session in a hosted Daytona sandbox, syncs the sandbox to a local branch named `opencode/<n>`, and produces live preview links when a server starts. Requires a Daytona account and API key. *Useful to us?* Maybe, for untrusted-code experiments; otherwise the firewall and container setups in this chapter's Firewall and Network Configuration section cover the same ground.
- [opencode-devcontainers](#) (MIT; 222 stars; release `v0.5.1` on 2026-08-24; pushed 2026-09-08). Runs several devcontainer instances at once, one per branch, with auto-assigned ports. *Useful to us?* Yes, for repositories that already ship a `.devcontainer/`.
- [opencode-worktree](#) (kdcokenny 2026d) (MIT; 726 stars; no tagged release; pushed 2026-08-17). Adds `worktree_create` and `worktree_delete` tools that create an isolated git worktree under `~/.local/share/opencode/worktree/`, sync configured files into it, and open a terminal with OpenCode running there. *Useful to us?* Yes; it is the OpenCode counterpart of the per-agent worktrees our Claude Code sessions use.

Orchestration and workflow bundles

- [oh-my-opencode](#) (now `oh-my-openagent`; about 68,800 stars; release `v5.0.0-beta.51` on 2026-09-09). Covered in Section 65.
- [opencode-workspace](#) (kdcokenny 2026c) (MIT; 586 stars; no tagged release). A one-install bundle of 16 components: four plugins (workspace management, async delegation, notifications, git isolation), two npm plugins, three MCP servers (documentation, web search, code search), four specialist agents (researcher, coder, scribe, reviewer), four skill modules, and one command interface. Its sibling [opencode-background-agents](#) (kdcokenny 2026b) (MIT; 382 stars) provides `delegate()`, `delegation_read()`, and `delegation_list()` tools whose results persist to disk under `~/.local/share/opencode/delegations/`, so they survive context compaction; only read-only sub-agents may run in the background, because background sessions sit outside OpenCode's undo and branching system. *Useful to us?* Worth a trial as a lighter alternative to Oh My OpenCode; the disk-persisted delegation results address the same loss-on-compaction problem our lab notebook convention exists for.
- [micode](#) (MIT; 483 stars) and [octto](#) (MIT; 493 stars), from the same author: a brainstorm-plan-implement workflow with session continuity, and a browser UI that turns an agent's clarifying questions into multi-question forms. *Useful to us?* Possibly `octto`, which does for OpenCode what `AskUserQuestion` does in Claude Code.

- [opencode-conductor](#) (Apache-2.0; 129 stars; last push 2026-03-02) ports the Context, Spec, Plan, Implement lifecycle described in Section 62 to OpenCode. *Useful to us?* Only if that lifecycle is adopted; the port looks dormant.
- [planannotator](#) (Apache-2.0; about 8,600 stars; release v0.27.12 on 2026-09-03) is a visual plan-and-diff annotation tool with an OpenCode plugin; it is agent-agnostic and also targets Claude Code. *Useful to us?* Yes, for reviewing agent plans before implementation; it addresses the plan-review step directly.
- [opencode-goal-plugin](#) (MIT; 252 stars; release v0.10.0 on 2026-09-07) adds a session-scoped `/goal` that keeps an objective in context and auto-continues until “evidence-gated completion”. *Useful to us?* Maybe, for unattended ARDI-style loops.

Safety and observability

- [opencode-vibeguard](#) (inkdust2021 2026) (MIT; 198 stars; no tagged release; last push 2026-03-01). Replaces configured secrets and personal data with placeholders of the form `__VG_<CATEGORY>_<hash>__` before each request leaves for the model provider, restores them locally when output completes, and restores them again before tool execution so shell commands still work. *Useful to us?* Yes in principle, for work touching participant data, but the redaction list is hand-configured and the project has been quiet for six months; test it before relying on it.
- [opencode-shell-strategy](#) (MIT; 136 stars) and [opencode-pty](#) (MIT; 571 stars) tackle the same hang: an agent running a TTY-dependent command with no terminal attached. The first injects instructions to use non-interactive flags; the second gives the agent a real pseudo-terminal it can write to and read from. *Useful to us?* Yes; the hang they fix is one we hit regularly with R and `quarto preview`.
- [opencode-sentry-monitor](#) (53 stars), [opencode-wakatime](#) (198 stars), and [opencode-helicone-session](#) (16 stars) send traces or usage to Sentry, Wakatime, and Helicone respectively. *Useful to us?* No; we do not run those services.

Notifications and quality of life

Four plugins do desktop notifications ([opencode-notifier](#), 810 stars, is the most-used; [opencode-notify](#), [opencode-notificator](#)), and the rest of the list is small conveniences: [opencode-md-table-formatter](#) cleans up model-generated Markdown tables, [opencode-zellij-namer](#) names Zellij sessions, [opencode-supermemory](#) (about 1,600 stars) adds cross-session memory through the hosted Supermemory service, and [opencode-scheduler](#) runs recurring jobs through `launchd` or `systemd`. Vendor plugins from JFrog, Firecrawl, and Tavily wrap their own CLIs. *Useful to us?* A notifier, yes; the rest only with a specific need.

Notable clients and integrations

- [CodeNomad](#) (Neural Nomads AI 2026) (Neural Nomads AI; MIT; about 2,600 stars; release v0.19.0 on 2026-08-24; pushed 2026-09-09). Describes itself as “The AI Coding Cockpit for OpenCode”: a SolidJS front end with a Node.js server that wraps an OpenCode

CLI already on your `PATH`. It ships as Electron and Tauri desktop builds for macOS, Windows (x64 and ARM64), and Linux, and as a password-protected local web server (`npx @neuralnomads/codenomad --password <password> --launch`) for remote or browser access. Features include multi-instance workspaces, session management, git worktrees, voice input, a file browser, a command palette, and “SideCars” that embed local web tools as tabs. Despite the ecosystem page’s “Desktop, Web, Mobile and Remote” description, the README documents no native mobile app; mobile access is through the web server. *Useful to us?* Yes, as the most complete graphical front end for OpenCode and a plausible answer to “I want several OpenCode sessions side by side without `tmux`”.

- [OpenChamber](#) (OpenChamber 2026) (now under its own organization; MIT; about 9,700 stars; release `v1.22.2` on 2026-09-05). A workspace for running and reviewing agent work on desktop, web, VS Code, iOS, and Android, with session goals, a “multi-run” mode that runs the same task across up to five models, a guided changes walkthrough for large diffs, scheduled tasks, and an end-to-end encrypted “Private Relay” for remote connections. Like CodeNomad it uses OpenCode as the engine and is not affiliated with the OpenCode team. *Useful to us?* Yes; the multi-model run is a direct fit for our habit of getting a second model’s review.
- [OpenWork](#) (different-ai 2026) (different-ai; about 23,400 stars; release `v0.18.44` on 2026-09-09). Positions itself as “an open-source alternative to Claude Cowork and Codex” for macOS, Windows, and Linux, sharing skills, MCP servers, and connected services across tools and machines. Code outside `ee/` is MIT; the organizational control plane under `ee/` uses a separate license that is free for up to five users. *Useful to us?* Compare against the collaborative workspaces in Section 68; the five-user free tier fits a lab, and the open core makes it auditable.
- [opencode.nvim](#) (Dyke 2026) (Nick van Dyke; MIT; about 3,800 stars; release `v1.0.0` on 2026-08-20) connects Neovim to a running OpenCode server (`opencode --port`), injects editor context (cursor, selection, diagnostics) into prompts, surfaces OpenCode’s server-sent events as Neovim autocommands, and reloads buffers when the agent edits files. A second, unrelated [sudo-tee/opencode.nvim](#) (Apache-2.0; 937 stars) is a full Neovim front end rather than a bridge. *Useful to us?* For the Neovim users in the lab, yes.
- [kimaki](#) (remorses 2026) (MIT; about 1,400 stars; release `kimaki@0.27.0` on 2026-09-01). A Discord bot in which each project is a channel and each session a thread; it queues messages, forks sessions, transcribes voice messages, shows diffs, and maps Discord roles to permission controls. *Useful to us?* No for now; our coordination runs through GitHub, not Discord.
- [portal](#) (hosenur 2026) (MIT; 798 stars; last push 2026-05-12) is a mobile-first web UI meant to be reached over Tailscale. [OpenCode-Obsidian](#) (MIT; about 1,100 stars) embeds OpenCode in Obsidian’s sidebar. *Useful to us?* Niche; CodeNomad and OpenChamber cover the remote-access case with more activity.
- [ocx](#) (kdcokenny 2026a) (MIT; 942 stars; release `v2.0.15` on 2026-08-17). An extension manager with portable, isolated profiles: `ocx profile add` installs a profile from a registry, `ocx oc -p <name>` launches OpenCode with it, and components are copied

into `.opencode/` (the shadcn model) rather than hidden in dependencies, with SHA verification of what is installed. *Useful to us?* Yes, if we standardize an OpenCode profile for the lab; it is the closest thing to the plugin marketplace we use for Claude Code (Section 63).

- [ai-sdk-provider-opencode-sdk](#) (MIT; 116 stars) exposes OpenCode’s configured providers to the Vercel AI SDK, and [opencode-plugin-template](#) (archived) was the scaffold for writing plugins. *Useful to us?* Only when building on the SDK.

Agent bundles

- [OpenAgentsControl](#) (listed as `opencode-agents`; MIT; about 4,800 stars; release v0.7.1 on 2026-01-30) is a plan-first framework with approval-gated execution and built-in test, review, and validation steps.
- [agentic](#) (MIT; 638 stars; last push 2025-09-02) is a context-engineering toolkit that appears dormant.

Useful to us? `OpenAgentsControl` overlaps heavily with Oh My OpenCode (Section 65) and `opencode-workspace`; pick one such harness rather than layering them.

OpenCode Zen and OpenCode Go

The ecosystem page does not describe them, but two first-party paid services sit alongside the community projects:

- **OpenCode Zen** (OpenCode 2026e) is the maintainers’ own model gateway: a curated set of models tested against coding-agent workloads, billed pay-as-you-go per million tokens from a prepaid balance (with optional auto-reload, by default \$20 whenever the balance drops below \$5). As of 2026-09-09 it lists six free models under limited-time trials, including Big Pickle, MiMo-V2.5 Free, and Nemotron 3 Ultra Free, beside paid Claude, GPT, Gemini, Grok, Qwen, DeepSeek, Kimi, and GLM models.
- **OpenCode Go** (OpenCode 2026c) is a \$10-per-month subscription to 30-plus open-weight coding models (Qwen, DeepSeek, Kimi, GLM, MiMo, Grok, and others) with usage caps expressed in dollar value: \$12 per five hours, \$30 per week, \$60 per month. When a cap is hit, an optional “Use balance” setting falls back to Zen credits. Only one member per workspace can hold the subscription.

Useful to us? Yes; OpenCode Go is already one of the lab’s delegation budgets, and the Zen free tier is a zero-cost way to try a new open-weight model before routing work to it (compare the `OpenRouter :free` models in Section 26).

Summary

Need	Start with	Also see
Graphical or remote front end	CodeNomad, OpenChamber	portal, OpenWork
Multi-agent orchestration	Oh My OpenCode (Section 65)	opencode-workspace, OpenAgentsControl
Cheaper tokens	OpenCode Go, Zen free models	opencode-dynamic- context-pruning
Isolation per task	opencode-worktree, opencode-devcontainers	opencode-daytona
Redaction of sensitive data	opencode-vibeguard	(none)
Plan review before implementation	plannotator, octto	opencode-conductor
Shareable lab configuration	ocx	(none)

Skip the subscription-bridge plugins.

67 Managing Gemini API Spend and Cost Optimization

This guide describes how to manage Google AI Studio and Google Cloud Gemini API spend caps, unpause paused API services, and optimize token consumption across local tools and GitHub Actions workflows.

Identifying Paused Projects (Project Numbers vs. Friendly Names)

When a project reaches its monthly budget limit, Google sends an email notification stating that Gemini API service has been paused. These email notifications identify the affected project using its internal **numerical GCP Project Number** (e.g. 156839315029).

In contrast, [Google AI Studio](#) lists projects by their **friendly display names** (such as `ai-config Project` or `gha-project`) and alphanumeric client IDs (`gen-lang-client-...`).

To find which project in AI Studio corresponds to the notification email:

1. Open [Google AI Studio Projects](#).
2. Locate the project corresponding to your client ID or spend alert.
3. Any project that has hit its monthly spend cap will have its API requests paused until the limit is updated.

Adjusting and Unpausing Spend Caps

To restore API access for a paused project:

1. Open [Google AI Studio Spend](#).
2. Increase the monthly spend cap dollar amount or set it to unlimited.
3. Save your changes. API requests will automatically resume within a few minutes.

If no manual action is taken, accumulated spend resets to **\$0 on the 1st of the next month**, and API service automatically resumes up to the configured cap.

Setting Up Google Cloud Budget Alerts

To receive early warnings before reaching a spend cap:

1. Open [Google Cloud Console Billing Budgets & Alerts](#).
2. Select your billing account and click **Create Budget**.
3. Name the budget (e.g., **Lab AI API Monthly Budget**) and select the relevant GCP projects.
4. Set your target monthly budget amount.
5. Configure trigger rules for email notifications at **50%**, **75%**, and **90%** of the budget.
6. Click **Finish**. You will receive email alerts before any project hits its spend cap.

Cost Optimization Best Practices

To maximize the efficiency of your API spend across local CLI sessions, subagents, and automated workflows:

- **Right-Size Model Selection:** For general Gemini API, Python SDK, or custom scripts, prefer Flash-tier models (such as `gemini-2.5-flash`) over Pro-tier models (`gemini-2.5-pro`). Flash models provide a substantially lower token cost for routine search, log parsing, and background processing. For Antigravity Agent workflows (`google-antigravity`), the agent defaults to `gemini-3.7-flash` (already a Flash-tier model). Supported Antigravity Agent model options include `gemini-3.7-flash`, `gemini-3.6-flash`, `gemini-3.5-flash`, and `gemini-3.5-flash-lite`.
- **Use Context Caching:** For workloads that repeatedly pass static context over 2,048 tokens (such as large reference docs, system prompts, or codebase indices), use Gemini Context Caching. Cached input tokens receive a substantial discount compared to standard input tokens.
- **Use the Batch API for Non-Realtime Tasks:** For offline batch processing, evaluation suites, or background doc updates, submit requests via the Gemini Batch API to receive a 50% discount on input and output tokens.

- **GitHub UI Diff Collapsing:** Mark dependency lockfiles (`*.lock`, `package-lock.json`, `yarn.lock`, `renv.lock`) and generated build artifacts as `linguist-generated=true` in `.gitattributes` to collapse them in GitHub's web diff view and exclude them from repository language statistics.

68 Collaborative AI Workspaces: Claude Cowork and Gemini Spark

The 2026 AI ecosystem has expanded beyond reactive chat windows and command-line coding orchestrators into collaborative workspace agents (measured 2026-09-01). These systems operate directly on multi-file workspaces, desktop applications, and cloud productivity suites to automate complex, multi-step analytical and administrative workflows.

Claude Cowork

[Claude Cowork](#) (Anthropic 2026r) (also discussed in [?@sec-claude-cowork](#)) is Anthropic's desktop-native agent architecture designed to collaborate directly within local folders and desktop application environments:

- **Local Sandboxed Execution:** Cowork runs in a sandboxed virtual environment on the host machine, reading, modifying, and creating local files (spreadsheets, markdown manuscripts, datasets) without requiring manual file uploads or cloud synchronizations.
- **Desktop Application and Browser Interaction:** Beyond static file manipulation, Cowork interfaces with local applications and browser sessions (via [Claude in Chrome](#)) to execute multi-application workflows.
- **Persistent Project Context:** Cowork organizes workspaces into persistent Projects that retain custom instructions, reference documentation, and project memory across sessions.
- **Best Use Cases:** Knowledge work with heavy local file management, local data hygiene, and multi-document manuscript preparation where code and data are managed directly on the workstation.

Gemini Spark

Gemini Spark (Google) represents a cloud-native, always-on agent paradigm deeply integrated into the Google Workspace ecosystem:

- **Asynchronous 24/7 Cloud Agency:** Unlike desktop agents that terminate when the terminal closes or the laptop sleeps, Gemini Spark executes continuously on Google Cloud infrastructure. It continues running scheduled workflows, monitoring inputs, and synthesizing outputs even when the user is disconnected.

- **Google Workspace Ecosystem Integration:** Spark natively interfaces with Gmail, Google Drive, Google Docs, Google Sheets, and Google Calendar to execute automated organizational workflows (e.g., aggregating weekly updates, auditing spreadsheets, and triggering reminder notifications).
- **Proactive Event Triggers:** Spark initiates actions based on calendar milestones, email arrivals, or document modifications rather than waiting for an explicit interactive prompt.
- **Best Use Cases:** Asynchronous operational monitoring, automated team coordination, and cross-document synthesis across cloud-hosted Google Workspace environments.

Related Collaborative Workspace Offerings

Similar collaborative workspace paradigms have emerged across other frontier ecosystems:

- **ChatGPT Work and OpenAI Canvas:** [ChatGPT Work](#) (OpenAI 2026e) (see [?@sec-chatgpt-work](#)) and Canvas provide side-by-side document and code editing with inline line-level revisions, interactive targeted edits, and multi-file artifact tracking.
- **Cursor and Google Antigravity Agent Workspaces:** Developer-centric workspace agents providing multi-agent delegation, worktree isolation, and structured planning workflows (such as Conductor extension spec-driven development, Section 62).
- **Notion AI and Microsoft Copilot Studio:** Enterprise knowledge graph agents designed for querying organizational wikis and automating business process workflows.

Comparative Taxonomy of Workspace Agents

Dimension	Claude Cowork	Gemini Spark	OpenAI Canvas / ChatGPT Work	Developer Workspace Agents (Antigravity / Cursor)
Primary Architecture	Desktop-native (sandboxed)	Cloud-native (always-on)	Cloud web interactive	Desktop IDE / CLI workspace
Execution Lifetime	Active session / local machine	24/7 continuous cloud	Interactive turn-by-turn	Session-scoped multi-agent
Data Locality	Local files (cloud sessions on some plans)	Google Workspace cloud	OpenAI cloud sandbox	Local repository worktrees
Primary Domain	File and document workflows	Cloud suite coordination	Document and text drafting	Codebase refactoring and review
Key Strength	Local file privacy and tool use	Autonomous background monitoring	Real-time targeted line edits	Deterministic builds, tests, git workflows

Practical Guidance for Research Teams

When selecting a collaborative workspace agent for academic and computational research:

1. **Evaluate Data Confidentiality and Cloud Transmission:** For private research datasets and unpublished manuscripts with confidentiality constraints, desktop-native environments operate on local files under user supervision, though any integration that calls cloud APIs or browser extensions still transmits prompt context to provider infrastructure.
2. **Use Cloud Agents for Asynchronous Administration:** For non-sensitive scheduling, lab meeting summaries, and routine administrative reminders, always-on cloud agents (like Gemini Spark) reduce cognitive overhead by automating background tasks.
3. **Keep Code and Statistical Analyses in Version Control:** While workspace agents are effective for document drafting, statistical pipelines and computational scripts should remain in dedicated git-managed repositories with automated CI validation and code review.

69 AI-Enabled Knowledge Workspaces: Notion and Alternatives

Notion and its competitors sell a hosted wiki, database, and document editor with an AI layer on top. This section reviews Notion’s AI features and the closest alternatives from the perspective of a lab that already keeps its notes in git-backed Quarto and Markdown (this site, and the [Benchbook](#) pattern). The question is not whether these products are good — most are polished — but whether any of them beats a plain-text wiki that a coding agent can already read, write, and version. Prices and feature lists below were read from each vendor’s own pages and are volatile (measured 2026-09-09).

Notion

Notion’s AI surface has three layers (Notion Labs 2026b):

- **Notion AI** in the editor: drafting, rewriting, summarizing, and auto-filling database properties.
- **Notion Agent** and **Custom Agents**: a chat agent that runs multi-step tasks across the workspace and connected apps (Slack, Google Drive, GitHub), plus scheduled or triggered agents that run without a user present. Custom Agents were free through 2026-05-03 and now bill by credit, at \$10 per 1,000 credits, on Business and Enterprise plans.
- **AI Meeting Notes** and **Enterprise Search** across connected sources.

For agent access from outside Notion there are two routes. The public REST API exposes pages, databases, blocks, users, and search under internal, OAuth, or personal access tokens (API version 2026-03-11) (Notion Labs 2026c). The Notion MCP server is a remote server

hosted by Notion, authorized by OAuth and scoped to what the user can already see, that Claude Code, Cursor, and Codex can connect to (Notion Labs 2026e). It exposes tools for search, fetch, page and database creation and update, comments, file uploads, and custom-agent sessions, rate-limited to 180 requests per minute per user (30 for keyword search), with some tools gated behind Business or Enterprise plans (Notion Labs 2026f).

Pricing: Free, Plus at \$10 per member per month, Business at \$20, Enterprise on request. Full Notion Agent, Custom Agents, and Meeting Notes require Business or above; Free and Plus get a limited AI trial (Notion Labs 2026g). Students and educators at accredited universities get the Plus plan free for a one-member workspace, re-verified annually (Notion Labs 2026d). That is a personal plan: a shared lab workspace with AI would be Business at \$20 per member per month.

Export: pages export as Markdown, databases as CSV plus Markdown subpages, and whole pages or workspaces as HTML or PDF. Comments survive only in HTML; only the current database view exports; “include subpages” for full exports is a Business or Enterprise feature; a workspace export can take up to 30 hours; and workspace owners can disable export entirely (Notion Labs 2026a).

Obsidian

Obsidian is the opposite design: a local Markdown folder with an editor on top. Notes are plain files on disk, so any coding agent can already read and write them with no API at all, and a vault can live in a git repository. Obsidian has no first-party AI features (measured 2026-09-09); the AI plugins are community-maintained (Obsidian 2026c). The official Obsidian CLI (Obsidian 1.12 or later, enabled in Settings, with the app running) exposes search, note creation, tasks, and developer commands, and the documentation names “agentic coding tools” as an intended user (Obsidian 2026a). MCP servers for Obsidian are community projects, several of which wrap that CLI.

Pricing: the app is free for personal and commercial use; a \$50 per user per year commercial license is optional. Sync (\$4 per month annual) and Publish (\$8 per month annual) are optional subscriptions, with a 40% discount for students, faculty, and nonprofit staff (Obsidian 2026b). Export is a non-question: the vault is the export.

Coda, now Superhuman Docs

Coda has been renamed Superhuman Docs; coda.io/pricing now redirects to superhuman.com/plans/docs (Superhuman 2026). The vendor pages are rendered client-side and could not be read in full for this review, so the numbers below come from a third-party summary (Breeze 2026): Pro at \$15 per Doc Maker per month (\$12 annual), Business at \$40 (\$33 annual), Free and Enterprise unchanged, with a rename date of 2026-07-08. The billing unit is the “Doc Maker”, so viewers and editors of existing docs are free. Treat the

product as in transition: a rename and a price increase in one quarter is a lock-in signal in itself.

Google Workspace with Gemini

Gemini is bundled into every Google Workspace plan rather than sold as an add-on: Starter (\$7 per user per month list) has limited Gemini in Gmail, Standard (\$14) and above have Gemini across Docs, Sheets, Meet, and Gemini Notebook (Google 2026g). For universities, Education Fundamentals is free and includes Gemini for Education and Gemini Notebook, though in-app Gemini in Docs and Sheets needs Education Plus (\$6 per user per year); the Google AI Pro for Education add-on is \$15 per user per month on an annual commitment (Google 2026a). Agent access is through the Docs, Sheets, and Drive APIs, which let a program create and edit document structure (Google 2026f); Google's pricing and API pages reviewed here mention no hosted MCP server for Docs (measured 2026-09-09). Google's always-on Gemini Spark agent is covered in Section 68. Lock-in is moderate: a Doc exports as a file, but comments, suggestions, and version history stay in Google's copy. Where the campus already licenses Workspace, the marginal cost to the lab is whatever that contract allows.

Microsoft Loop and OneNote with Copilot

Copilot in Loop offers preset prompts (Create, Brainstorm, Blueprint, Describe) and page summaries inside Loop workspaces (Microsoft 2026d), and Microsoft positions Loop alongside Copilot Pages (canvases for saving AI output) and Copilot Notebooks (grounded research spaces) (Microsoft 2026a). All of it requires a Microsoft 365 Copilot license on top of a qualifying base plan. Copilot Business is \$18 per user per month (annual, reduced from \$21) for small-business plans; the enterprise add-on is listed separately (Microsoft 2026g). Academic Copilot licenses exist for A1, A3, and A5 plans through volume licensing (Microsoft 2026f). None of the Microsoft pages reviewed here offers an MCP server for Loop or OneNote, and neither page describes a Markdown export, which makes this the weakest export story in this table. The .docx route for reviewer edits, described earlier in this chapter, remains the practical bridge to Word users.

Confluence with Roov

Confluence Cloud is free for up to 10 users; Standard and Premium are per-user and include Roov (search, chat, and agents) with 25 and 70 Roov credits per user per month respectively (Atlassian 2026e, 2026d). Credits are pooled at the organization level, reset monthly, and overage billing at \$0.01 per credit starts 2026-12-03 (Atlassian 2026d). Atlassian's academic licensing gives qualifying institutions 50% off cloud list price (Atlassian 2026a). The Atlassian Roov MCP Server is hosted at mcp.atlassian.com, authorizes by OAuth 2.1 or API token, and lets Claude, ChatGPT, GitHub Copilot CLI, and Gemini search, summarize, create, and update Confluence pages and Jira work items within the user's existing permissions (Atlassian

2026c). Export is per page to Word or PDF, and per space to PDF, CSV, HTML, or XML (Atlassian 2026b); there is no Markdown export, so leaving Confluence means converting HTML.

Anytype

Anytype is a local-first, source-available (Any Source Available License 1.0) knowledge base with zero-knowledge encrypted peer-to-peer sync (Any Association 2026b). It has no built-in AI. Instead it ships a Local API on `localhost`, authenticated by an API key generated from a challenge in the desktop app (Any Association 2026d), and an “Agents’ Skill” that lets Claude Code, Cursor, Gemini CLI, or GitHub Copilot run sandboxed JavaScript against that API to read, search, create, and bulk-edit objects (Any Association 2026a). Community MCP servers wrap the same API. Membership is a free tier plus paid storage tiers, with a 50% student discount (Any Association 2026e). Export is Markdown, PDF, HTML, or Anytype’s own Any-Block format, in JSON or Protobuf (Any Association 2026c).

Craft

Craft is a native document editor (macOS, iOS, Windows, web) with a credit-metered AI assistant: 15 credits on Free, 50 per month on Plus (\$6.40 per month annual), with a Team plan at \$50 per month for up to 10 seats (Craft Docs 2026a). It offers a hosted MCP endpoint per connection with user-defined scope, which Claude Desktop, Claude Code, ChatGPT, Cursor, VS Code, and other clients can attach to, and which since version 3.3.5 can create and update documents, not only read them (Craft Docs 2026c). Export is Markdown, `TextBundle`, PDF, and image on every platform, plus `.docx` on macOS, iOS, and web (Craft Docs 2026b). No education discount is listed.

Comparison

Product	AI features	Agent / MCP access	Export and lock-in	Academic pricing	Verdict
Notion	Notion AI, Notion Agent, Custom Agents (credits), meeting notes, enterprise search	Hosted MCP server (OAuth); REST API	Mark- down/CSV/HTML/ comments and views partly lost; export can be disabled by admins	Plus free for student/edu- cator; shared AI workspace is \$20 per member per month	Best hosted option; still a walled garden

Product	AI features	Agent / MCP access	Export and lock-in	Academic pricing	Verdict
Obsidian	None first-party; community plugins	Local files; official CLI; community MCP servers	Vault is plain Markdown on disk	Free app; 40% off Sync/Publish	Closest to what we already do
Superhuman Docs (Coda)	Coda AI, in transition	Packs and API	Per-doc export; rebrand and repricing in 2026	None listed	Avoid until it settles
Google Workspace + Gemini	Gemini in Docs/Sheets/ Gmail; Gemini Notebook, Spark agent	Docs/Drive APIs; no hosted MCP documented	File export; comments and history stay behind	Fundamentals free; Plus \$6 per user per year; AI Pro add-on \$15 per month	Use what the campus already licenses
Microsoft Loop / OneNote + Copilot	Copilot prompts and summaries; Copilot Pages and Notebooks	No MCP server documented	No Markdown export documented; weakest export	Academic Copilot via A3/A5 volume licensing	Only if the collaborators live in Teams
Confluence + Rovo	Rovo search, chat, agents (credit-metered, overage from 2026-12)	Hosted MCP server (OAuth 2.1)	Word/PDF per page, HTML/XML per space; no Markdown	50% off cloud list	Fine for Jira shops; poor for prose
Anytype	None built in	Local API with key; agent skill; community MCP	Mark-down/HTML/PDF/Block; local-first, encrypted	50% student discount	Interesting, niche
Craft	Credit-metered assistant	Hosted MCP with scoped connections	Mark-down/TextBundle/PDF/.docx	None listed	Nice editor, small ecosystem

Useful to us?

Mostly no, and for a structural reason rather than a feature gap. Every product above adds AI by putting a model *inside* the vendor’s editor and then, more recently, adding an MCP server so outside agents can reach in. A git-backed Quarto or Markdown wiki inverts that: the agent is already a first-class editor of the files, every change is a reviewable diff, CI renders and spell-checks the result, and the wiki is the export. [Benchbook](#) is the same idea with a written contract for the agent; this site is the same idea with a publishing pipeline. Neither needs an MCP server to let Claude Code read the lab’s notes, and neither can be locked, metered, or renamed by a vendor.

What the hosted products do better, and a plain repository does not:

- **Non-technical collaborators.** Notion, Google Docs, and Loop have editors that a clinician or a first-year student can use without learning git. The `.docx` round-trip described earlier in this chapter covers review comments, but not live co-editing.
- **Databases with views.** Notion and Coda tables with filters, rollups, and forms have no clean Markdown equivalent; a CSV in a repo is the honest substitute.
- **Meeting capture.** Notion’s AI Meeting Notes and Copilot’s transcription are convenient; the output is text and can be committed afterward.
- **Always-on agents.** Notion Custom Agents, Gemini Spark, and Rovo agents run on a schedule without a machine of ours awake. A GitHub Actions cron running a coding agent does the same for a repository.

Recommendations (measured 2026-09-09):

- Keep lab knowledge in git-backed Markdown or Quarto, and let coding agents edit it directly.
- If a hosted editor is needed for a specific collaborator group, prefer the one the campus already licenses (Google Workspace with Gemini) over a new per-seat subscription, and treat its content as a draft that gets committed.
- Among the standalone products, Obsidian is the only one whose data model is already ours, so a lab member who wants a graphical editor over the same repository should start there.
- Notion is the best of the hosted walled gardens, and the Notion MCP server makes it reachable from Claude Code; but the AI features that distinguish it are Business-plan and credit-metered, and export loses structure, so budget for the exit before entering.
- Re-check prices and plan boundaries before acting on any row of the table; several changed within the last quarter.

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